

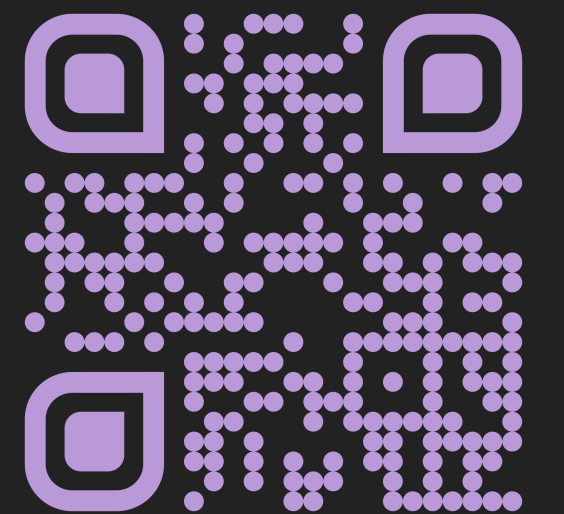
A/B-Testing

...Bayesianand Bandits

RSSS - 25

Morten Arngren
Head of Data Science

Code and deck
available on GitHub



[https://
github.com/
Arngren](https://github.com/Arngren)



Morten Arngren

Head of Data Science at Anthill Agency
(GenAI within marketing and pharma)

AnthillTM

PhD in Machine Learning in 2011

ex-VML

ex-Adform

ex-Issuu (My first Rec. Engine)

ex-Nokia

Dad | Snowboarder

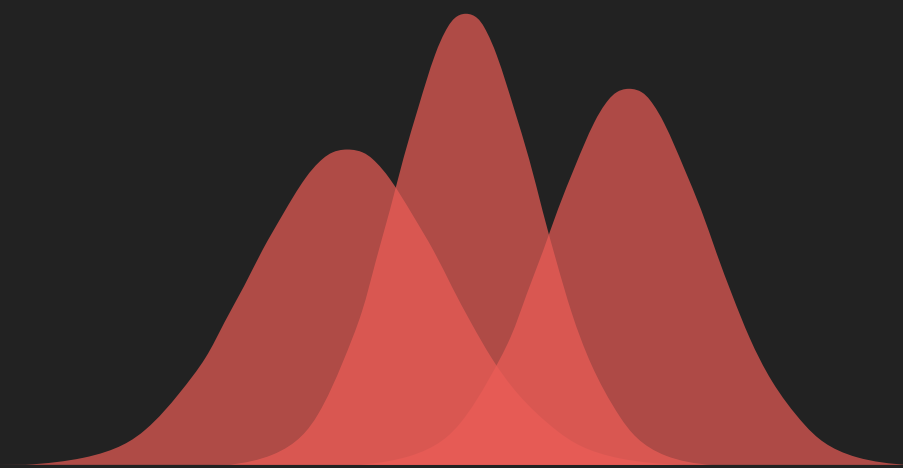




Classic & Complex
A/B testing

Hypothesis testing and
the infamous

p-value



Bayesian A/B testing

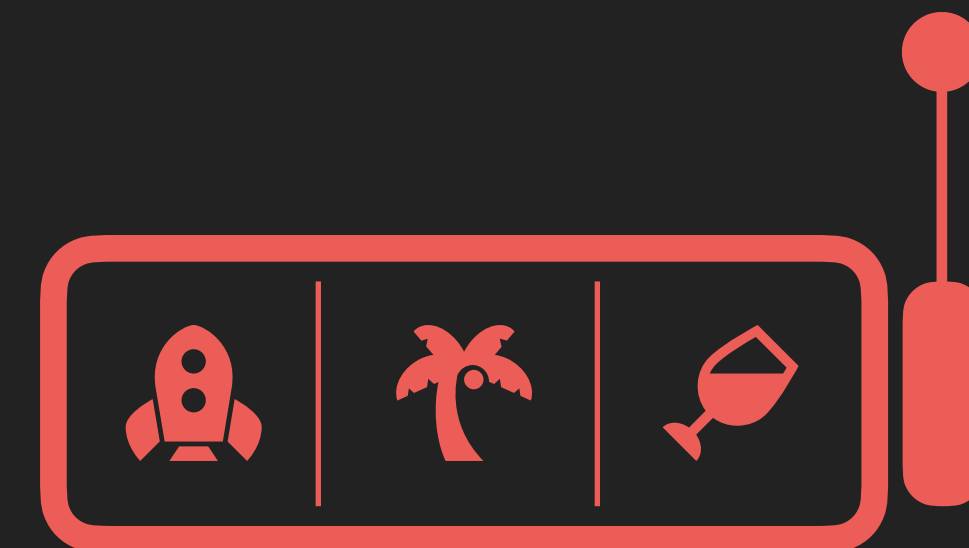
A/A-testing

Bayesian evaluation



Campaign

Simulation



Bandit testing

Dynamic A/B-testing



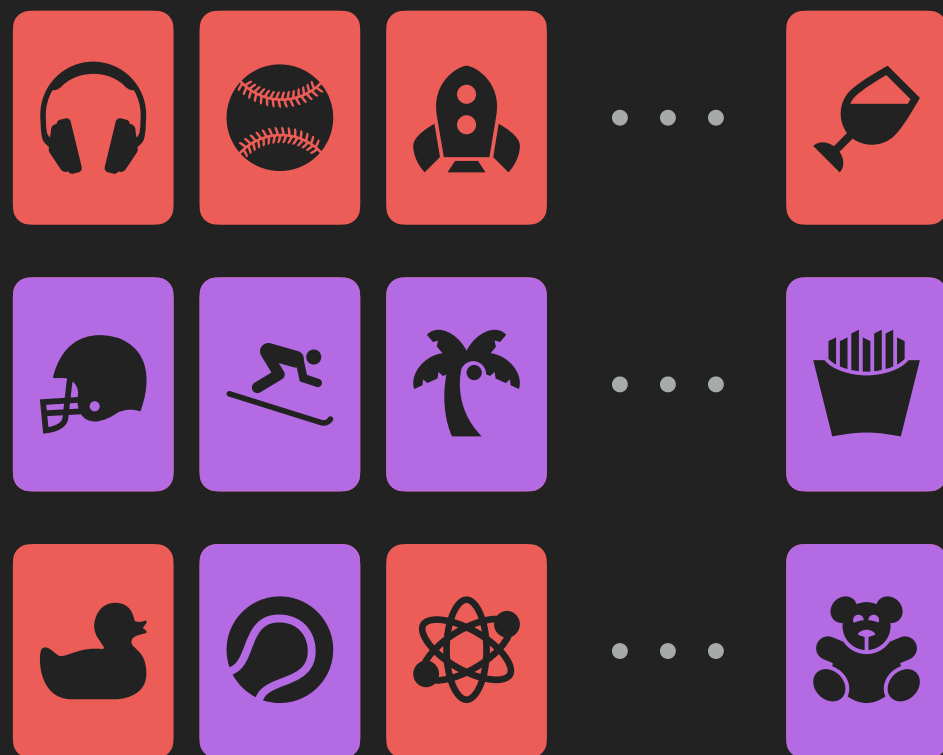
The stage

The stage

Two Rec. Engines

Movie Streaming

Recommendations



Slates



Running in production

New developed Rec. Engine



which has
the best
performance
?

News site

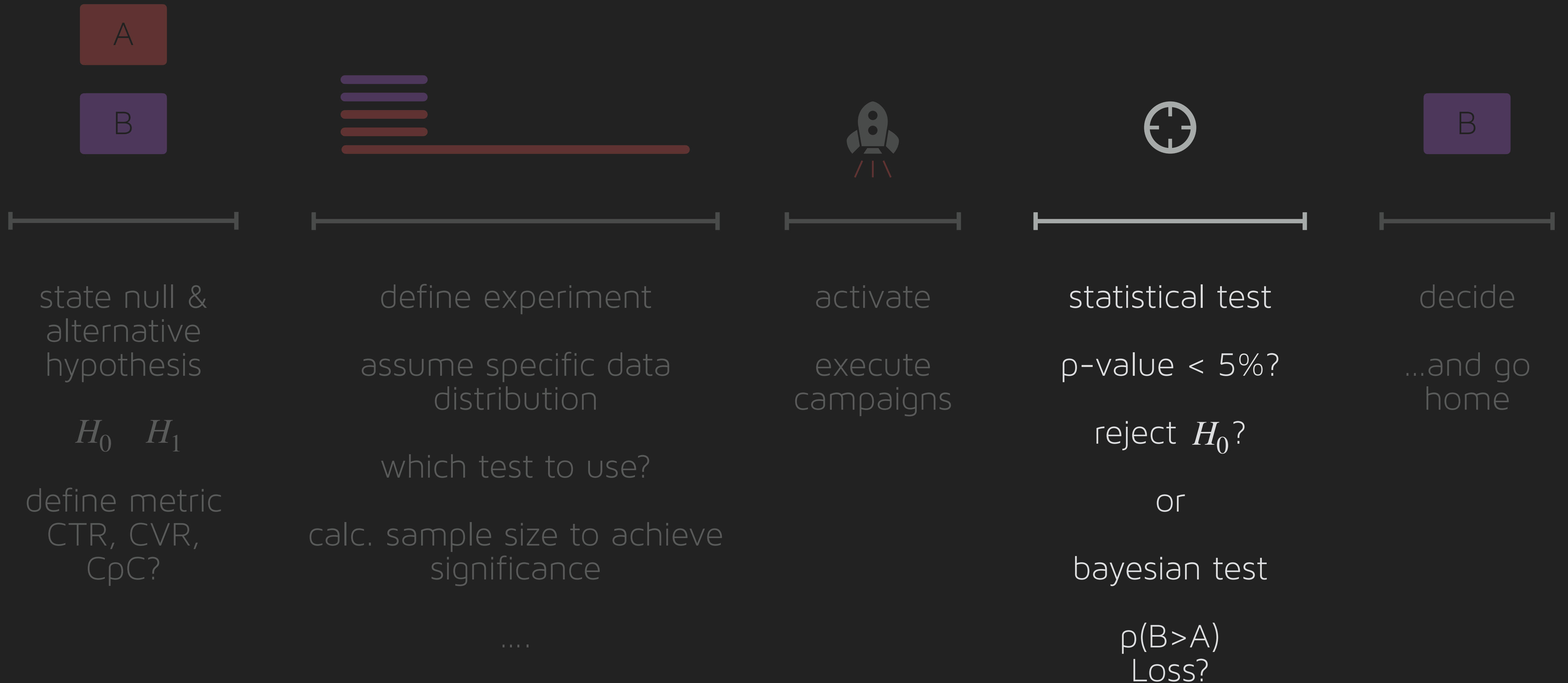
Advertisement



Advertisements

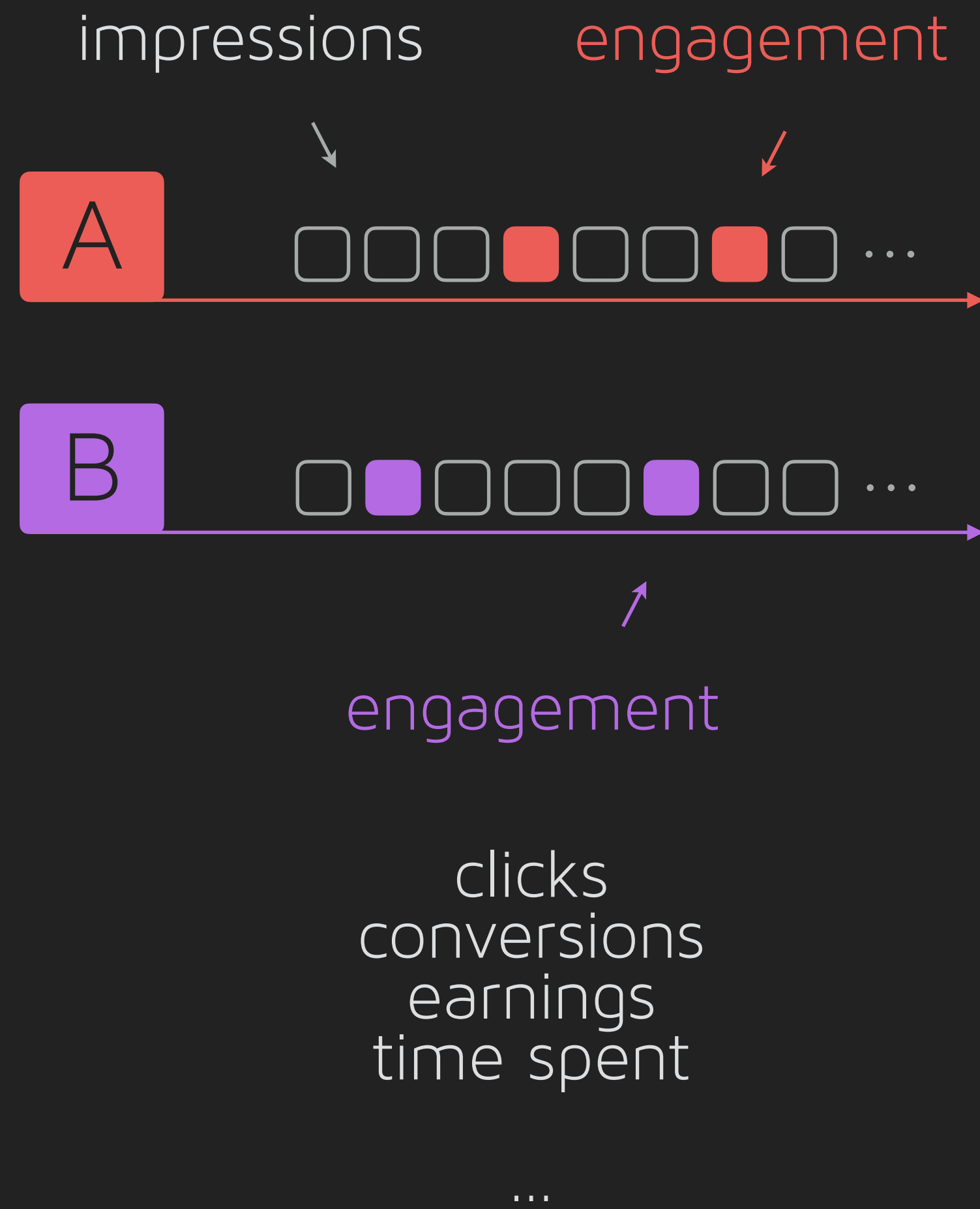
A/B testing

Classic approach



The stage

logs



business metrics

(Simple)

CTR	Click-Through-Rate Fraction of clicks? [%]
CVR	Conversion Rate Product buys [%]
CpC	Cost-per-Click Price of a click [\$]
???	Webshop margins Earnings [\$]
...	

Attributions can be difficult

Clicks can leave eco system...

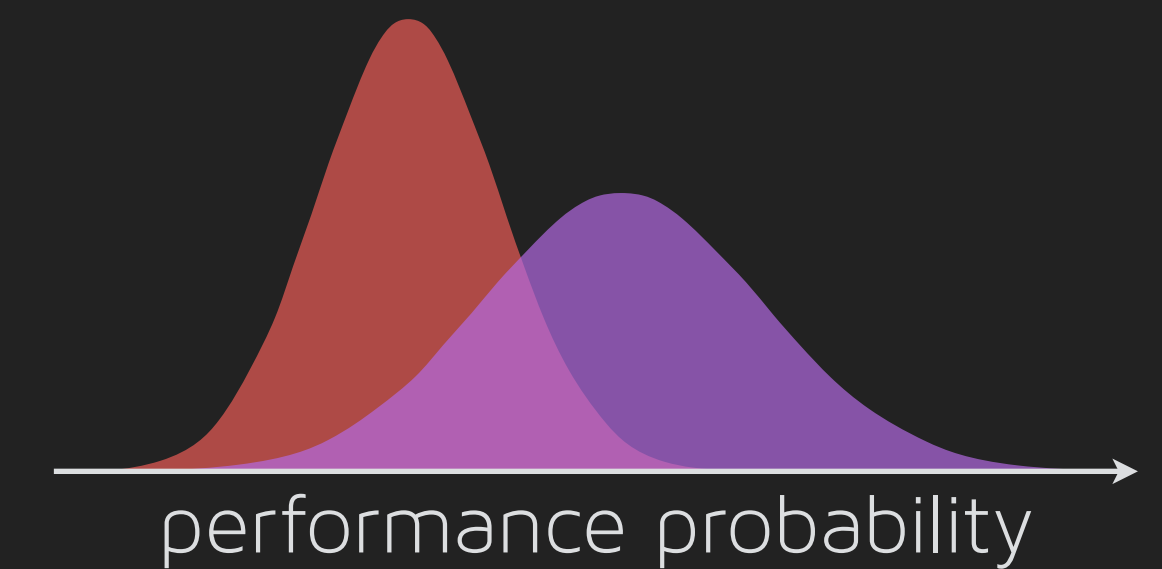
Privacy ensure no tracking

evaluation

Hypothesis testing
 $p\text{-value} < 5\%$

or

Bayesian testing

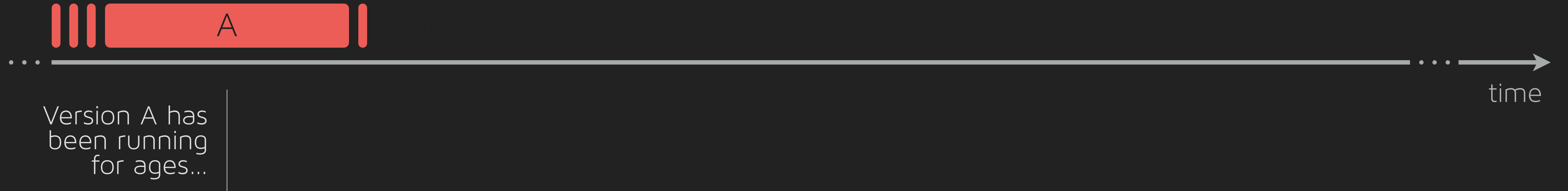




Introduction

Classic A/B-test

Timeline...



“What would be the most basic approach?”

Complex A/B-test

...or the more right way...

~20%

Reserve

don't spend it all on testing,
what if B sucks....

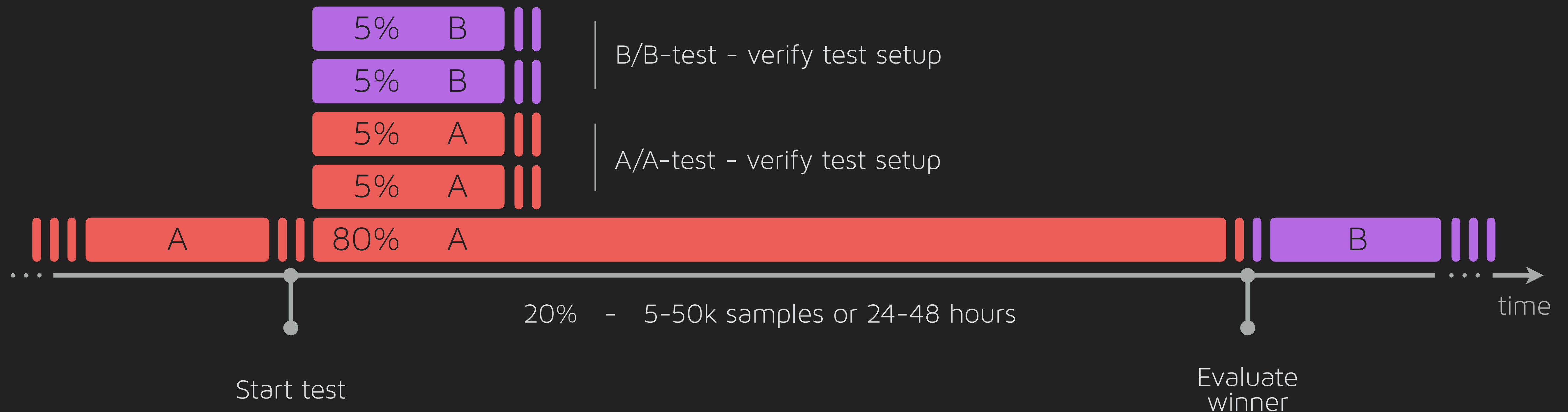
consider business requirements
and price...!

A/A test

Verify

split into several smaller tracks

conducting an A/A test to verify
the whole test setup



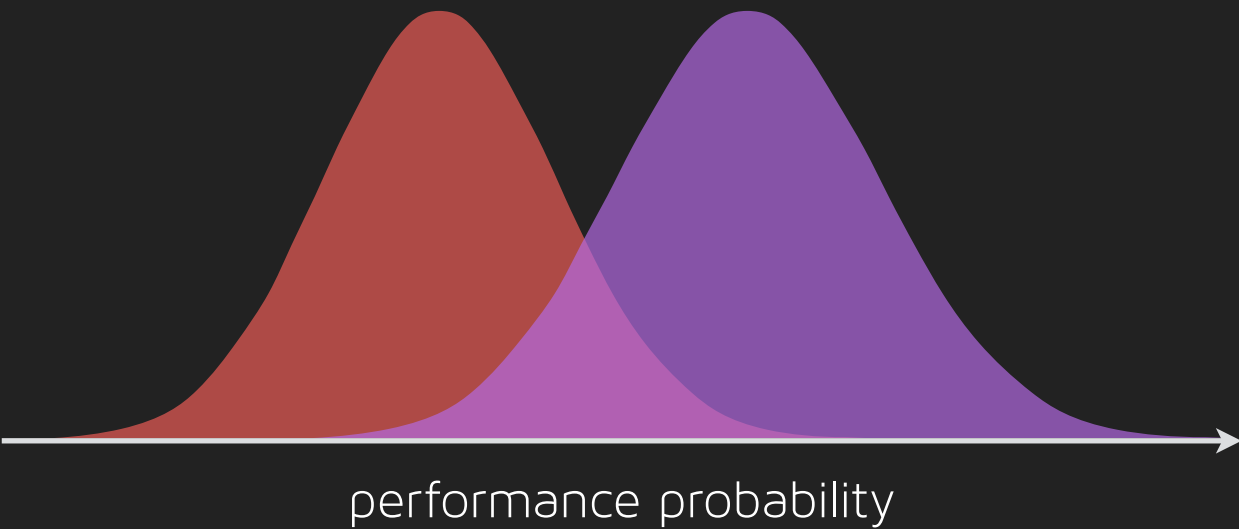
Bayesian A/B-test

Bayesian evaluating - probability of winner

models performance as probability distributions

captures the epistemic uncertainty

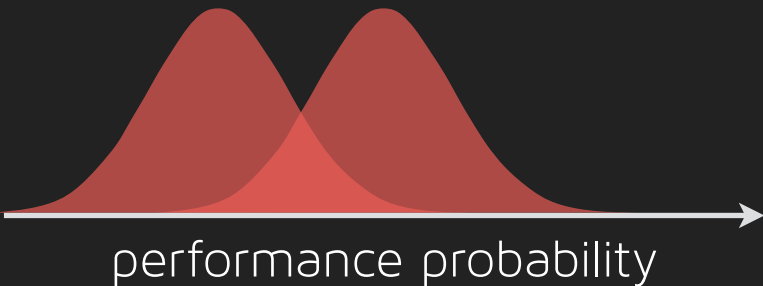
allow us to quantify the confidence



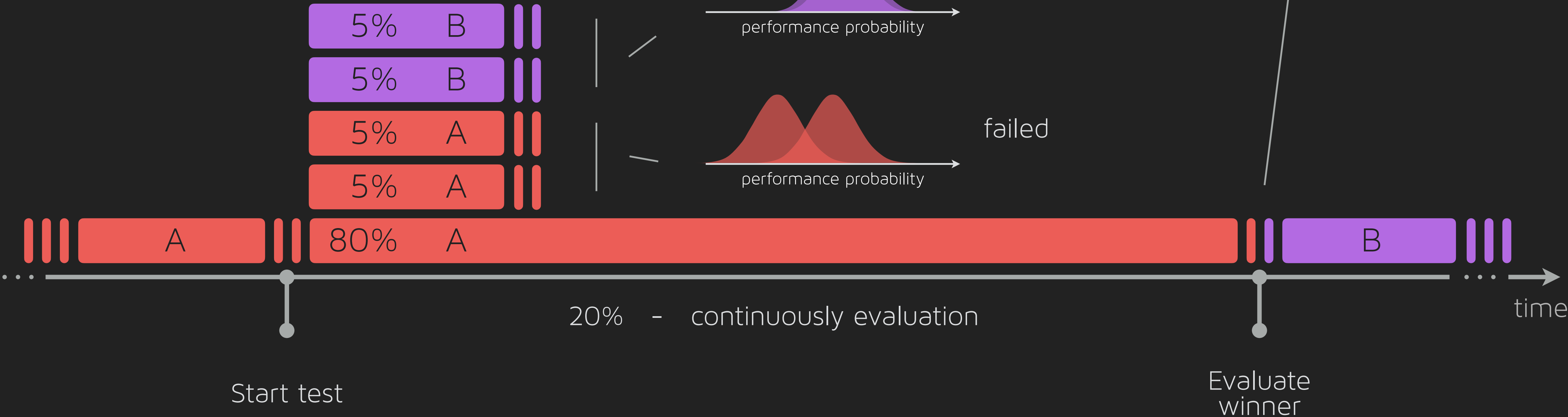
A/A-tests



successful



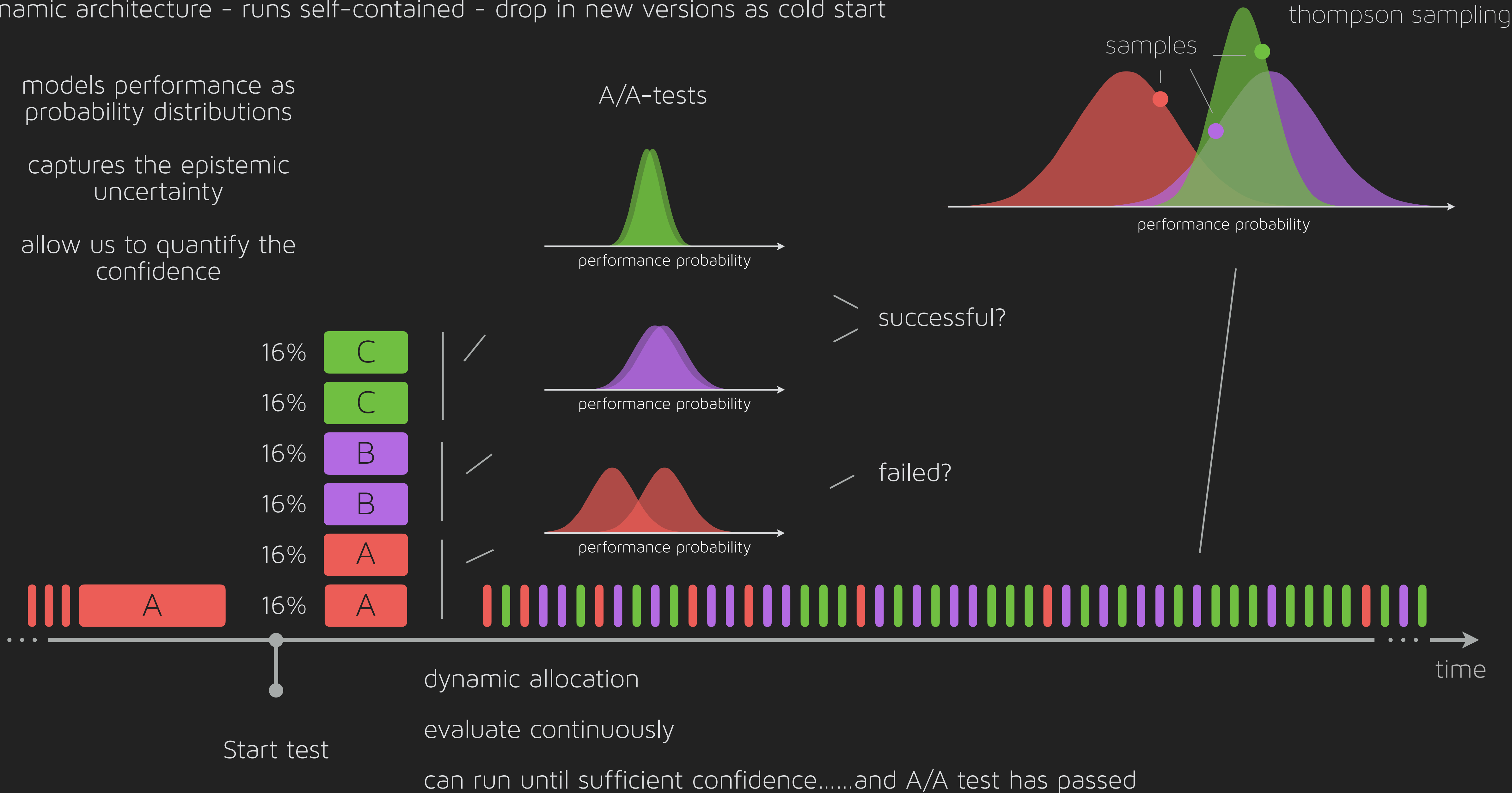
failed



Bandit A/B-test

Dynamic architecture - runs self-contained - drop in new versions as cold start

- models performance as probability distributions
- captures the epistemic uncertainty
- allow us to quantify the confidence

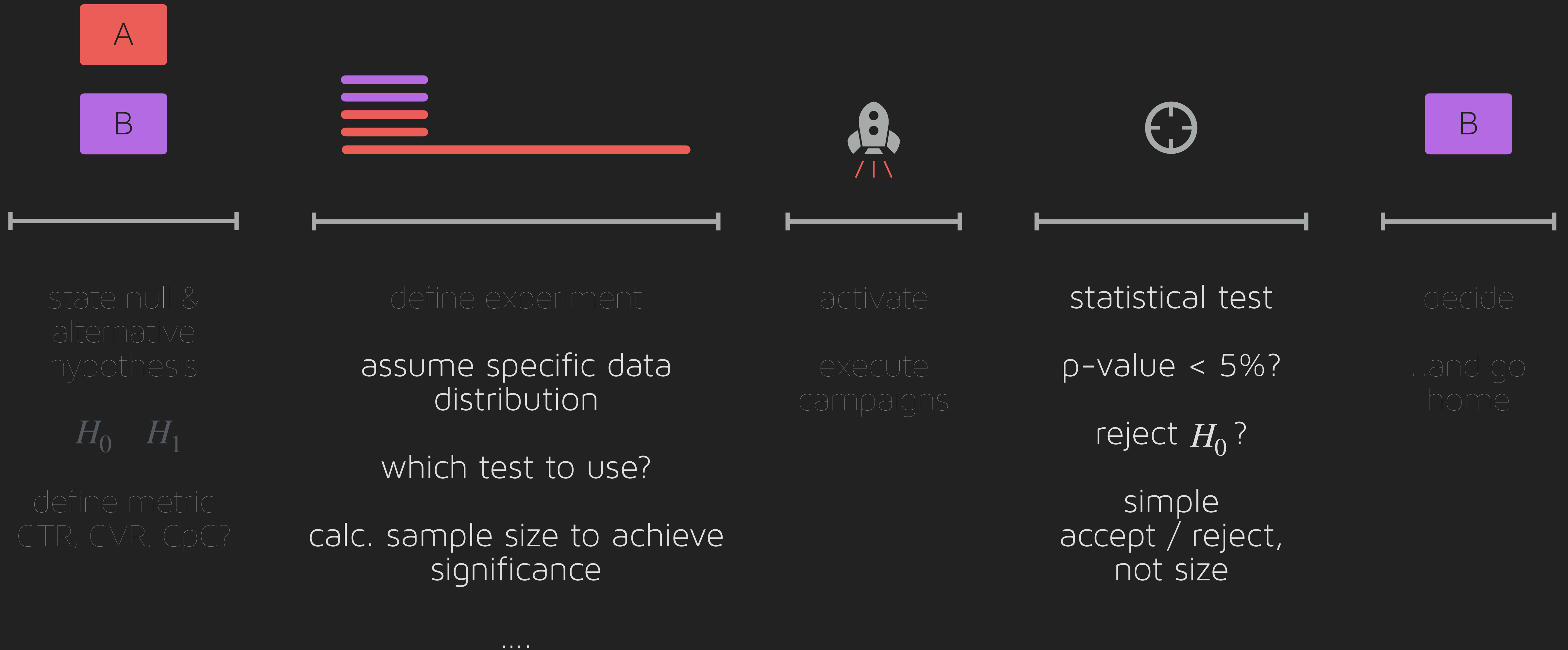




Hypothesis Testing

Hypothesis significance testing

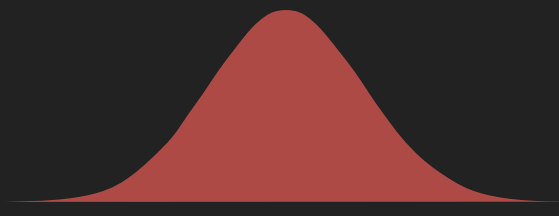
What's specific to it...?



Hypothesis significance testing

Popular hypothesis tests...

Students T-Test



Assumptions

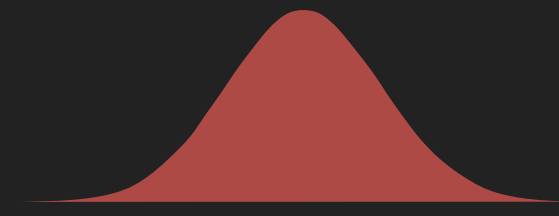
Student-t distributions

Very few samples

Mean of two identical distributions

https://en.wikipedia.org/wiki/Student%27s_t-test

Z-test



Assumptions

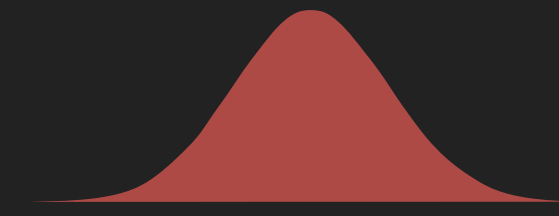
Gaussian distributions

Homogeneity of variance

Mean of two identical Gaussian distributions

<https://en.wikipedia.org/wiki/Z-test>

Fisher's exact test



Assumptions

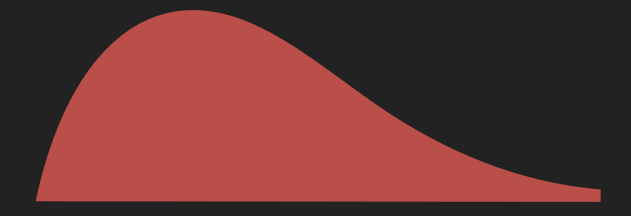
Contingency tables

Small sample sizes

Count data evaluating ratios

https://en.wikipedia.org/wiki/Fisher%27s_exact_test

Chi-squared test



Assumptions

Contingency table

Large sample sizes

Count data evaluating ratios

Chi-squared distribution

Generalisation of Fischer test

https://en.wikipedia.org/wiki/Chi-squared_test

The infamous p -value

....but what is it?

Hypothesis significance testing

Classic approach

Test up against alternative hypothesis

Prove by rejecting null-hypothesis
being responsible for the observation

p-value

Probability of seeing the
observed if the null
hypothesis is true...

Convention:
Reject null-hypothesis if p-value < 5%

R. Fischer - "Lady tasting tea" - 1935

Milk or tea first?



Dr. Murial Bristol claimed she could...

The null hypothesis is that she has no
ability to distinguish the teas.

Dr. Murial Bristol nailed all of them...

4 successes has 1 chance
out of 70 ($\approx 1.4\% < 5\%$)

= p-value

Hypothesis significance testing

Power Analysis - how many samples?

Sample size required to correctly reject a false null hypothesis...

$$n = \frac{(Z_{\alpha} + Z_{\beta})^2 \cdot (p_1 \cdot [1 - p_1] + p_2 \cdot [1 - p_2])}{(p_2 - p_1)^2}$$

have to assume a certain lift -
what if we are wrong in our
assumptions?

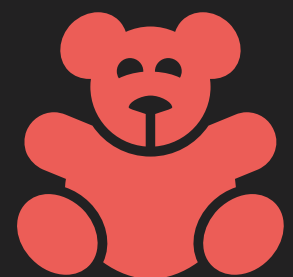
Significance	$Z_{\alpha} = 1.96$ $Z_{\beta} = 0.84$	Reliability of 5% Power of 20% (80% to reject H_0 if false)
Expected CTR values	$p_1 = 0.1$ $p_2 = 0.11$	CTR of 10% for A - expected... CTR of 11% for B - expected...
Sample size	$n = 14.752$	Samples required

Single-sided or double sided test?



What are we testing for?

$$H_1 \quad \boxed{A} < \boxed{B}$$
$$H_1 \quad \boxed{A} <> \boxed{B}$$



Exercise



```
streamlit run src/ui/AB_Test.py
```

A

1000
—
impressions

100
—
clicks

B

120
—
clicks

129
—
clicks

?
—
P-value

?
—
P-value

?
—
Sample size

Hypothesis testing challenges



General

can't really interpret probabilities or belief

if $p\text{-value} > 5\%$,
then whole campaign is thrown out

blackbox to most....

tells nothing of the difference between A and B

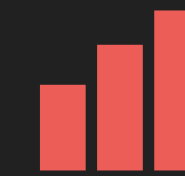


Assumptions

data distribution?
Gaussian, t-student? etc.

one-sided or double sided?

a lot of assumptions -
MUST be correct



Sample Size

hypothesis testing requires to estimate sample size first

must guess the performance of each variant. Not easy for B!

relies on estimating the significance when samples is reached



Peeking

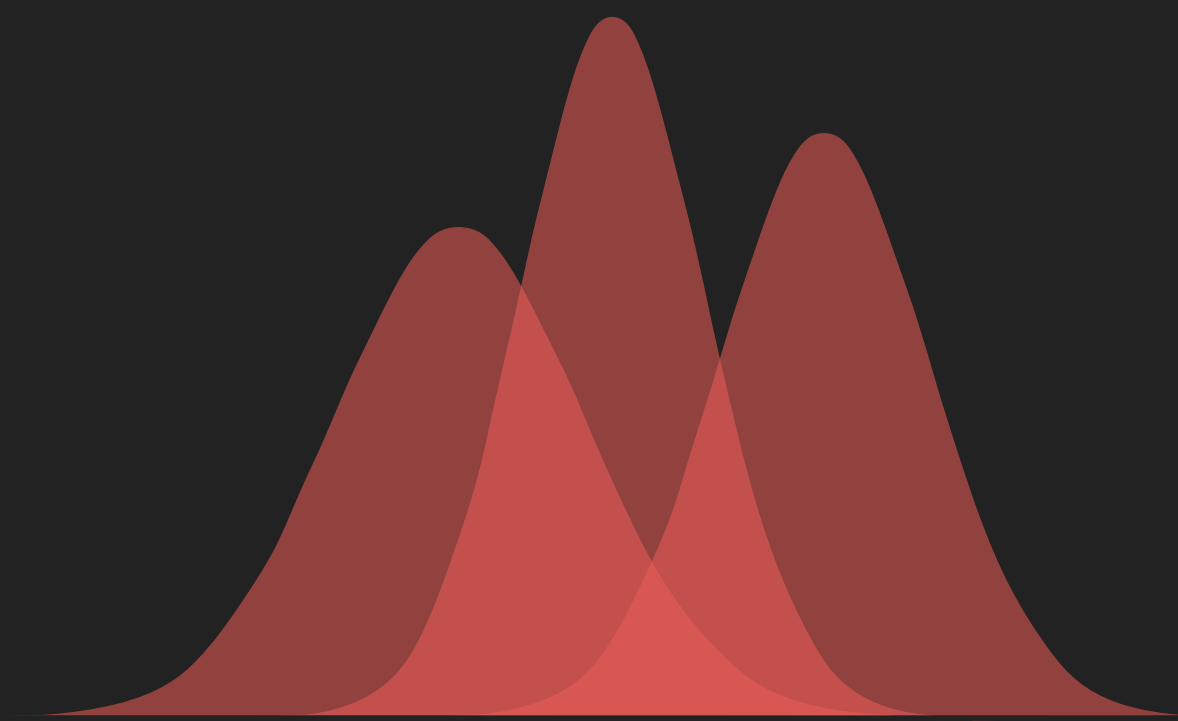
"...checking A/B results before the test is over..."

very tempting....

significance assessment only occurs when samples size is reached....



Questions



Bayesian A/B testing

Bayes equation definition

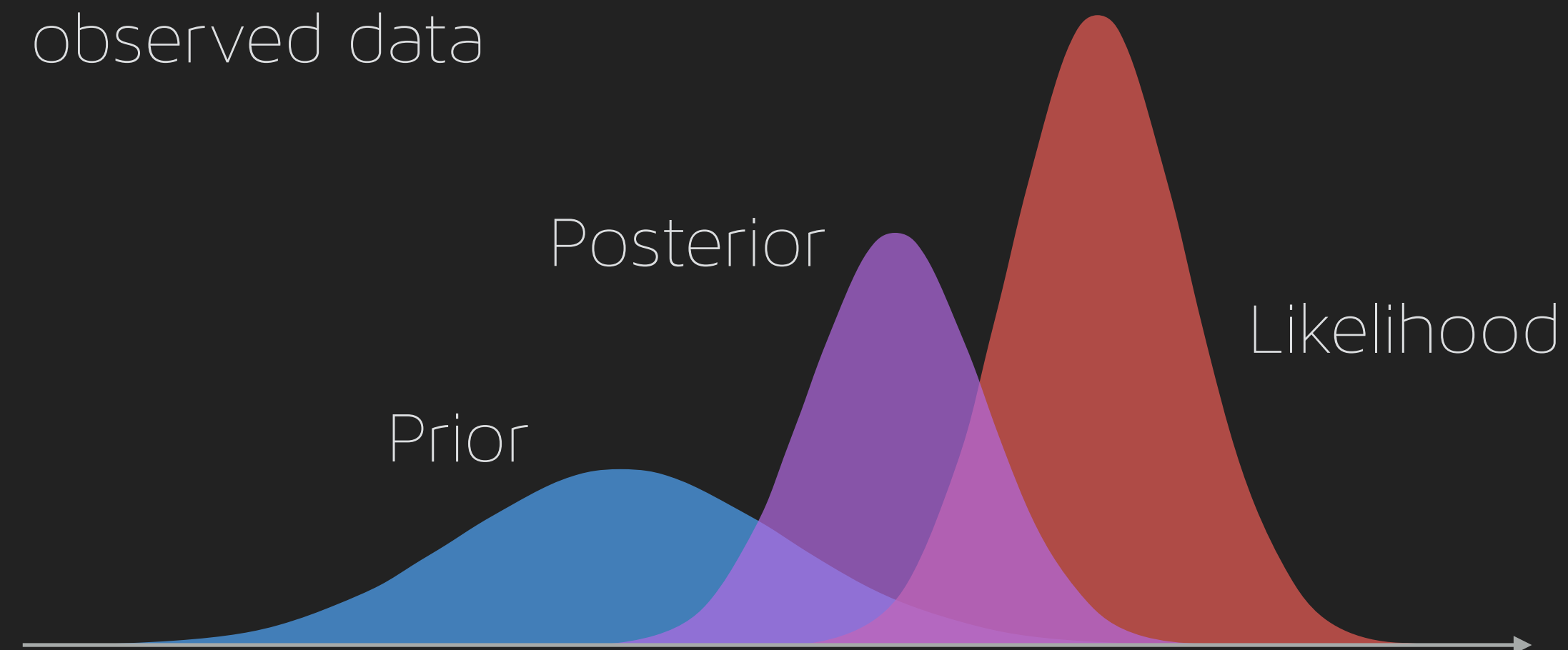
$$\underbrace{p(\theta | x)}_{\text{Posterior}} = \frac{\overbrace{p(x | \theta)}^{\text{Likelihood}} \cdot \overbrace{p(\theta)}^{\text{Prior}}}{\underbrace{p(x)}_{\text{Evidence}}} = \frac{p(x | \theta) \cdot p(\theta)}{\int p(x | \theta) \cdot p(\theta) d\theta} \propto p(x | \theta) \cdot p(\theta)$$

what we believe before the data

our updated belief after the data

prob. of the observed data

Normalisation



Types of Uncertainty

two main types

aleatoric
uncertainty

natural stochasticity in
observations (noise)

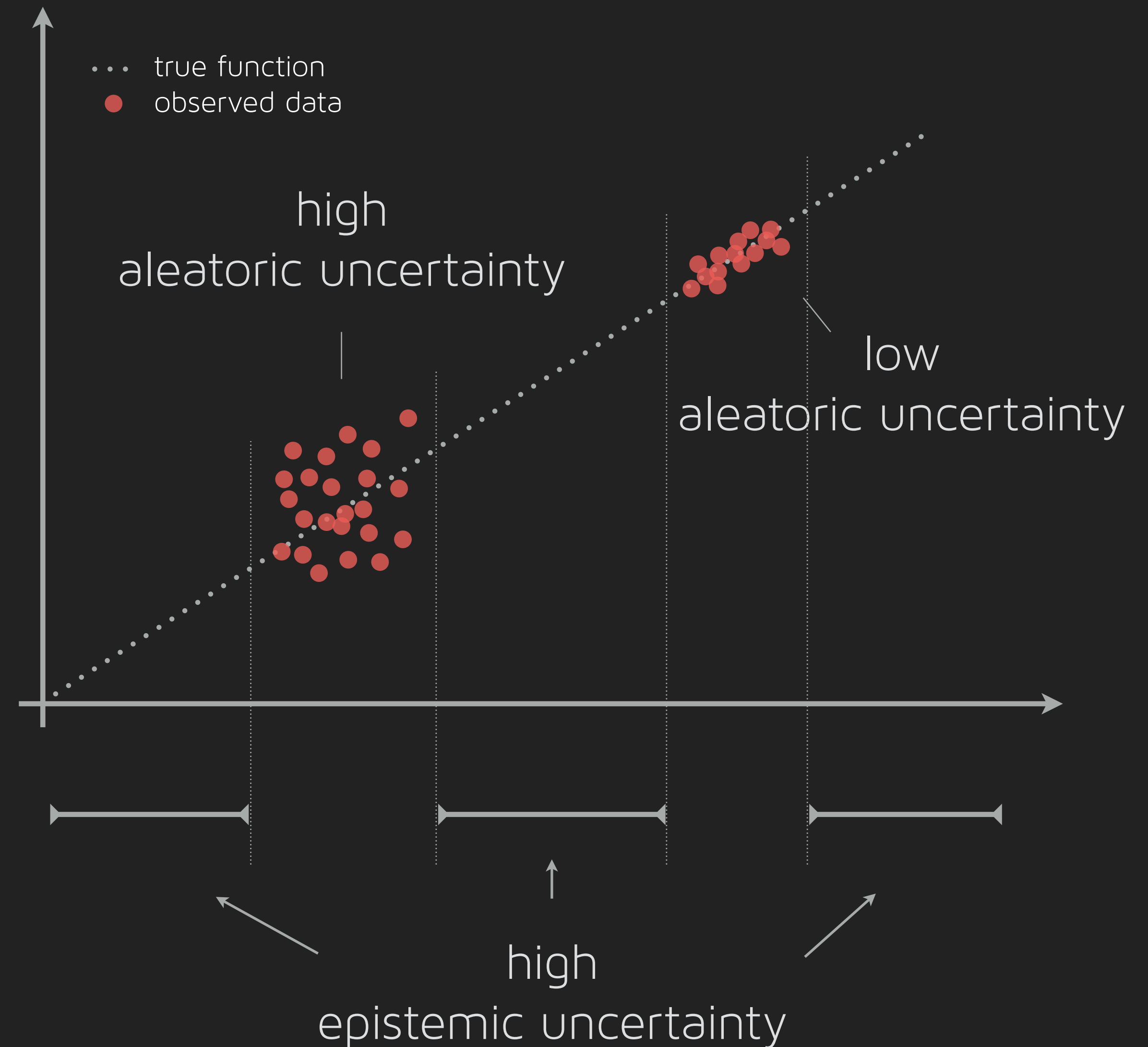
can't be reduced with
more data

epistemic
uncertainty

uncertainty around
too few data points

out-of-distribution
outliers

example



Being Bayesian....what does that mean in our case?

Few impressions

10	1	→ 10%
impressions	clicks	CTR

Point estimate
...but based on 10 samples

Could also be 6% or 14%?

Many impressions

10.000	1000	→ 10%
impressions	clicks	CTR

Point estimate
...now based on 10k samples

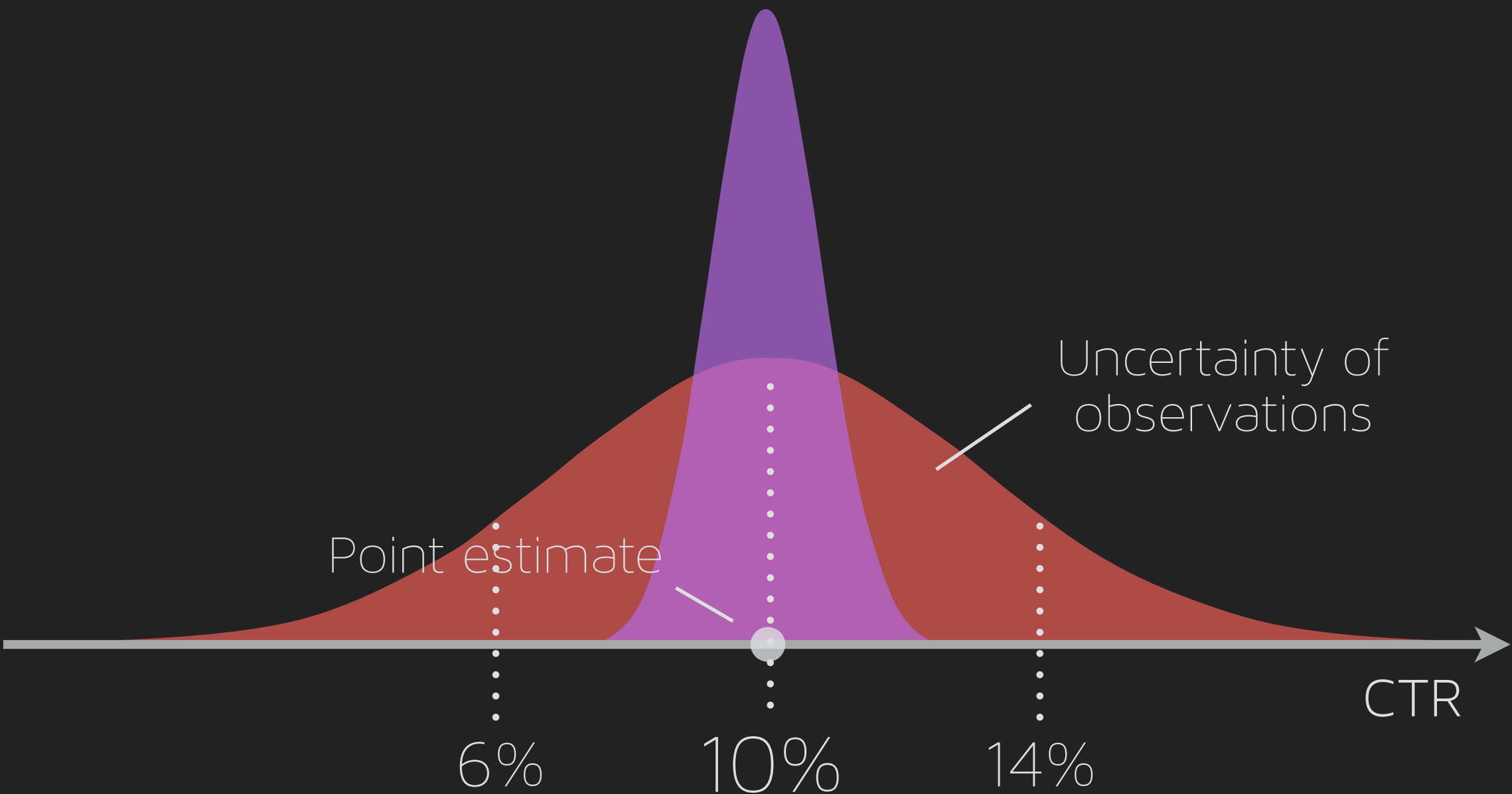
Could 9.5% or 10.5%?

Being Bayesian means including the uncertainty

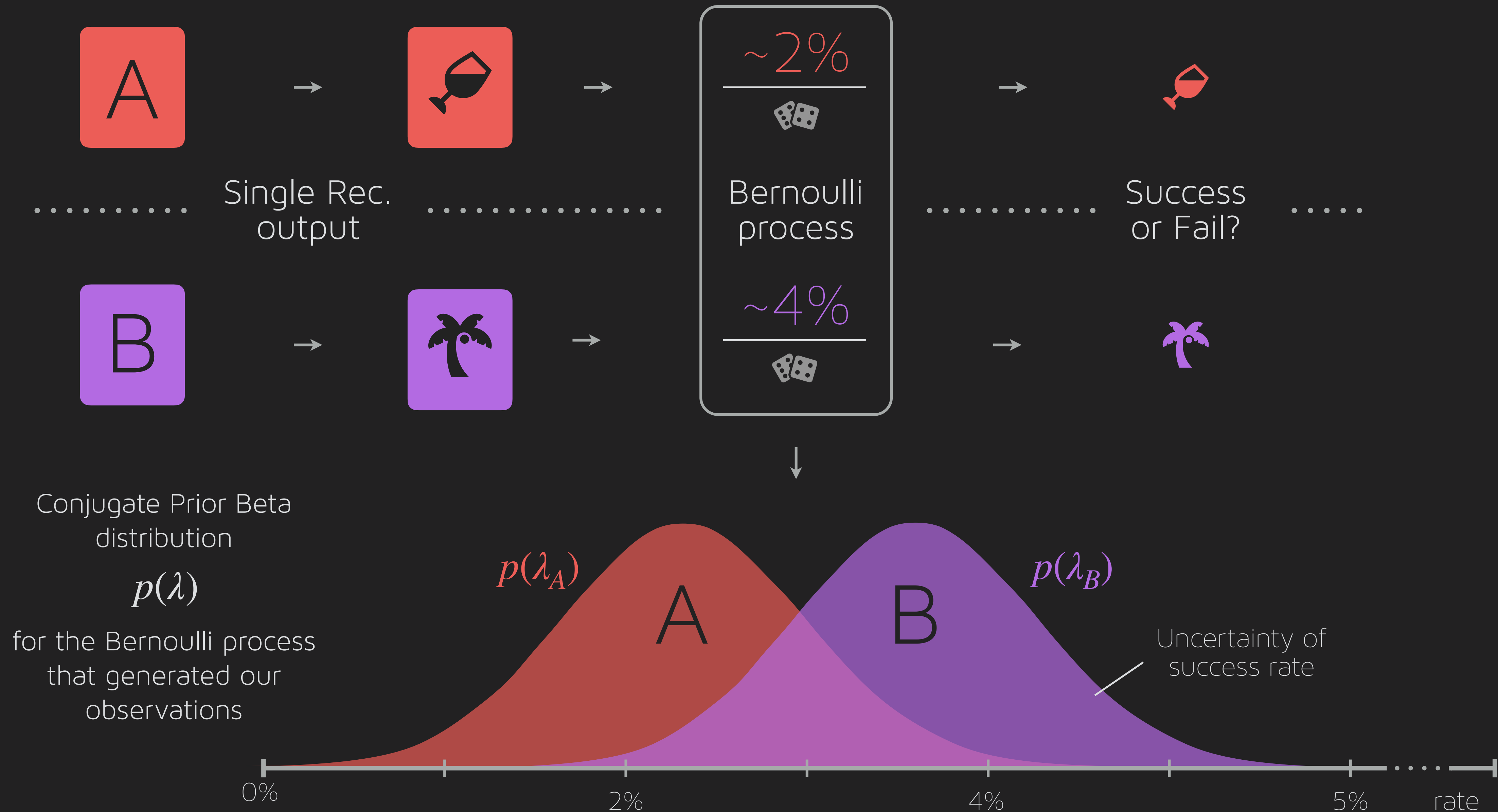
Operating on distributions instead

Point estimates + uncertainty

Math often more difficult
not in our case of A/B-testing
though



Stochastic process behind the scenes...



Bayesian stochastic modelling / testing

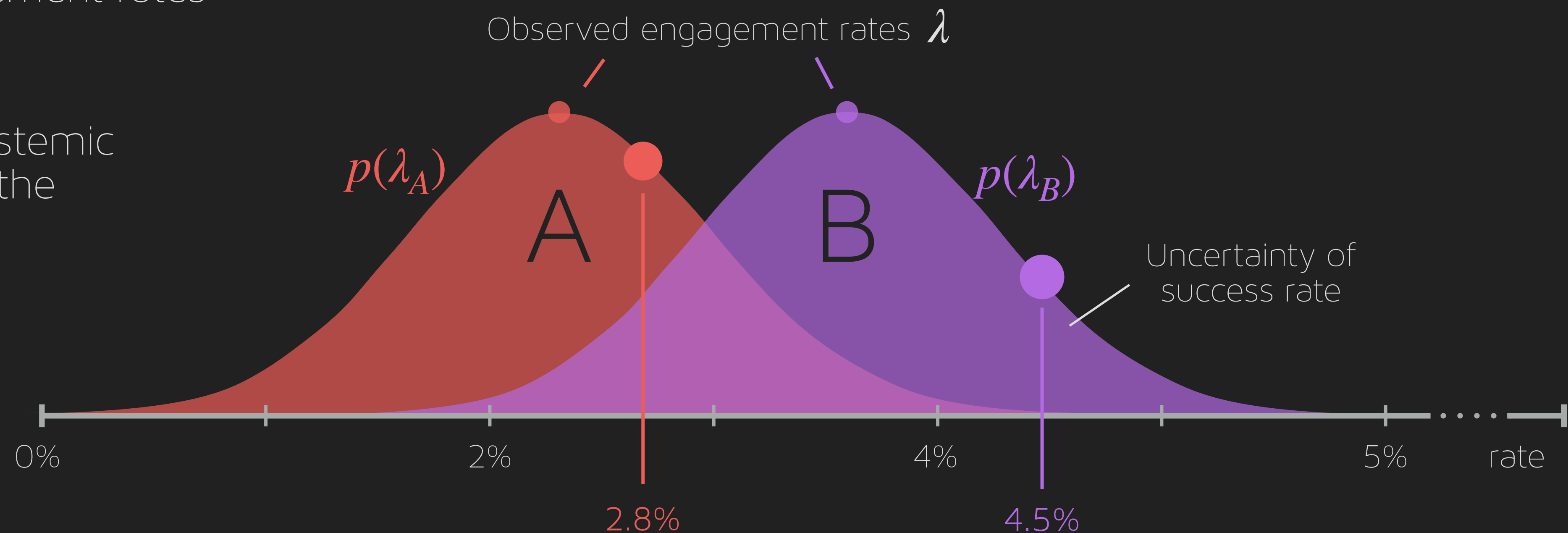
Model events as stochastic Bernoulli engagement events.

The engagement rate for each candidate can then be modelled as a probability distribution to capture the uncertainty.

Example of two candidates and their engagement rates with uncertainty.

Captures the epistemic uncertainty, not the aleatoric in this case.

The uncertainty allows us to quantify
how much do we believe B
is in fact the winner
...and evaluate the likelihood of many
other scenarios.



Example:

The case, where the single engagement rate sample tells us B is better than A.

Bayesian stochastic modelling / testing

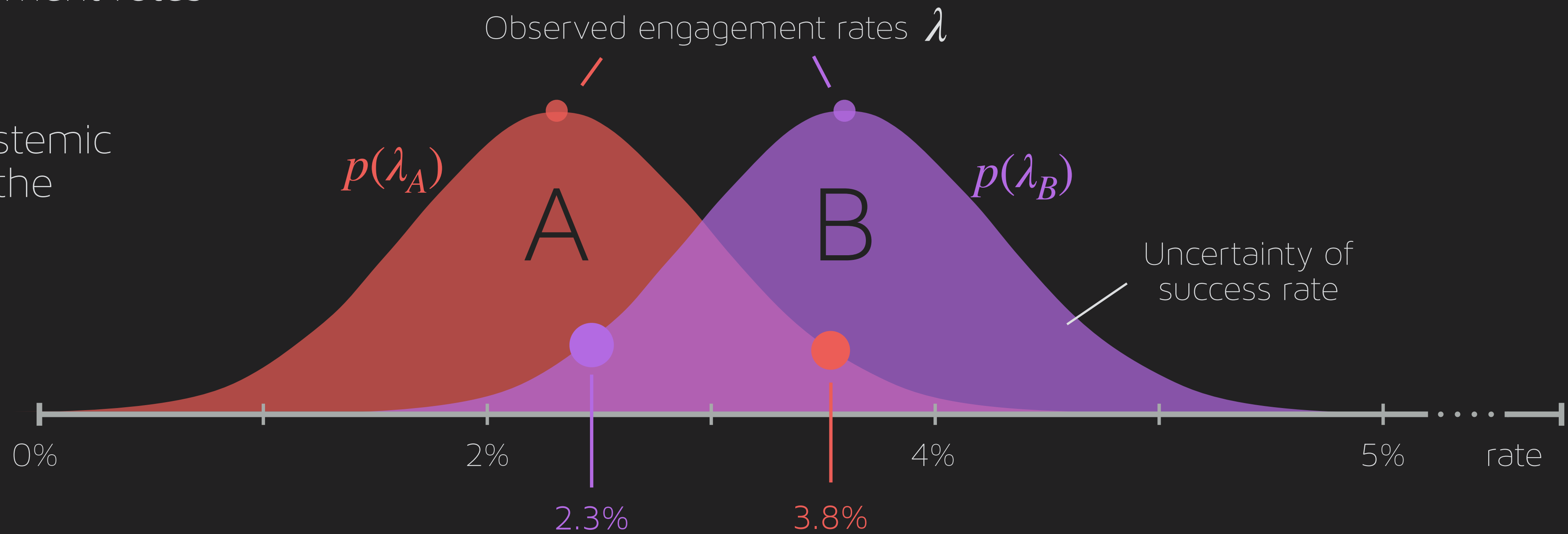
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Bayesian stochastic modeling / testing

Practical probability distributions to model engagement

clicks / impressions

beta distribution

$$p(x|\alpha, \beta) = x^{\alpha-1}(1-x)^{\beta-1} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$$

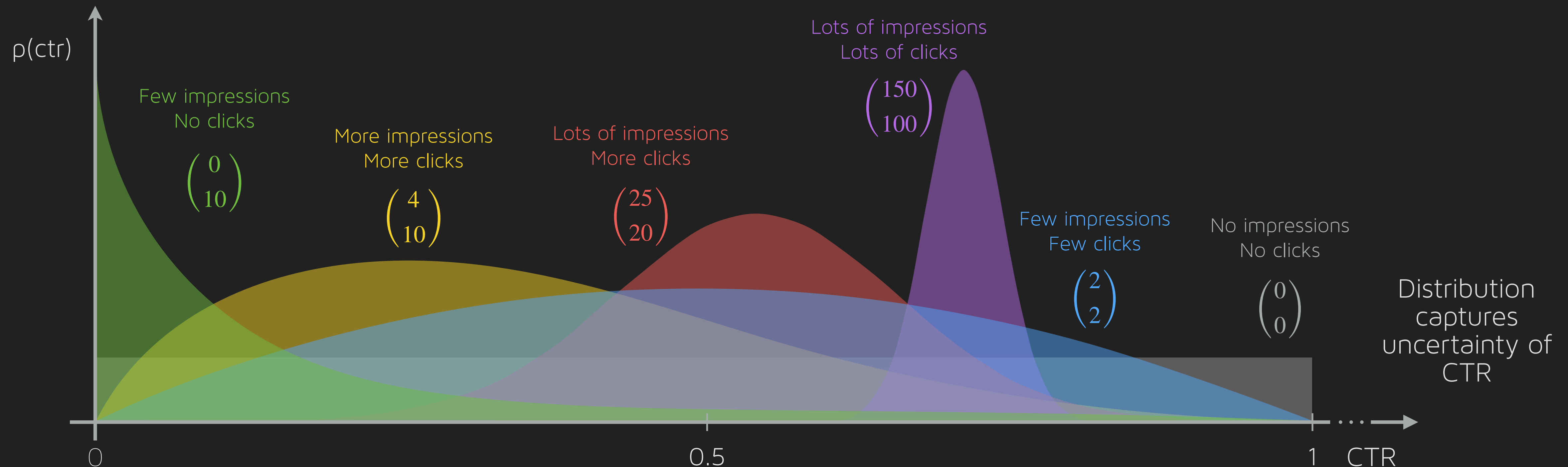
$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \text{clicks} \\ \text{impressions} - \text{clicks} \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

cost / click

gamma distribution

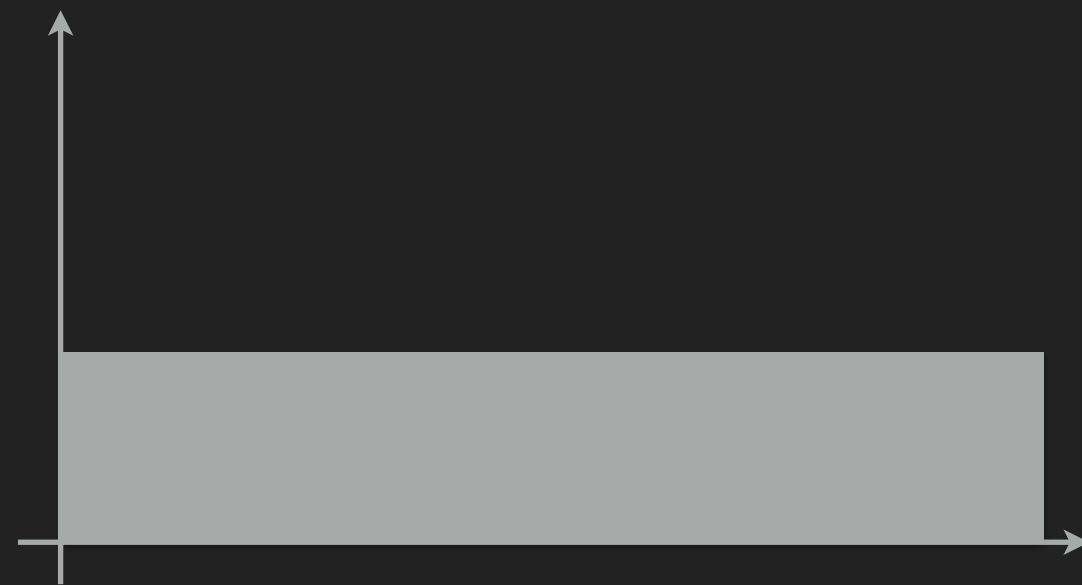
$$p(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \text{cost} \\ \text{clicks}^{-1} \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

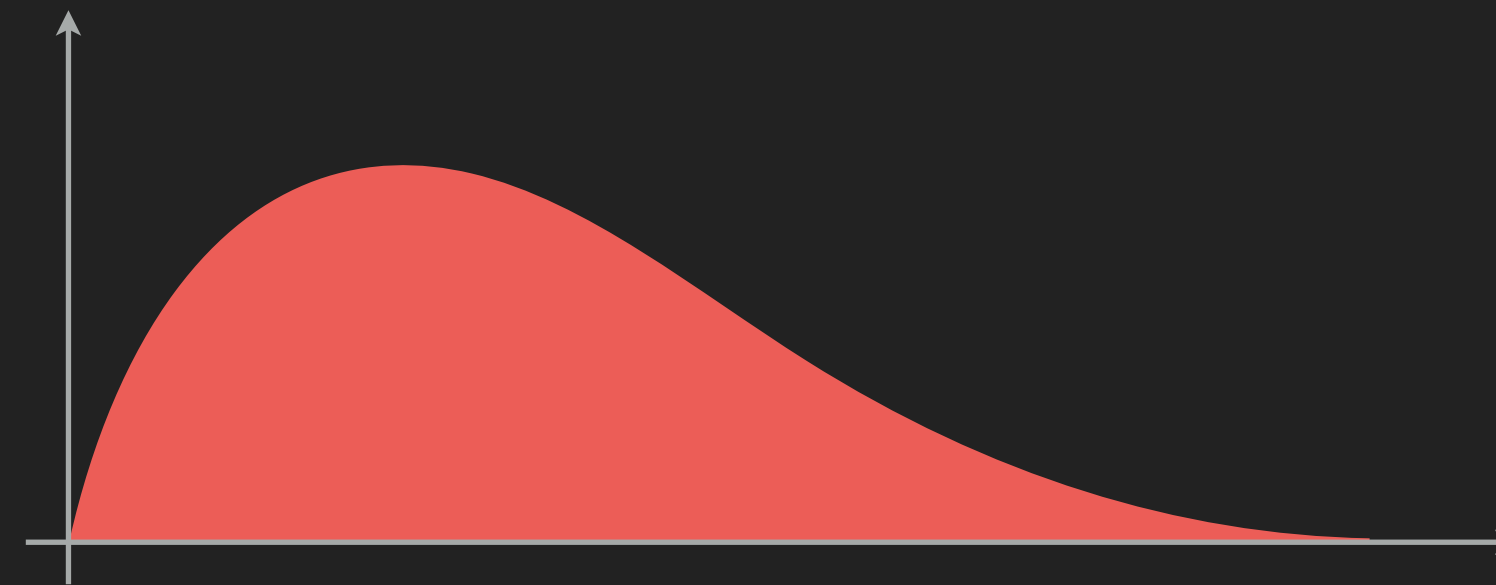


Bayesian stochastic modelling / testing

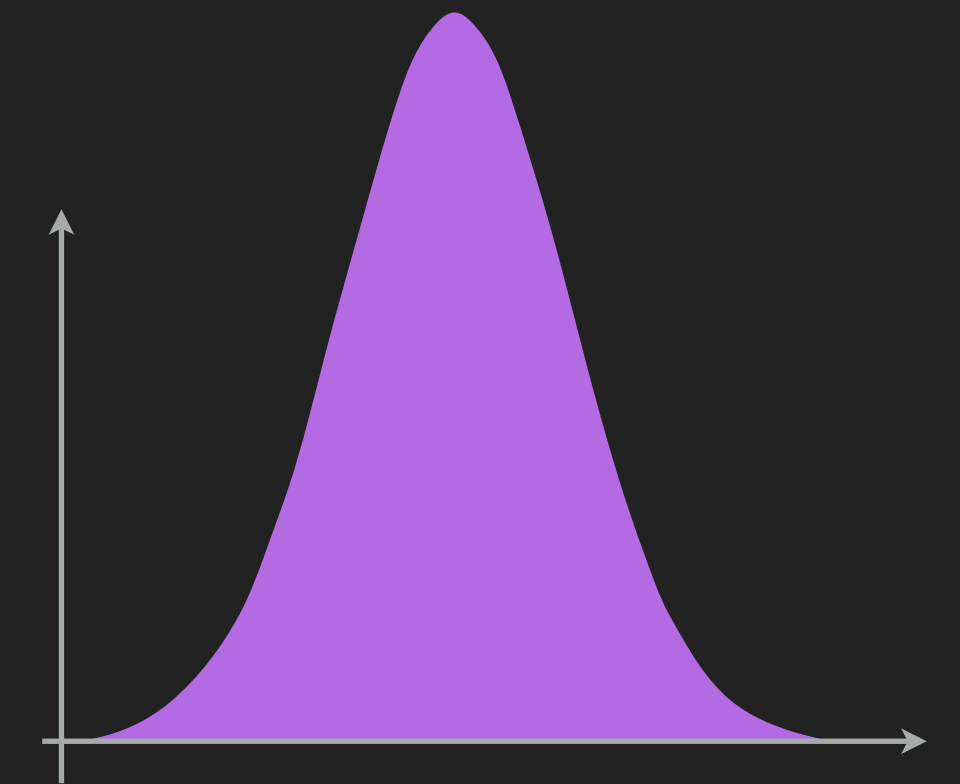
Why the Gaussian assumption is often not optimal....



Uniform



Beta / Gamma



Gaussian
(Central Limit Theorem)

Experiment just started

Early stages

Late stages

Know nothing

Know a little bit

Know everything

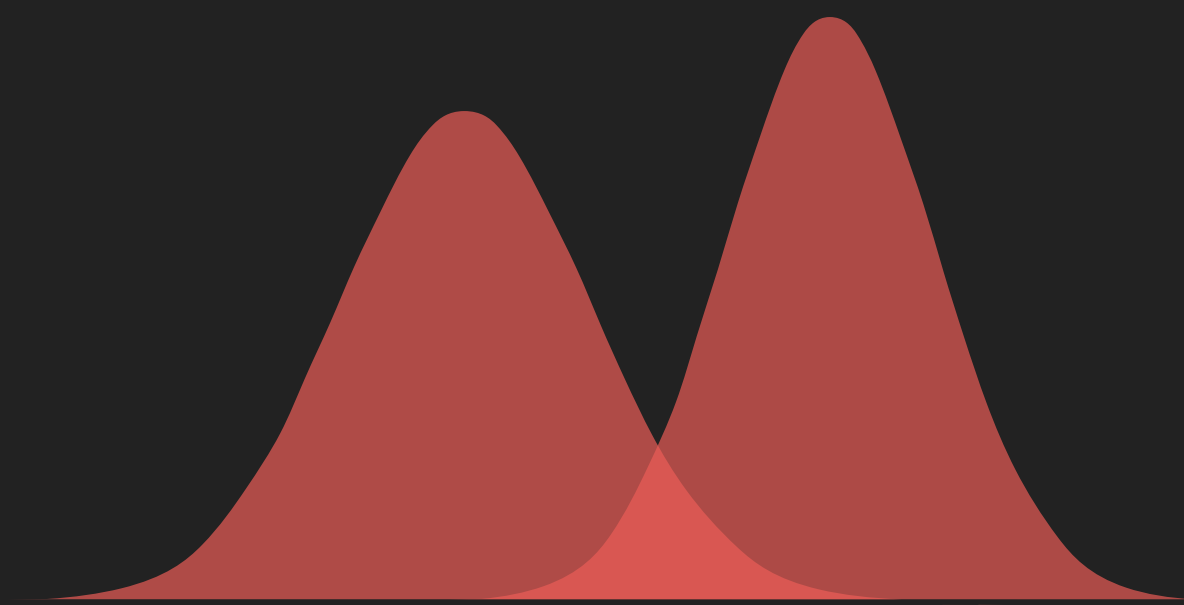
Operate on non-
Gaussian data

Time matters - to save
impressions and \$\$\$...

(This is where we mostly operate)

Evaluation expects
Gaussian distributed data

(So you have wait for this)



Bayesian Metrics
A/A testing

Bayesian same/same test

Are A/A the same distribution?

Selected metrics for measuring differences between two probability distributions

Kolmogorov-Smirnov

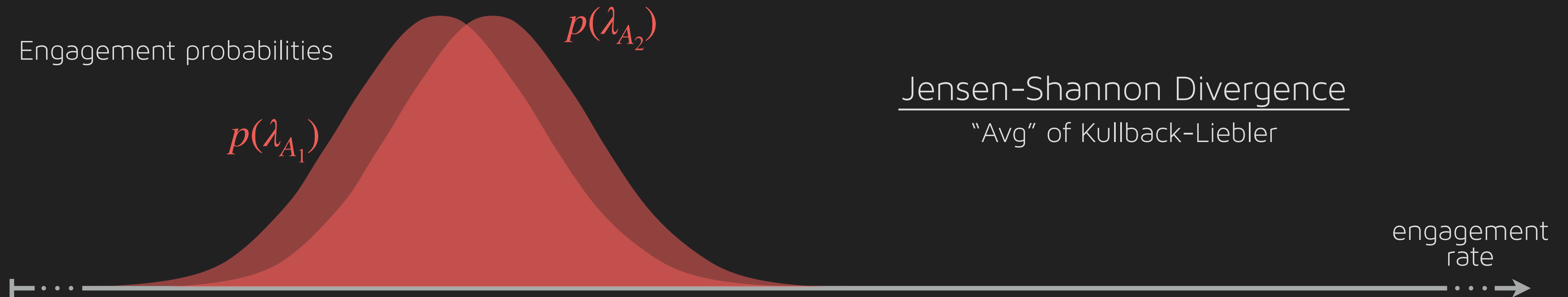
How much are the distribution overlapping?

Wasserstein

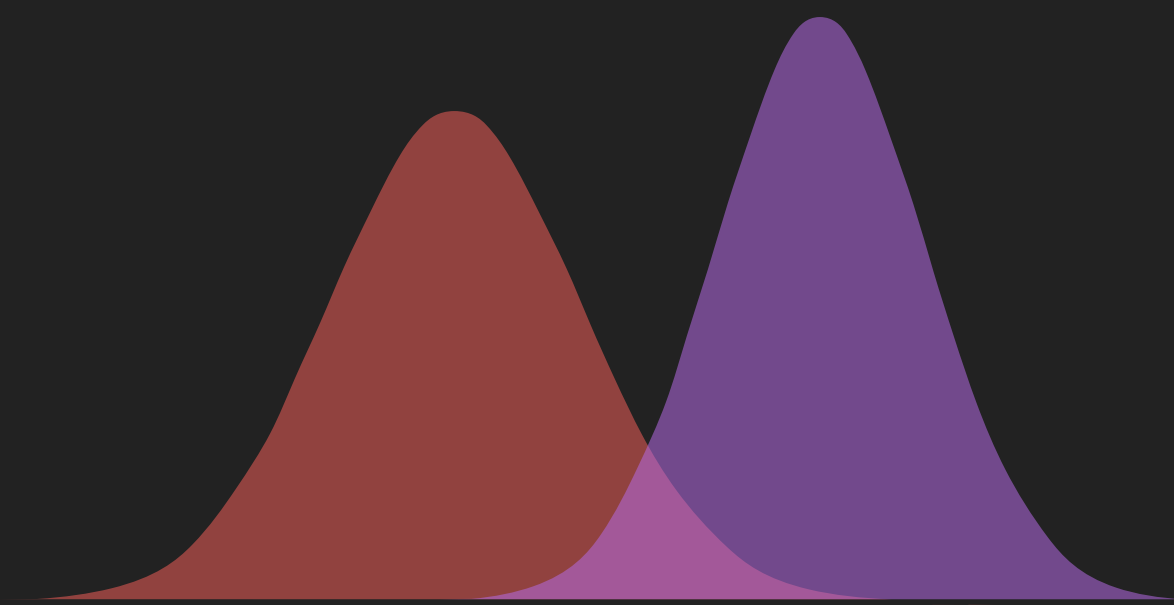
How much mass is moved?
(1 - wasserstein)

Jensen-Shannon Divergence

"Avg" of Kullback-Liebler



Example: A1 and A2 are close enough to each other....or what?



Bayesian Metrics
A/B testing

Bayesian Metrics - Probability of winner

Probability of selecting the right winner

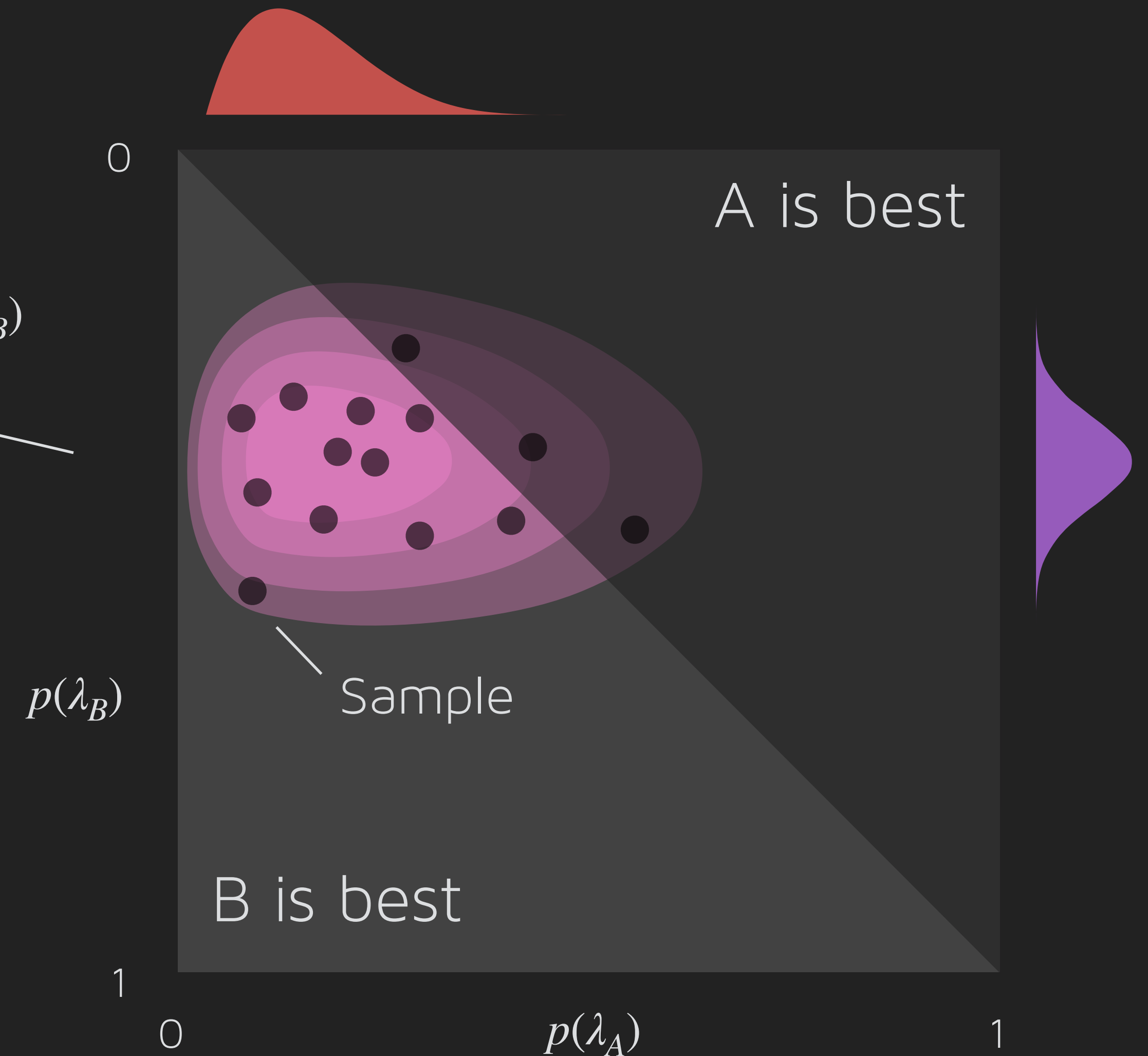
This can be deducted from the joint probability $p(\lambda_A, \lambda_B)$...and corresponds to the area, where $p_{\lambda_B} > p_{\lambda_A}$

This marginalisation can be estimated via Monte Carlo sampling

....or calculated analytically from for the Beta distribution

$$\begin{aligned} P(p_B > p_A) &= \int_0^1 \int_0^{\lambda_A} P(\lambda_A, \lambda_B) d\lambda_B d\lambda_A \\ &= \sum_{i=0}^{\alpha_B-1} \frac{B(\alpha_A + i, \beta_A + \beta_B)}{(\beta_B + i)B(1 + i, \beta_B)B(\alpha_A, \beta_A)} \end{aligned}$$

Joint probability of engagement-rates



Bayesian Metrics - Expected Loss

Average
performance loss for
wrong choice

Evaluation of loss if choosing wrong

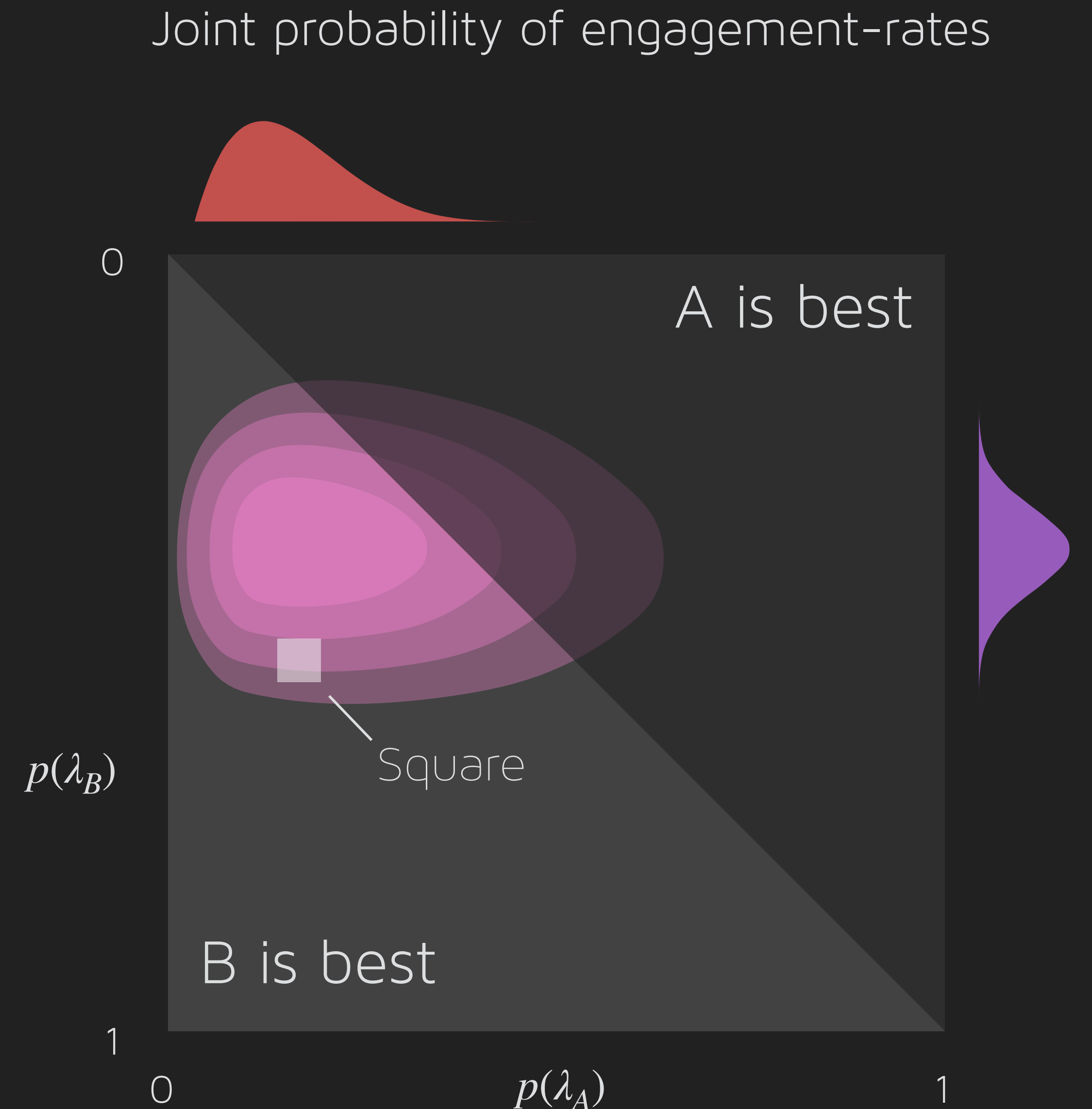
$$\mathcal{L}(\lambda_A, \lambda_B, A) = \max(\lambda_B - \lambda_A, 0)$$

$$\mathcal{L}(\lambda_A, \lambda_B, B) = \max(\lambda_A - \lambda_B, 0)$$

Expectation of loss by integrating across
joint distribution

$$E[\mathcal{L}](?) = \int_0^1 \int_0^1 P(\lambda_A, \lambda_B) \mathcal{L}(\lambda_A, \lambda_B, ?) d\lambda_B d\lambda_A$$

Unit will be λ - in some cases close to \$, as we shall see...





Single Example

Example....

Observations



1000
impressions

100
clicks

\$11
Cost



1000
impressions

120
clicks

\$19
Cost

Hypothesis chi2-testing - p-values....

Let's see what the numbers are after 1000 impressions...

Example

Impressions	1000
Clicks / Cost A	100 / \$11
Clicks / Cost B	120 / \$19

$$\lambda_A = \frac{100}{1000} = 10\% \quad \rightarrow \quad \frac{\text{B}}{\text{winner}} \quad ?$$
$$\lambda_B = \frac{120}{1000} = 12\%$$

1000
#samples

18%
p-value

Uuh, this is not good....
....Peeking

3026
#samples
actually required

1.4%
p-value

Aaaah, much better.....

Bayesian Metrics - Probability of winner & lift analysis

The uncertainty allows for reasoning and exploring the probability of various scenarios.

Example

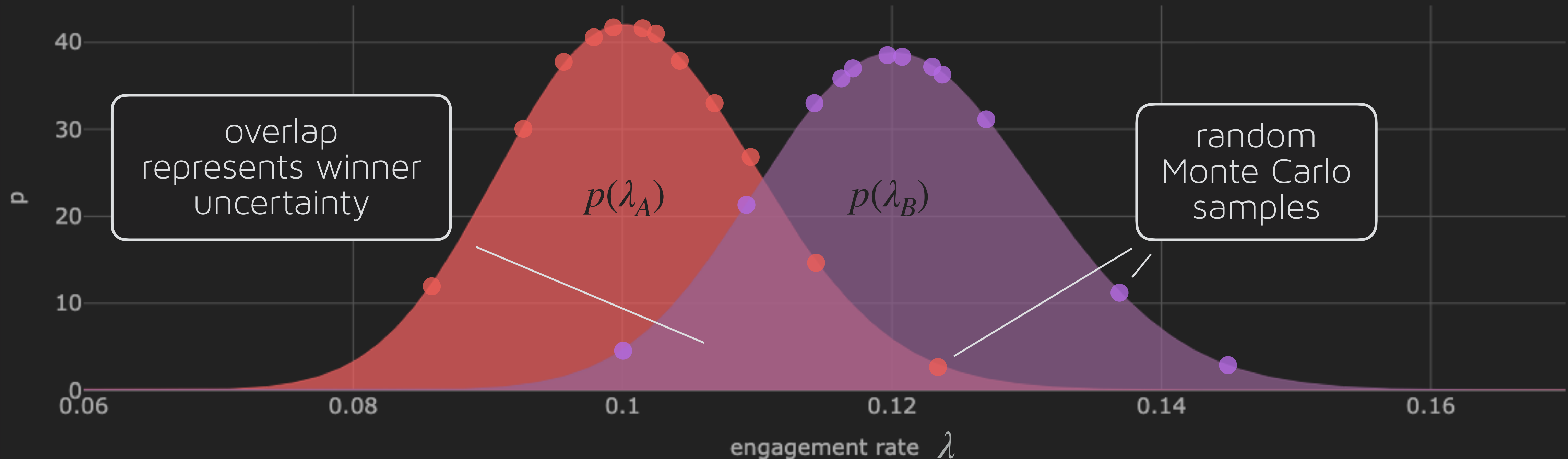
Impressions	1000
Clicks / Cost A	100 / \$11
Clicks / Cost B	120 / \$19

$$\lambda_A = \frac{100}{1000} = 10\%$$

$$\lambda_B = \frac{120}{1000} = 12\%$$

→ $\frac{B}{\text{winner}}$?

Beta distributions of CTR



Bayesian Metrics - Probability of winner & lift analysis

The uncertainty allows for reasoning and exploring the probability of various scenarios.

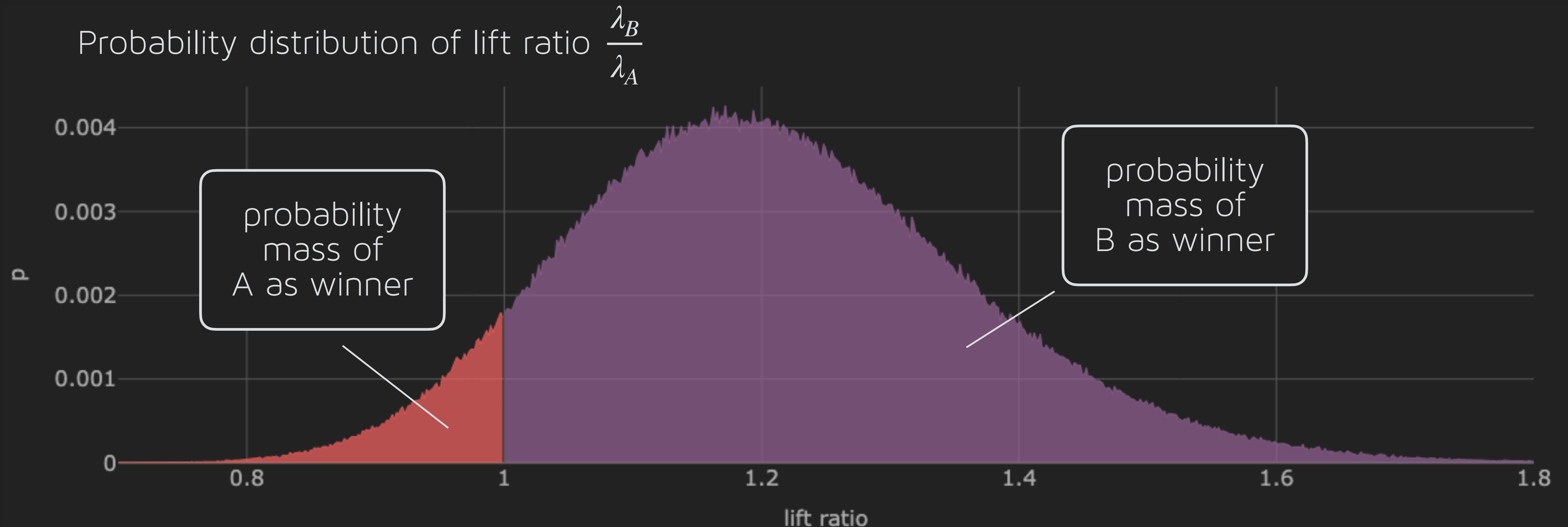
Example

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$$\lambda_A = \frac{100}{1000} = 10\%$$

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→ $\frac{B}{\text{winner}}$?



Bayesian Metrics - Probability of winner & lift analysis

The uncertainty allows for reasoning and exploring the probability of various scenarios.

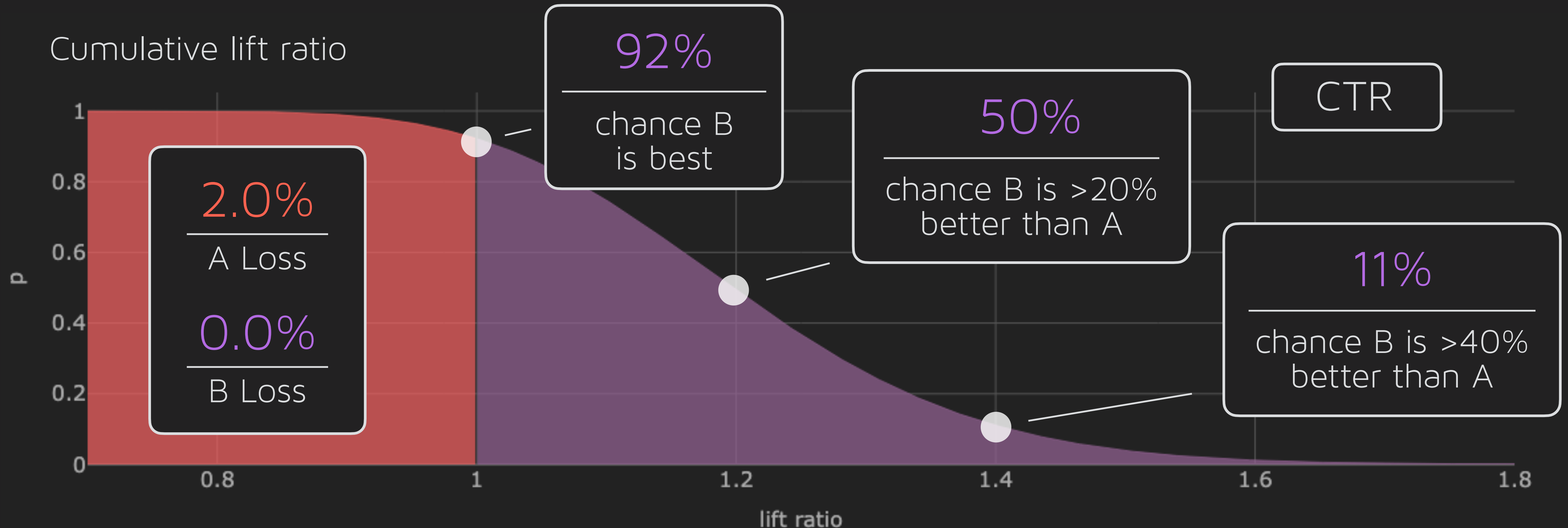
Example

Impressions
Clicks / Cost A
Clicks / Cost B

1000
100 / \$11
120 / \$19

$$\lambda_A = \frac{100}{1000} = 10\%$$
$$\lambda_B = \frac{120}{1000} = 12\%$$

→ B
winner ?



Bayesian Metrics - Probability of winner & lift analysis

The uncertainty allows for reasoning and exploring the probability of various scenarios.

Example

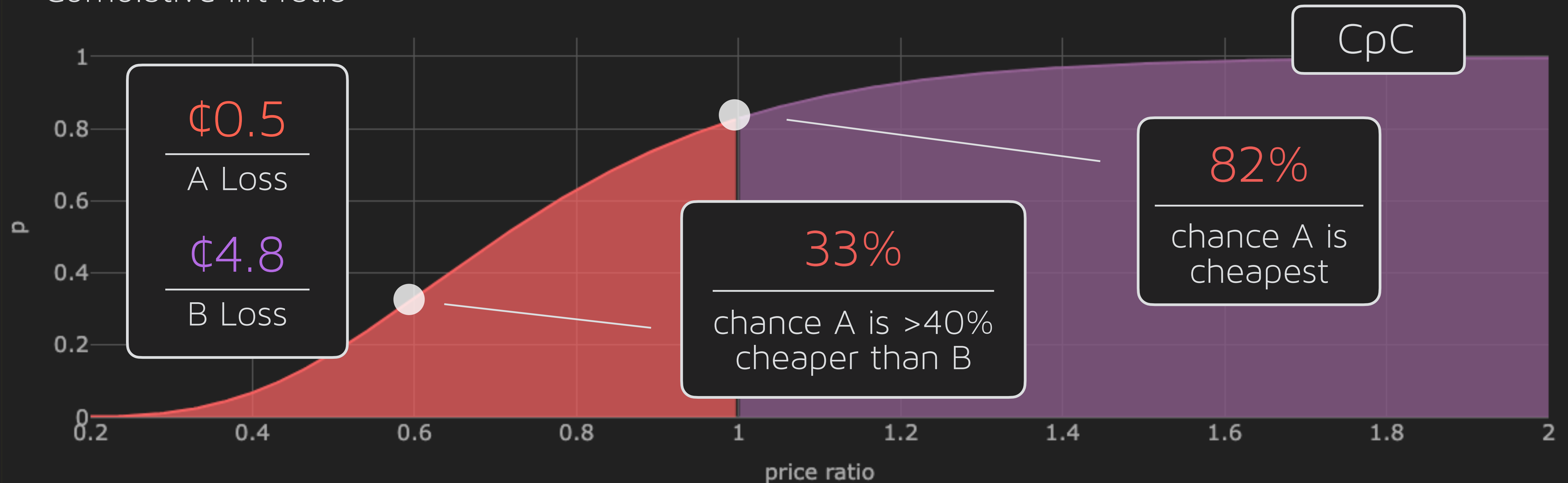
Impressions	1000
Clicks / Cost A	100 / \$11
Clicks / Cost B	120 / \$19

$$\lambda_A = \frac{\$11}{100} = \$0.11$$

$$\lambda_B = \frac{\$19}{120} = \$0.16$$

→ $\frac{A}{\text{winner}}$?

Cumulative lift ratio





Power Analysis

Hypothesis test – Power Analysis

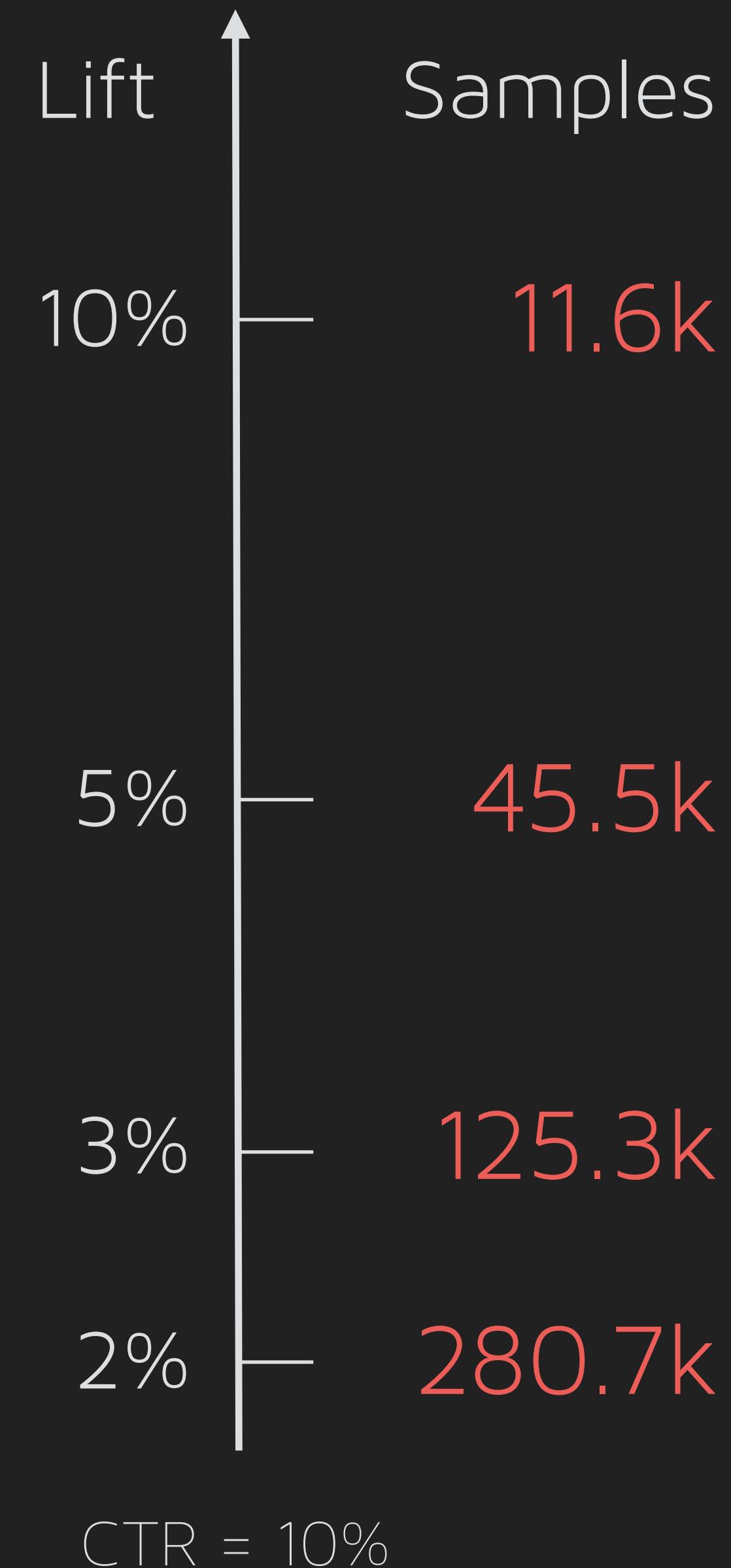
How much traffic should we spend to achieve a sufficient winning probability?

Power

determine the sample size needed for a study to achieve a desired level of statistical power:

the probability of correctly rejecting a false null hypothesis...

Does not tell the number of samples required to get good results...
 $p\text{-value} < 5\%$



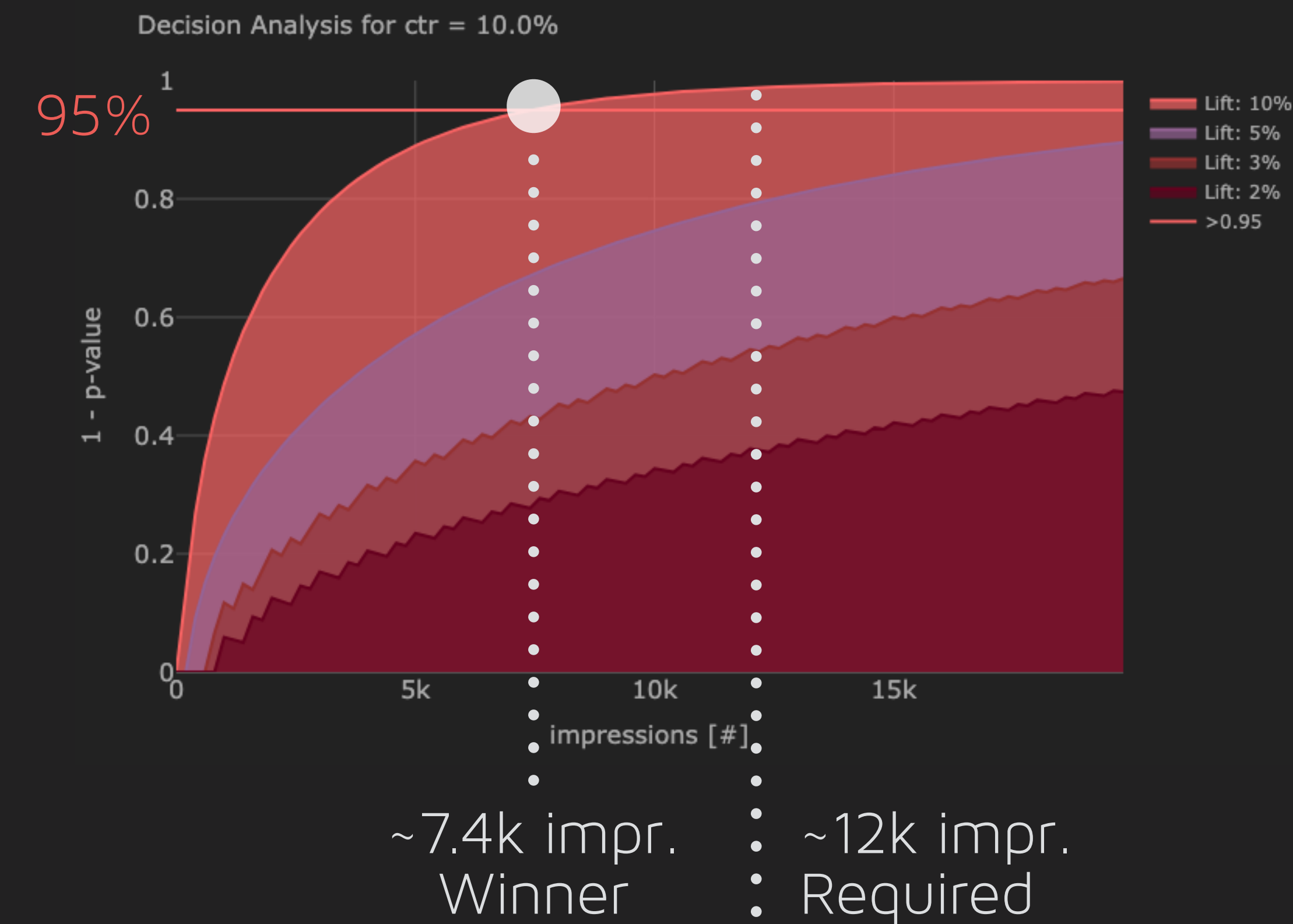
Decision Analysis

How much traffic should we spend to achieve a sufficient winning probability?

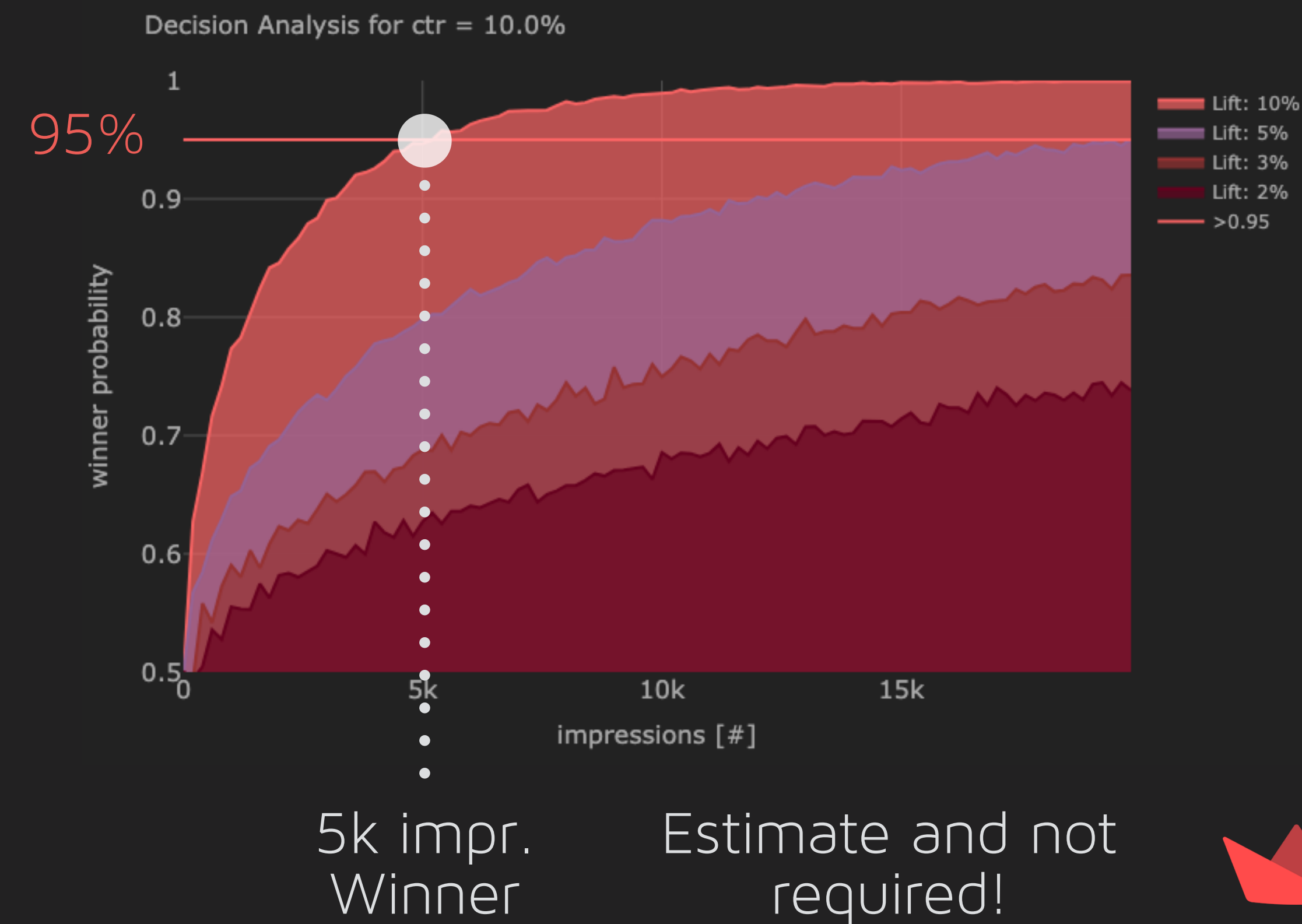
For 10% click rates, there is a reduction of required impressions of

~7k impr.

Hypothesis testing



Bayesian testing





Campaign

A/B Testing - Campaign simulation

...



10%
CTR A

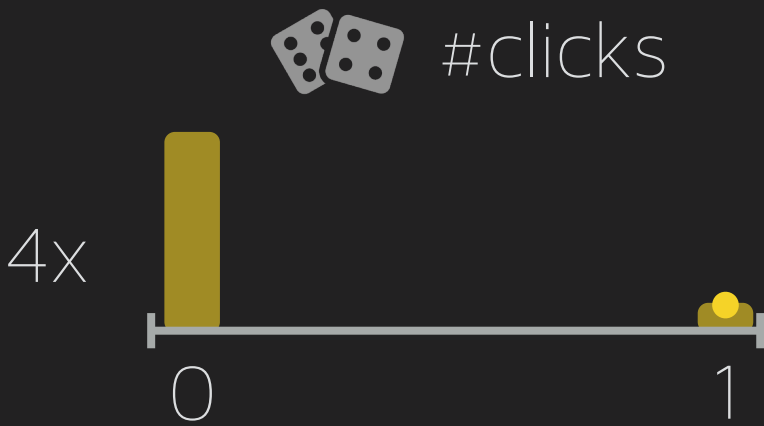
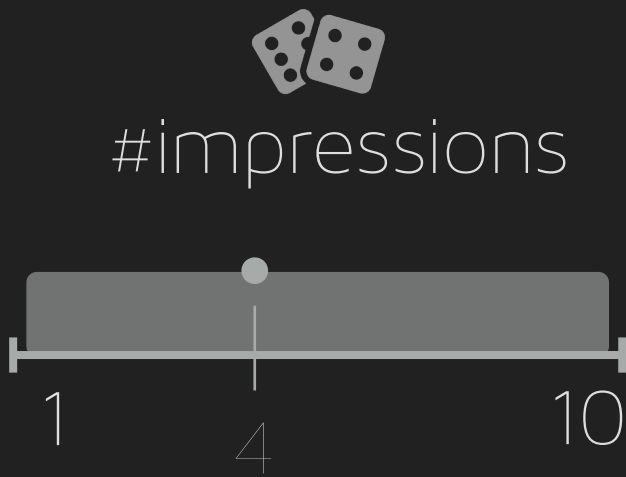
\$11
CPM A



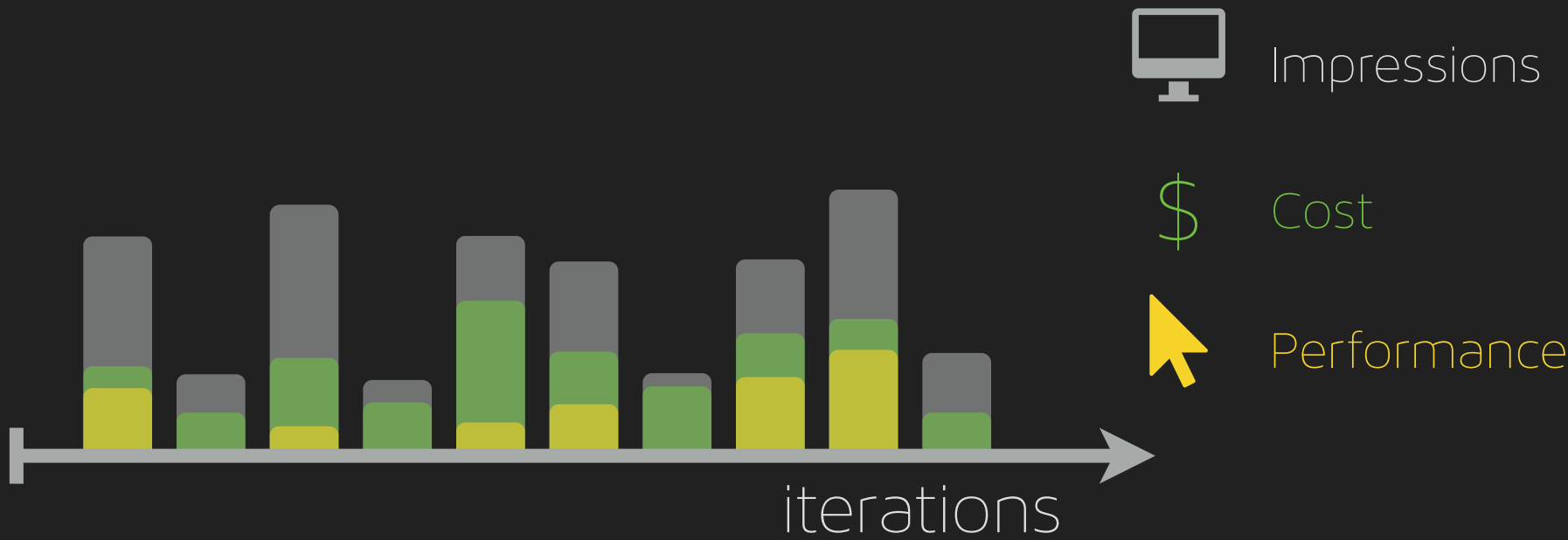
12%
CTR B

\$19
CPM B

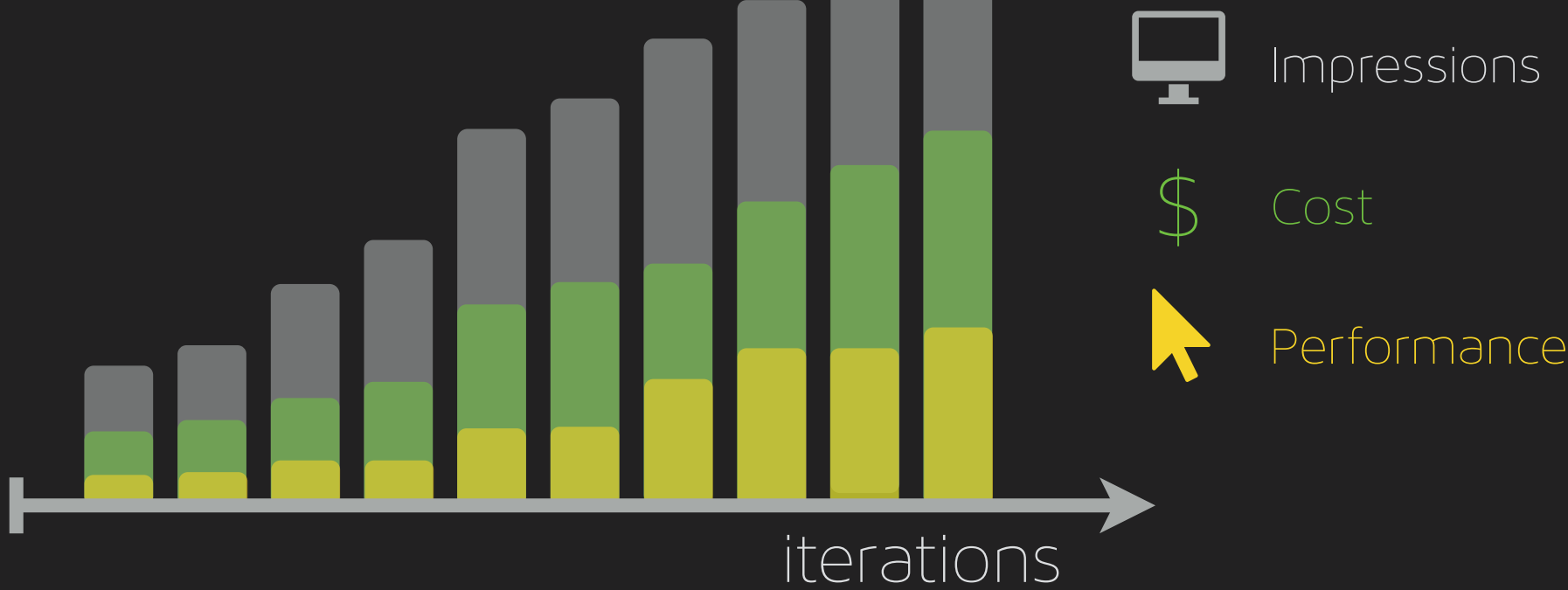
Simulate impression,
clicks and cost

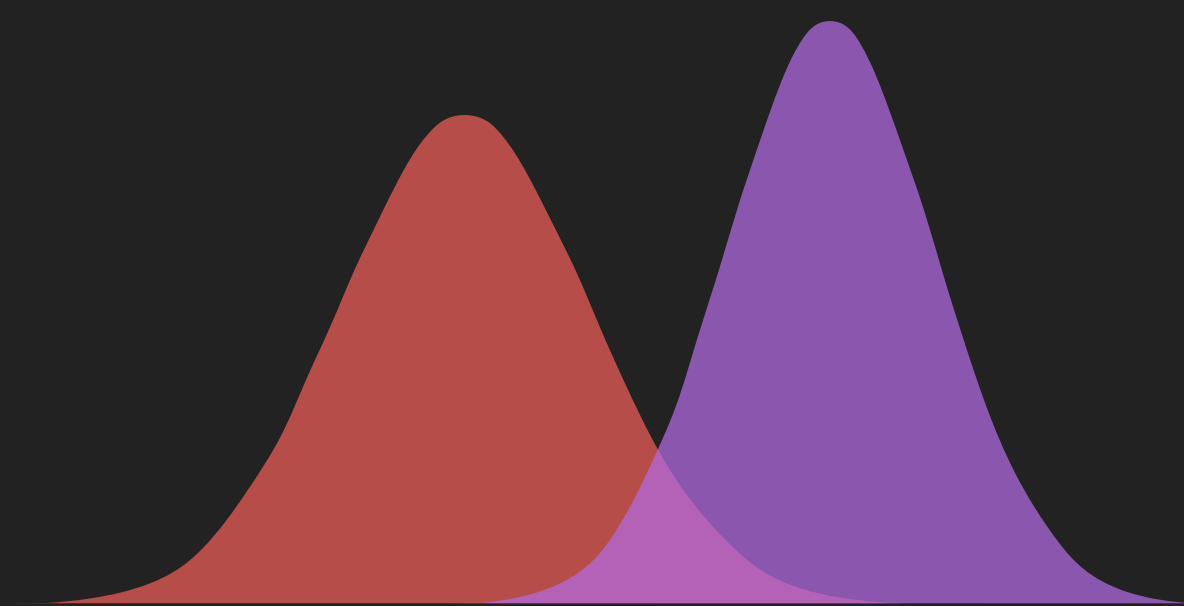


logs



accumulated





Bayesian A/B testing

Bayesian A/B Testing - Campaign simulation

A/B-test - CTR

A

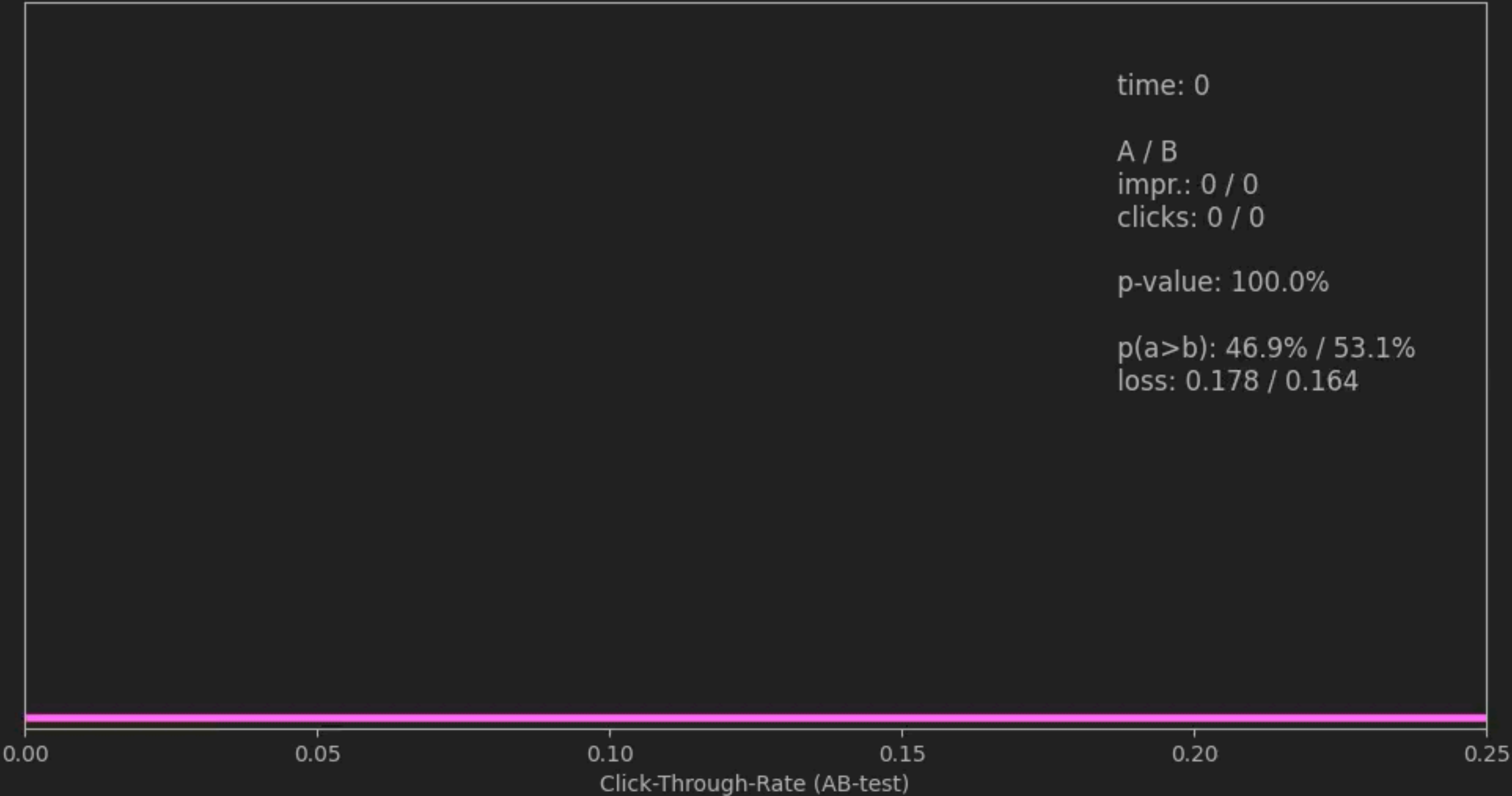
10%
CTR A

\$11
CPM A

B

12%
CTR B

\$19
CPM B



Hypothesis A/B Testing - Campaign simulation

A/B test

A

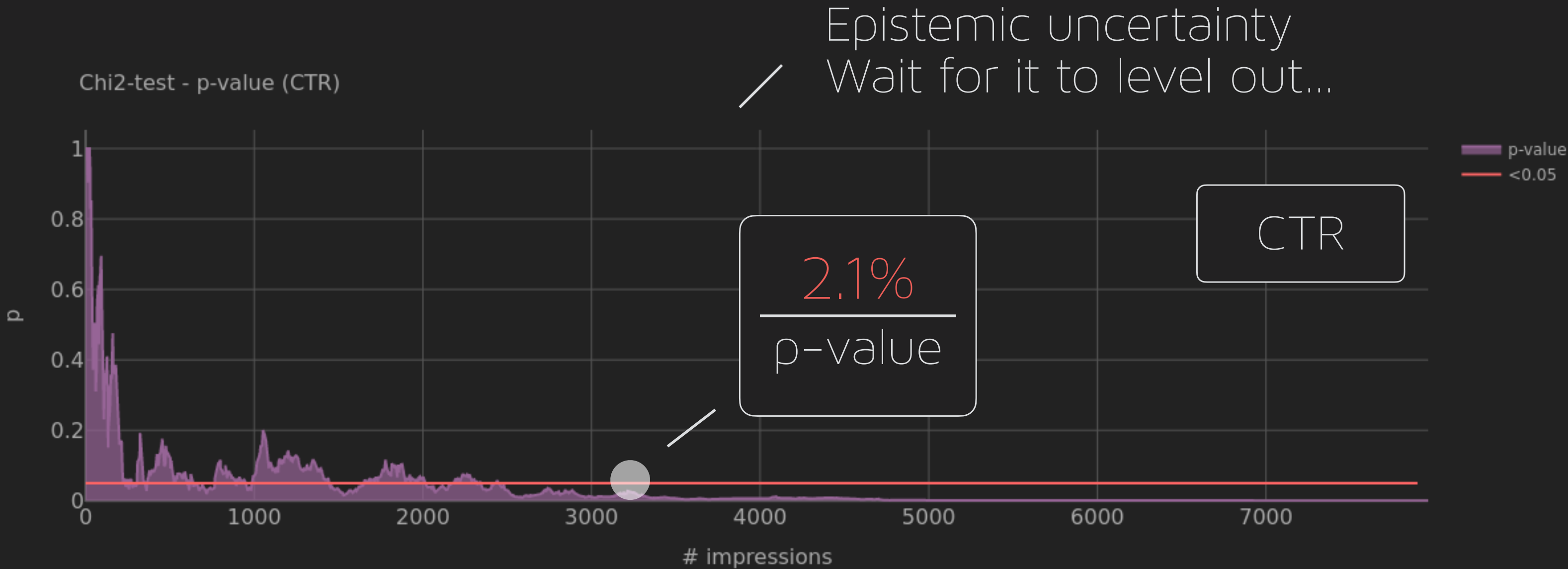
10%
CTR A

\$11
CPM A

B

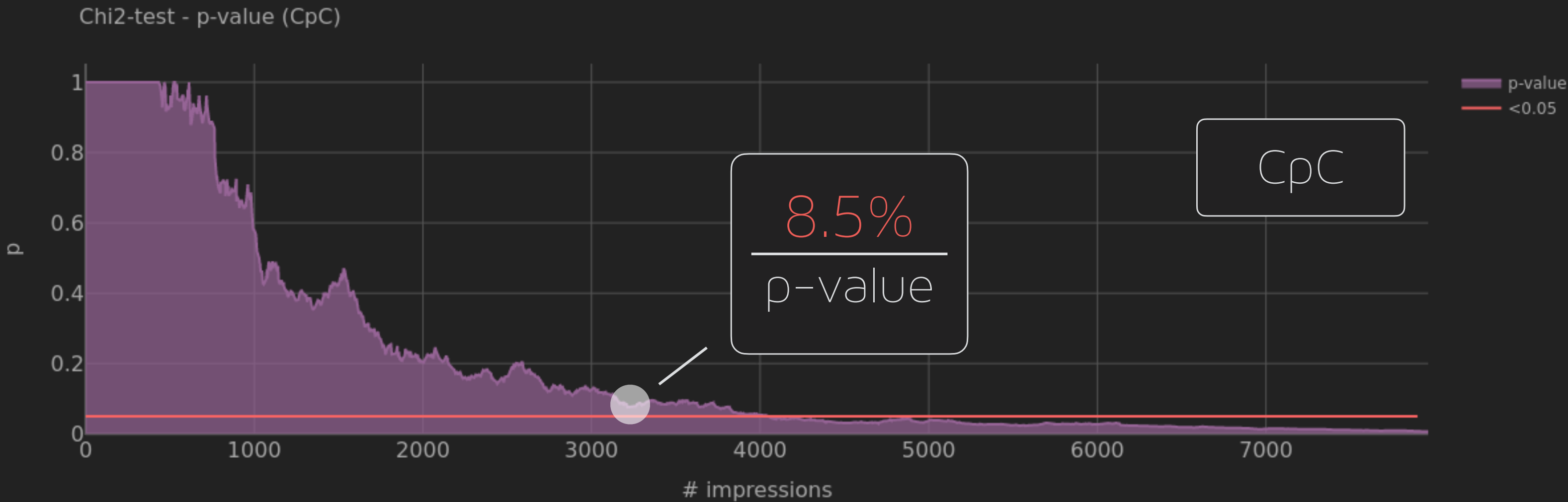
12%
CTR B

\$19
CPM B



Required
number of
samples

3206
#samples



Hmmm, B didn't
pass...

Higher cost did
not outweigh
better click rate...

Bayesian A/B Testing - Campaign simulation

A/B test - $P(B > A)$

A

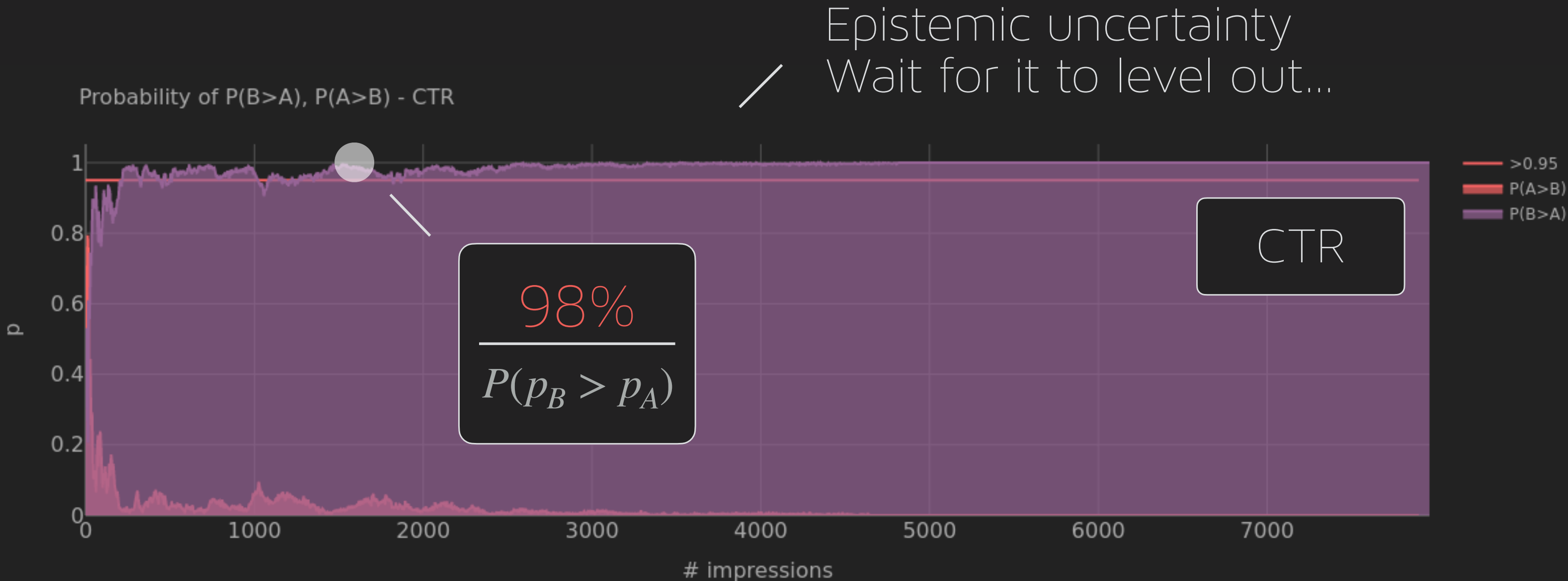
10%
CTR A

\$11
CPM A

B

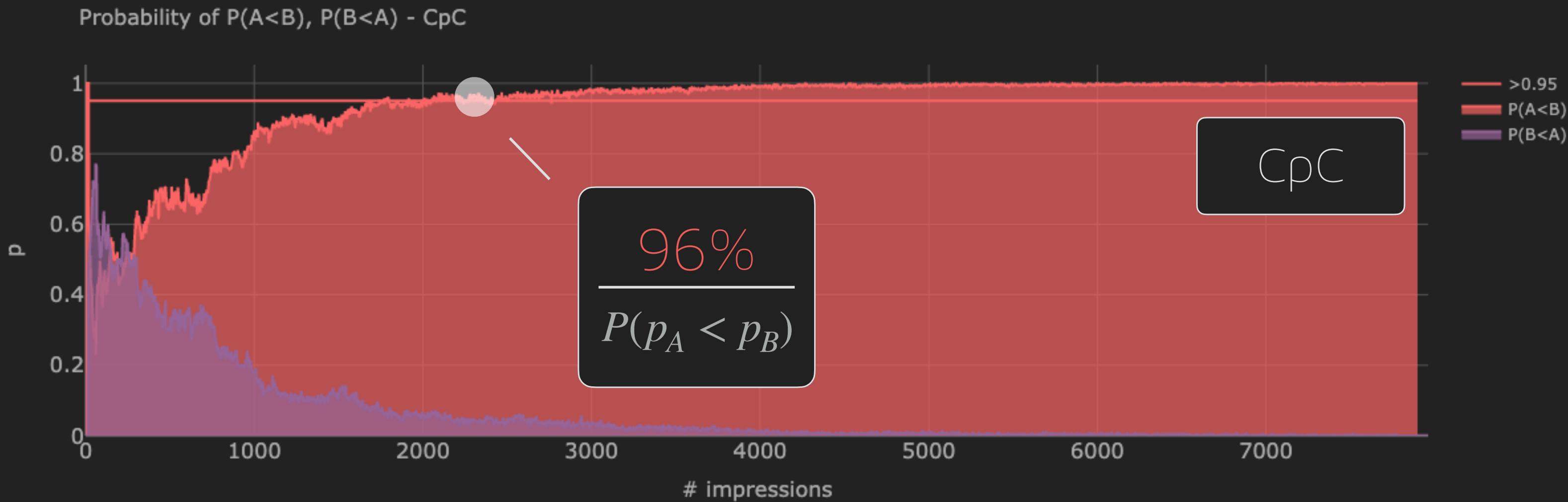
12%
CTR B

\$19
CPM B



Could we get results earlier?

~500-2500
#samples



Epistemic uncertainty messes with US...

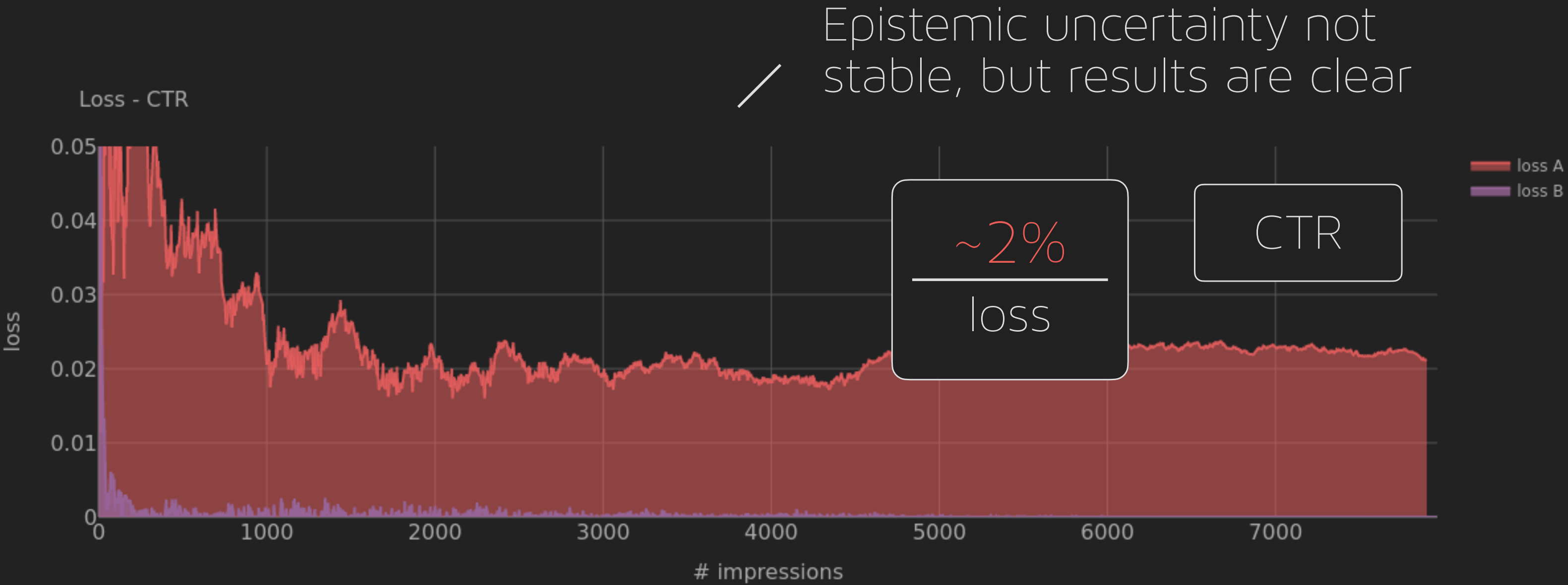
Bayesian A/B Testing - Campaign simulation

A/B test - Losses

A

10%
CTR A

\$11
CPM A

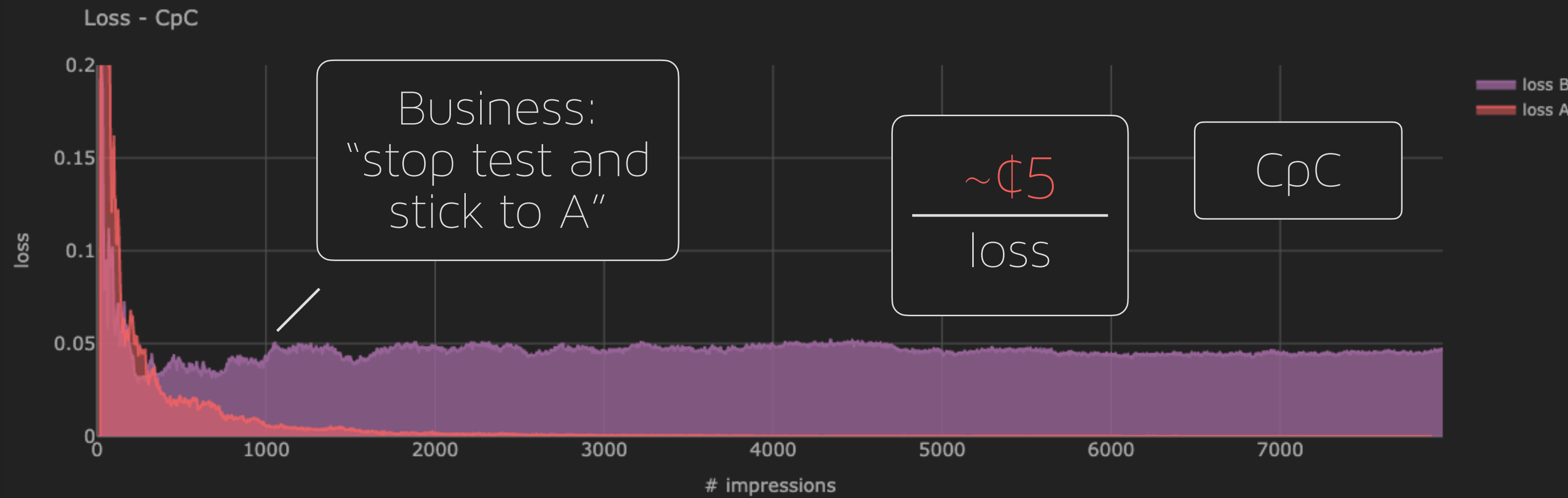


Choosing A is pretty expensive....

B

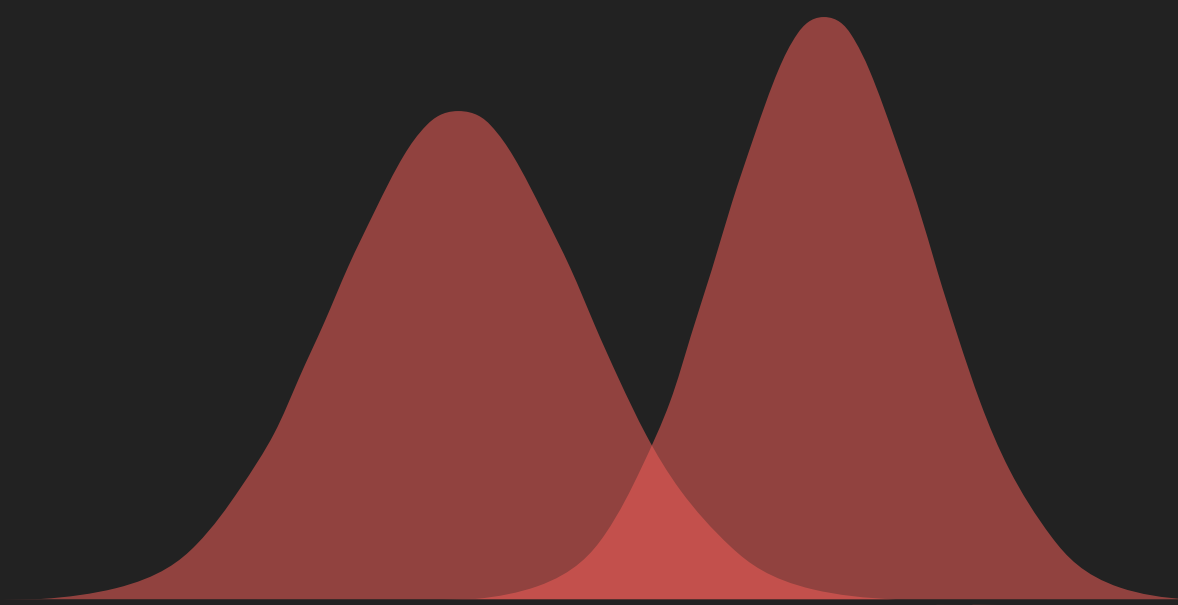
12%
CTR B

\$19
CPM B



Choosing B is pretty expensive....

This time in real \$\$\$



Bayesian A/A testing

Hypothesis A/A Testing - Campaign simulation

Same / Same test - CTR

A

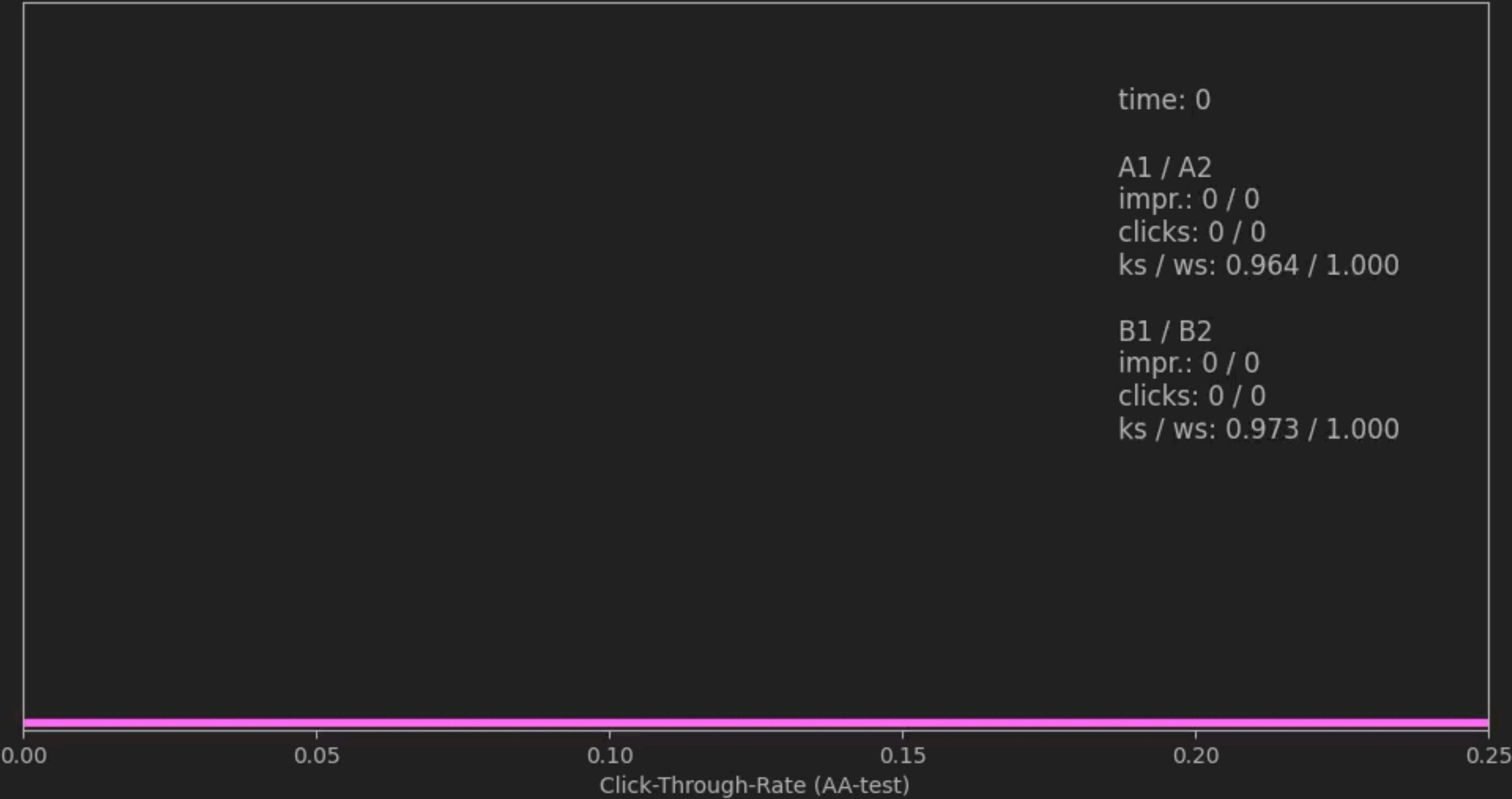
10%
CTR A

\$11
CPM A

B

12%
CTR B

\$19
CPM B



Hypothesis A/A Testing - Campaign simulation

Same / Same test - CTR

A

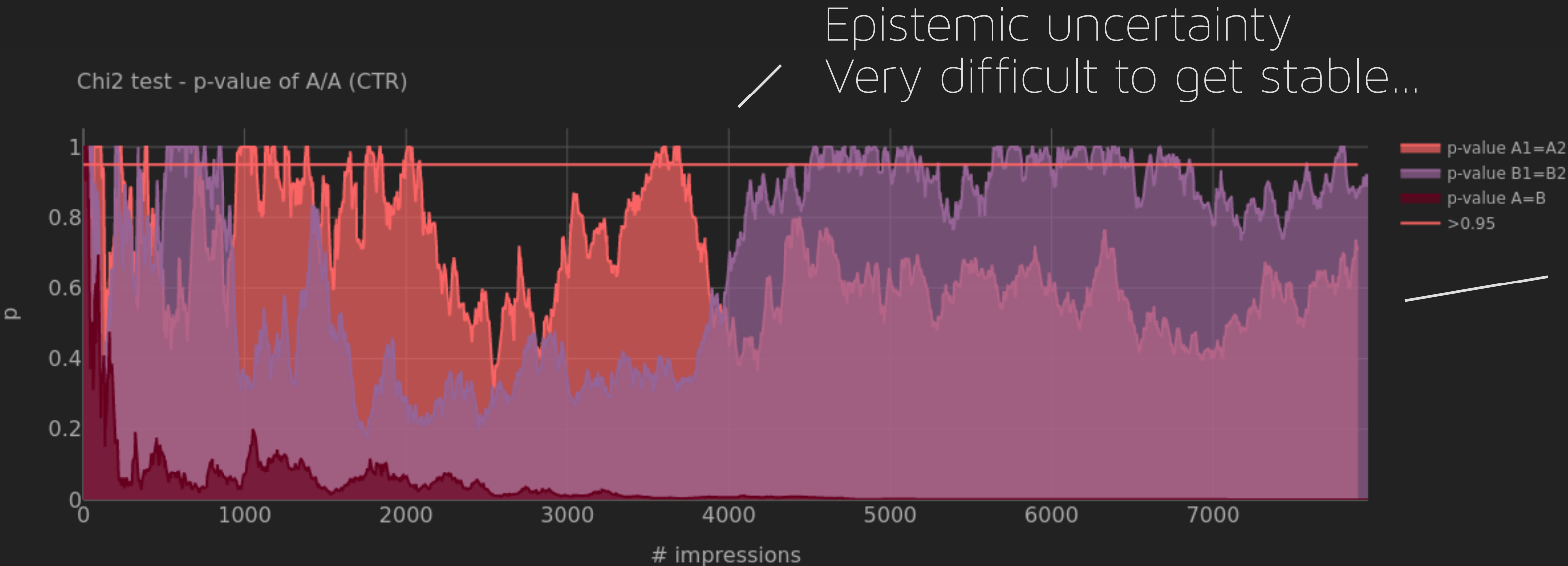
10%
CTR A

\$11
CPM A

B

12%
CTR B

\$19
CPM B

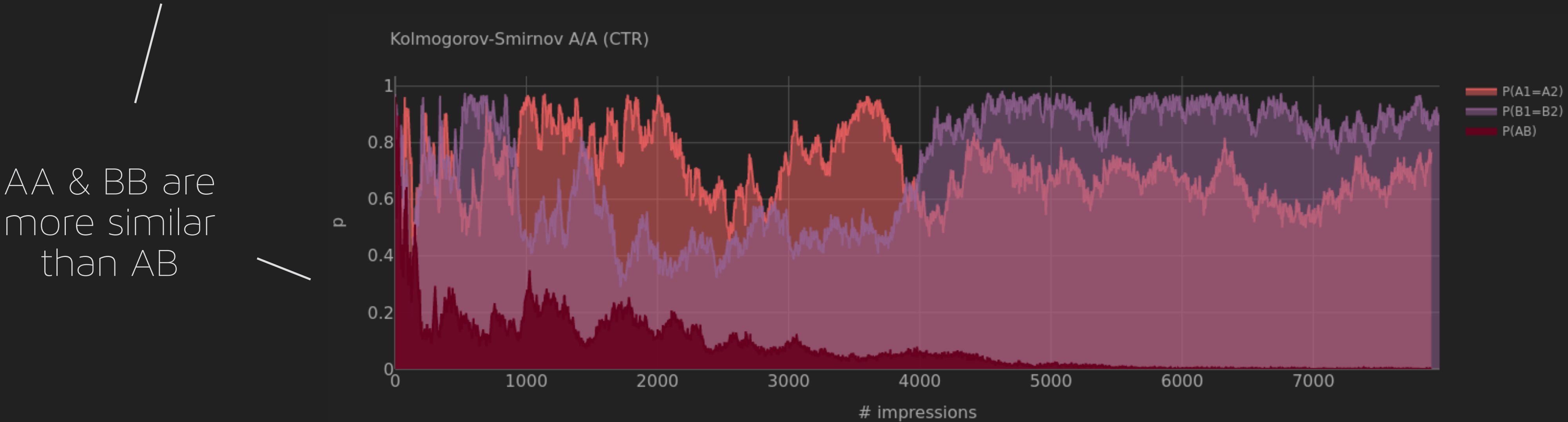


A didn't make it...

65%
p-value

97%
p-value

Does it make sense to use hypo. test for this?



Hypothesis A/A Testing - Campaign simulation

Same / Same test - CpC

A

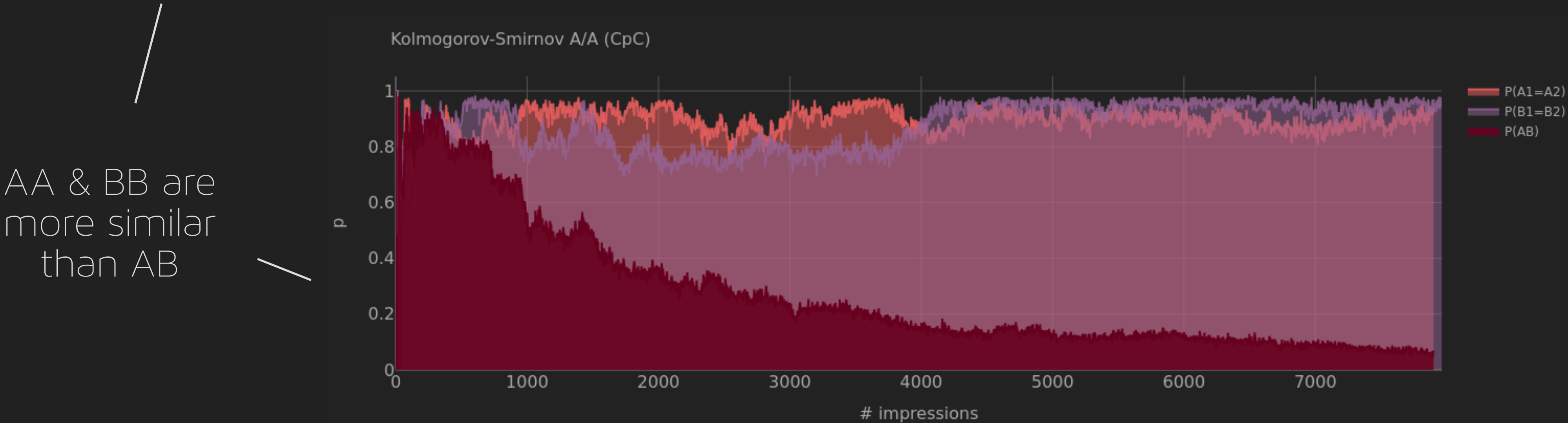
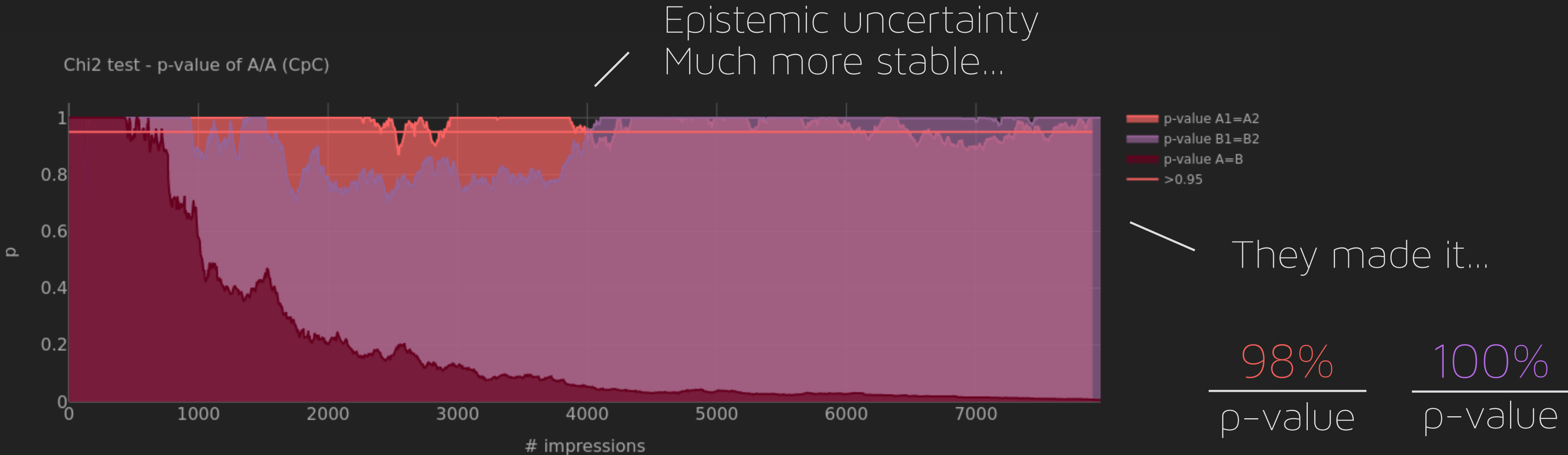
10%
CTR A

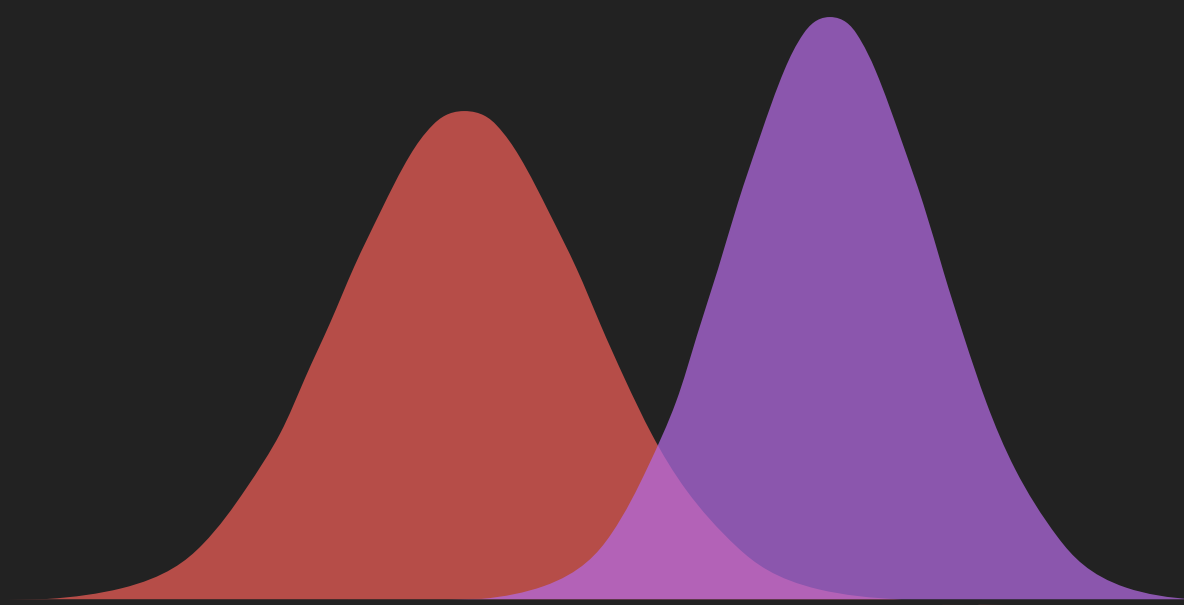
\$11
CPM A

B

12%
CTR B

\$19
CPM B





Practical Considerations
(...in theory)

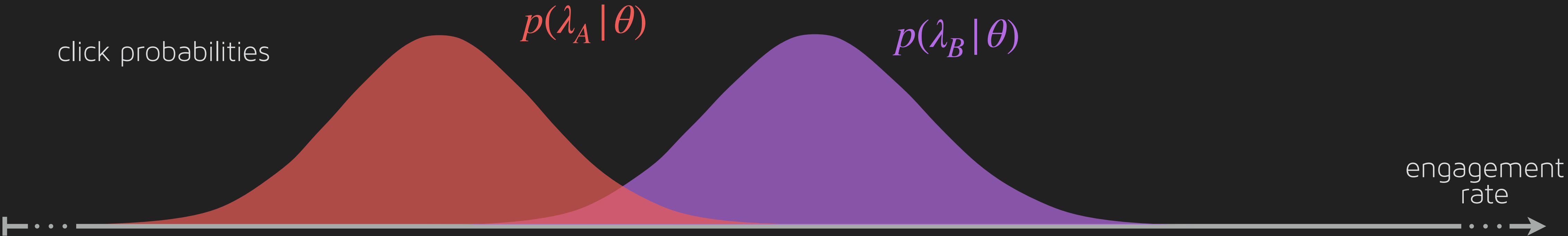
Bayesian Conditioning...

Condition on certain scenarios - eg. reel position / category...

How? Redo calc. on filtered subset of data. Uncertainty will tell you, if you can decide - just as before...

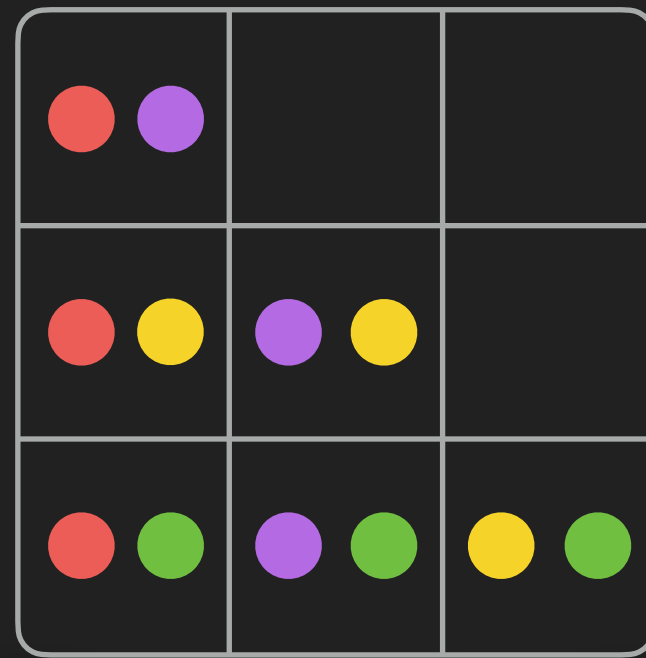
$P(\lambda_B > \lambda_A \mid \theta), \quad \theta =$

	Example	
Item category	"Vacation"	"Cars"
Time Geography ...	12:00 London ...	14:00 Paris ...
Slate location	3	2

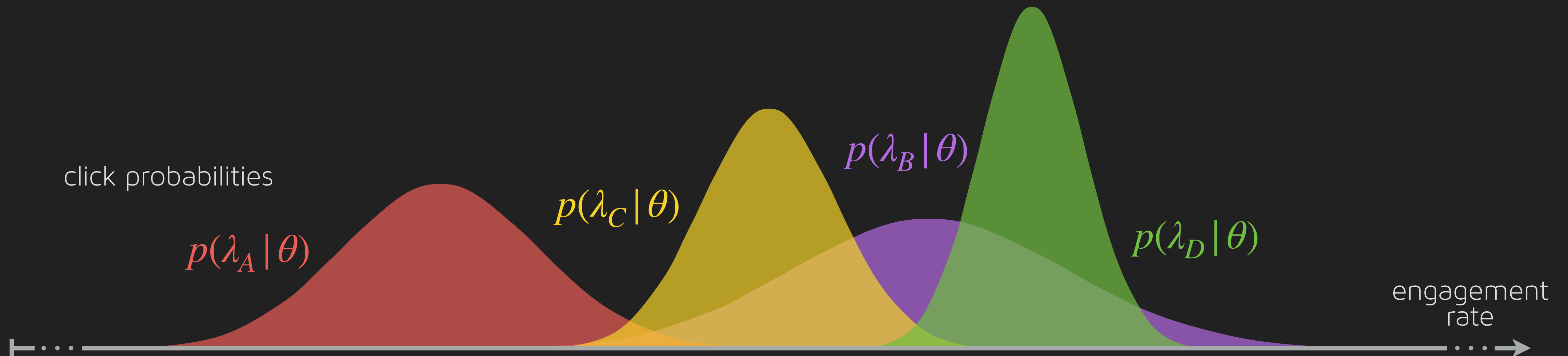


Bayesian A/B/C/D-Testing

Framework for evaluating pairs,
but could have many
combinations...



How to handle
multiple
variants?



Some practical considerations

How long should you wait for the noise to clear and decide...?

Balance
between noise
and decisions



Prime observations - Bayesian Prior

our observations what we believe before the data

Posterior Likelihood Prior

$$p(\theta | x) \propto p(x | \theta) \cdot p(\theta)$$

Potential faster conversion

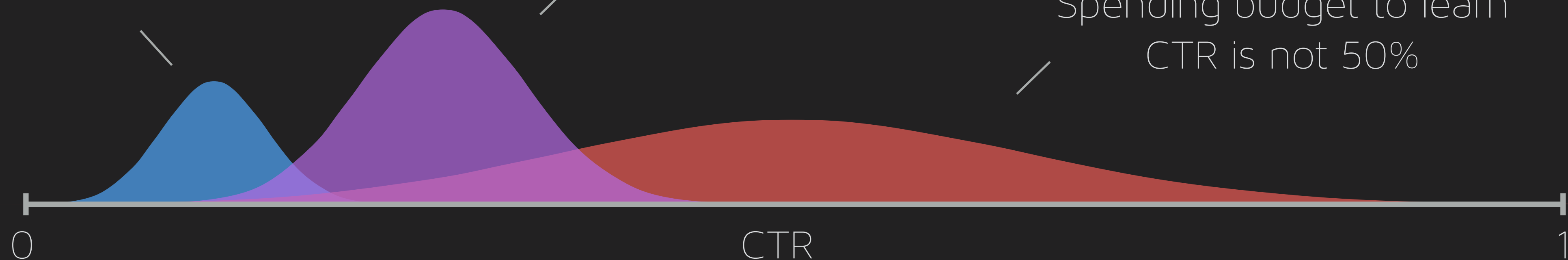
...but, don't overdo or it requires many samples to fix...

Prime
impressions / clicks
add small numbers
to observed count

Assisted or
updated belief
of the true CTR
value

Our observations in an
early campaign.

Spending budget to learn
CTR is not 50%



Using the Gaussian distribution
...If we must

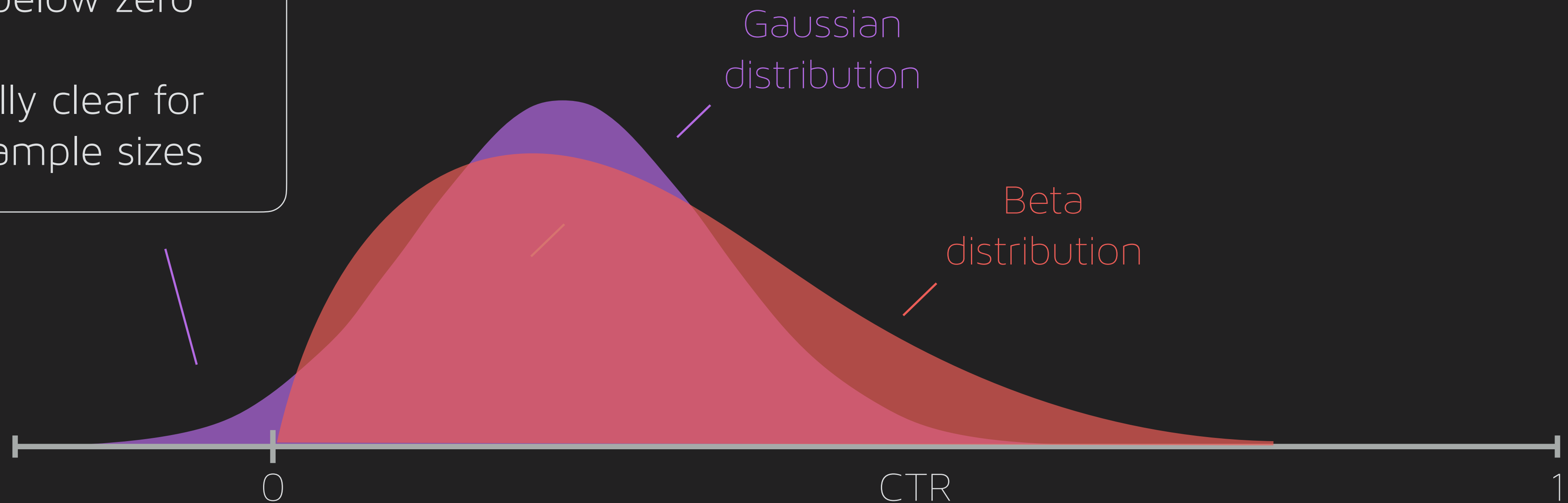


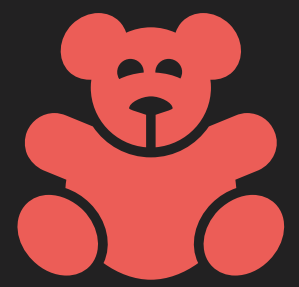
```
streamlit run src/ui/AB_Test.py
```

Gaussian Distribution

Unbounded
Leaks below zero

Especially clear for
small sample sizes





Small quiz

What
mistakes did I
make...?

Bayesian A/B-Testing End-of-Level summary

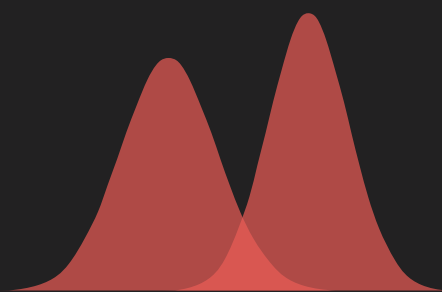
Practical



Know what you are doing!
You are the king/queen...



Be critical - visualise the
progression and assess,
not just the final numbers



A/A-test - don't forget

Simple tools to evaluate - but not
so simple to decide...?

Better to be confused, than blind....

Business



Business intuitive
Interpretable metrics.
Not only a DS decision



Faster conversion

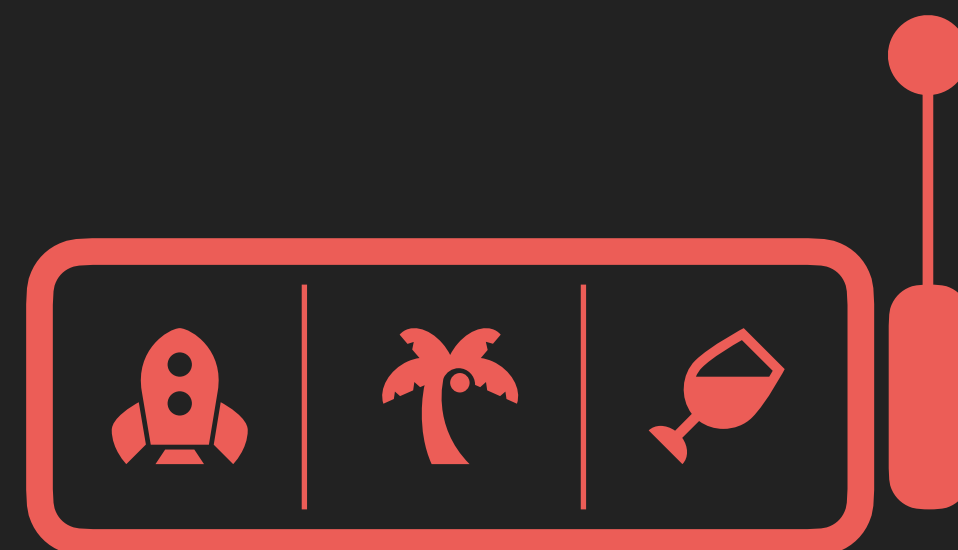
More granular decisions -
40% chance B is 10% better



Loss in real \$



Questions



Bayesian Bandit Testing

A/B-test challenges....

\$\$\$

Suboptimal for
23.5 hours

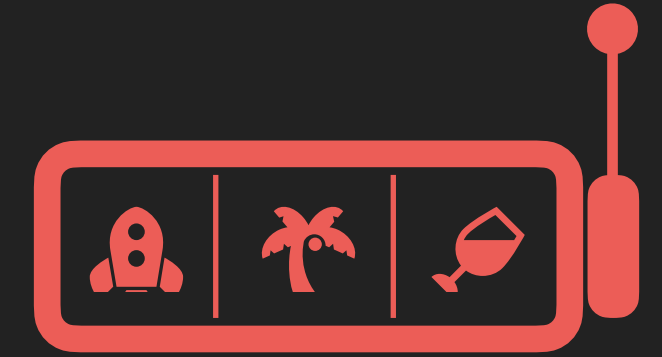
Test keeps running
even after winner is
found...



non-stationary
environment

Fails test on
Monday / Tuesday

But better on
weekends



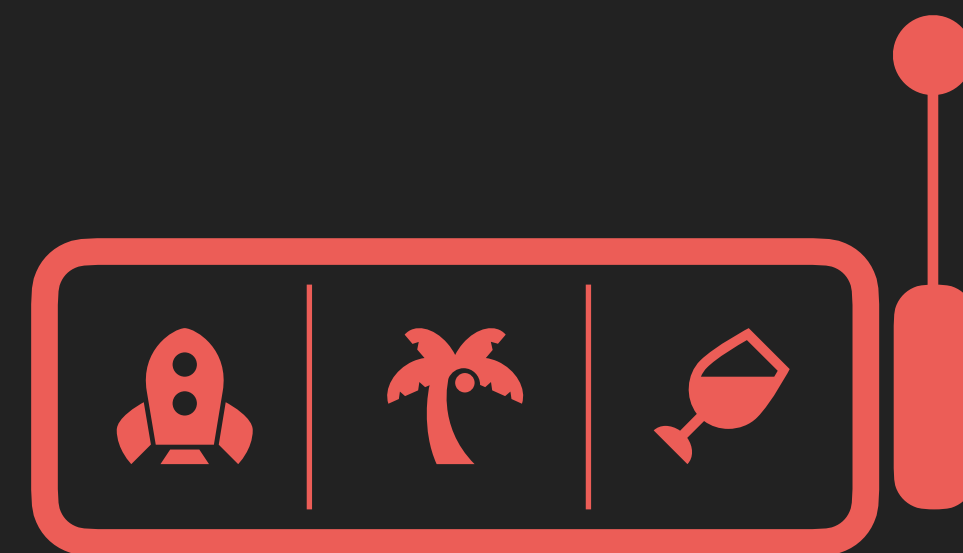
Bayesian Bandits

A bandit test system can
handle soft activation /
deactivation of all variants

....while you sleep :-)

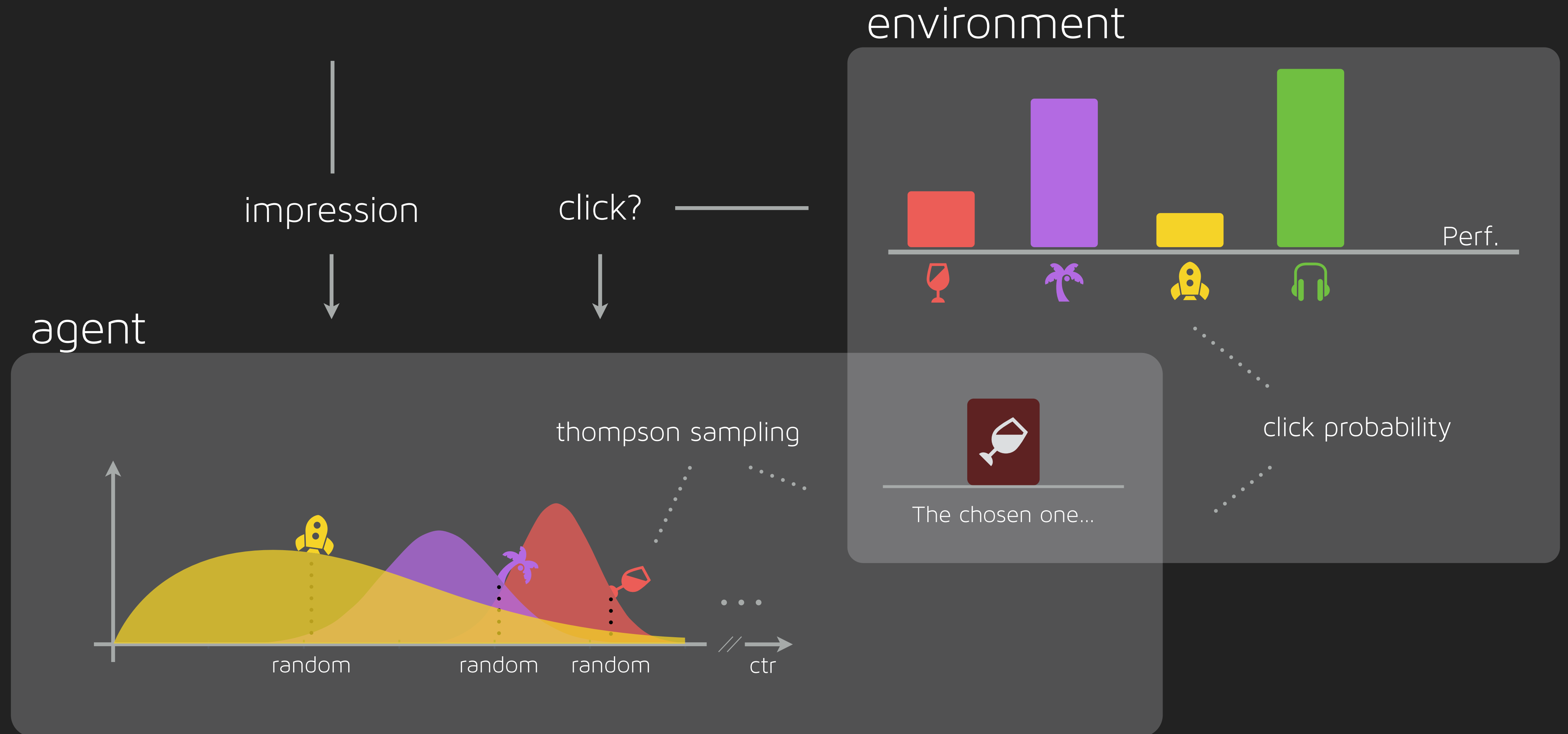
Just put it in and let it run....

Opportunity to rule out
losers...



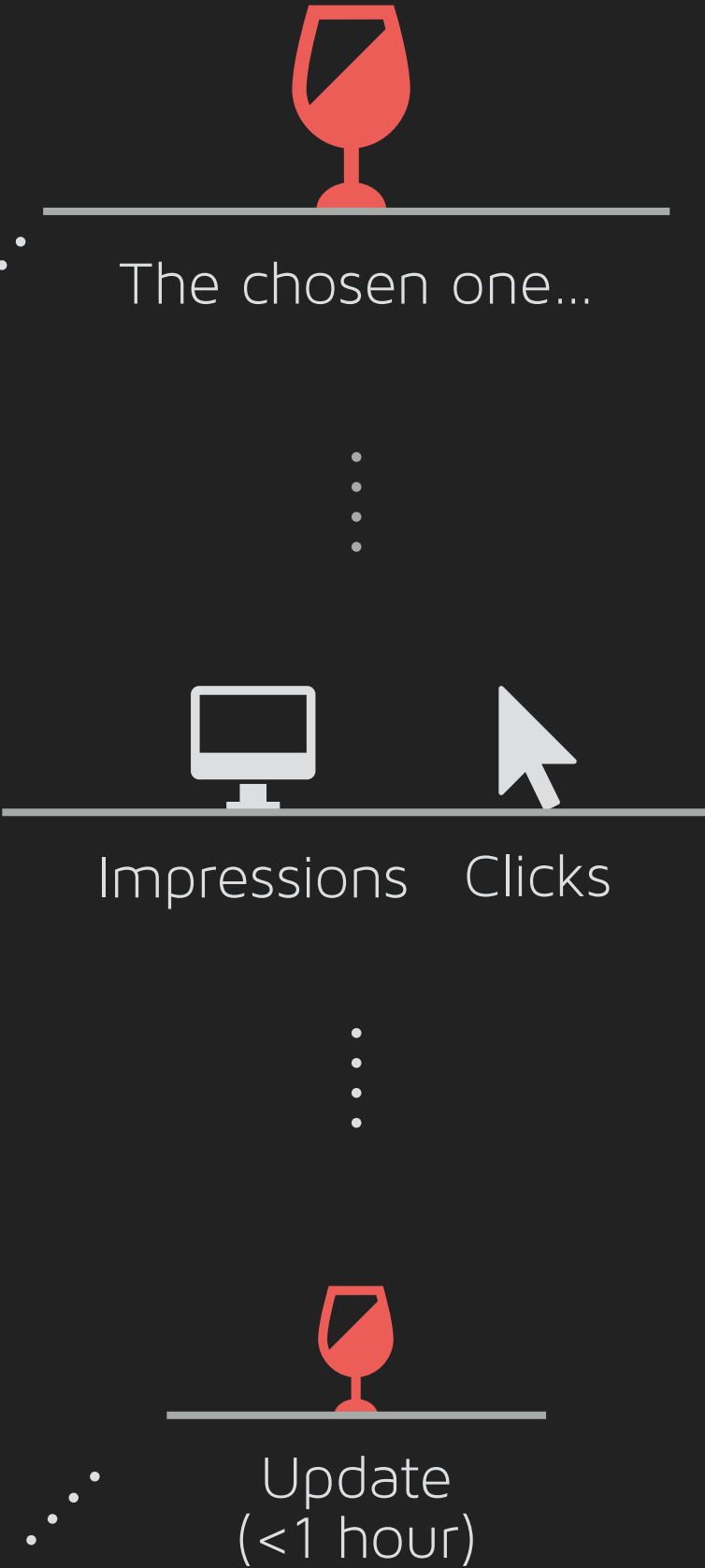
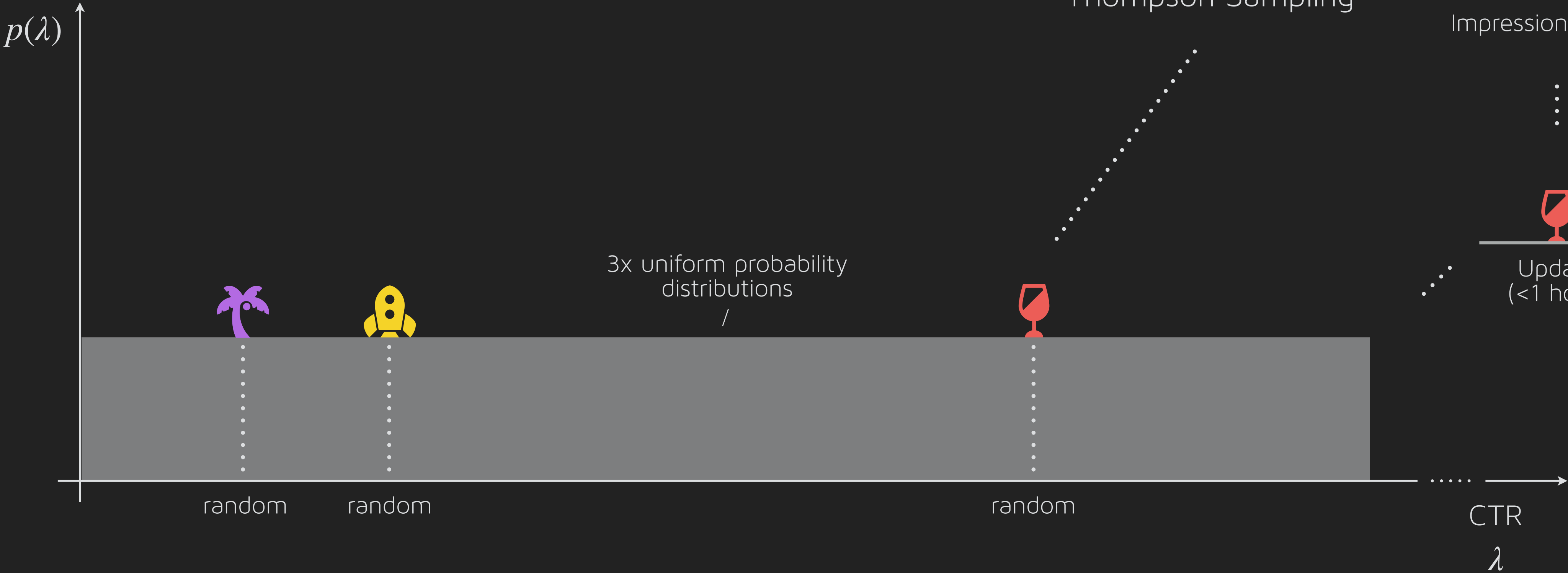
The Process

Bandit Architecture



Bayesian Beta Bandits

Zero impressions



Bayesian Beta Bandits

Few impressions



Variants



The chosen one...



Impressions

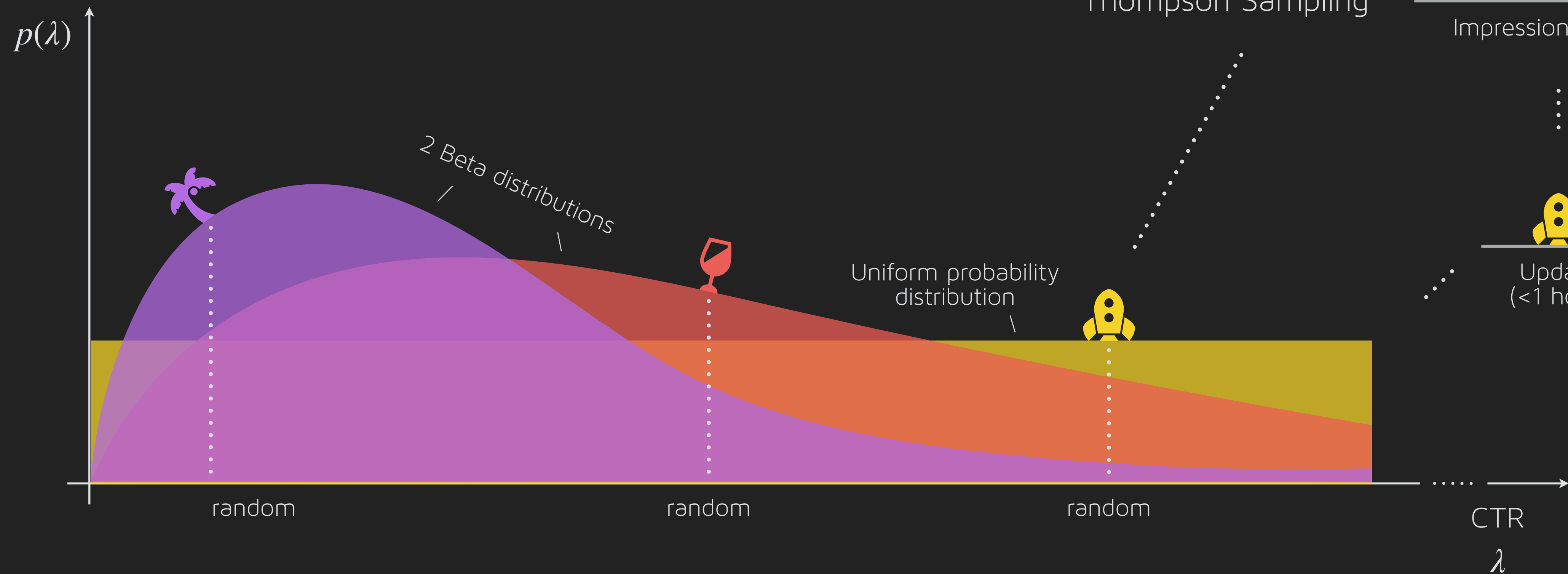


Clicks

Thompson Sampling

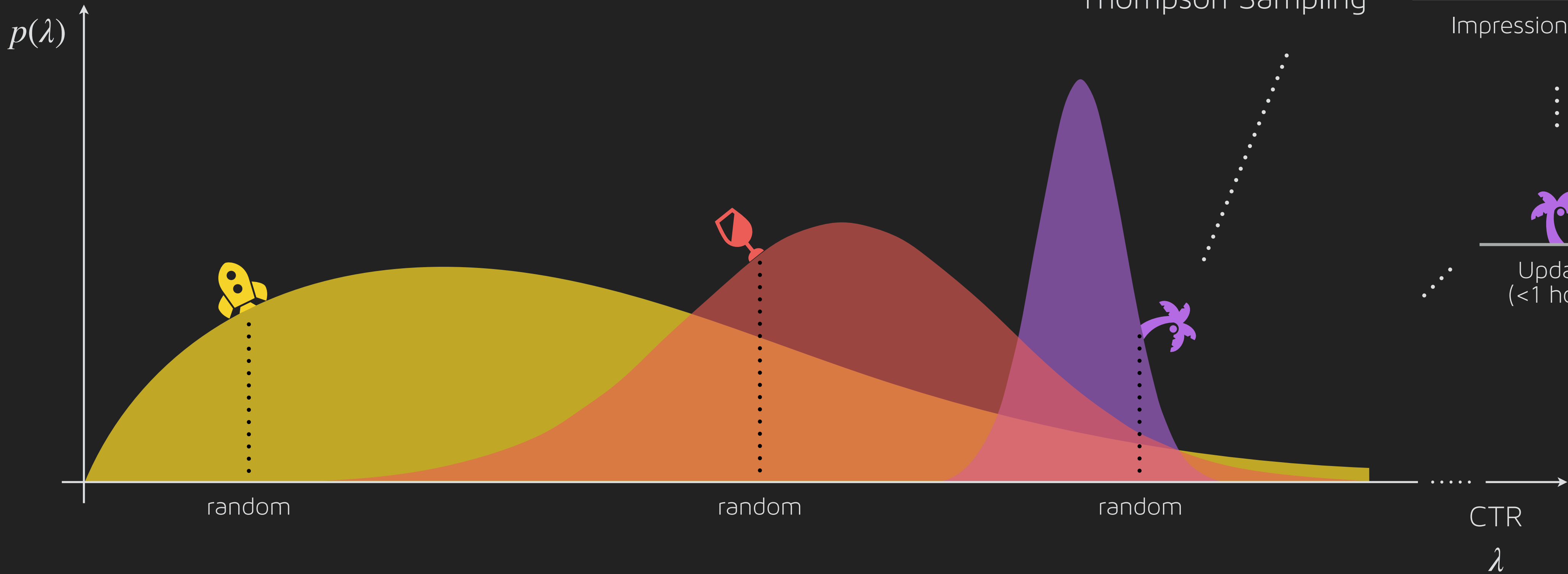


Update
(<1 hour)

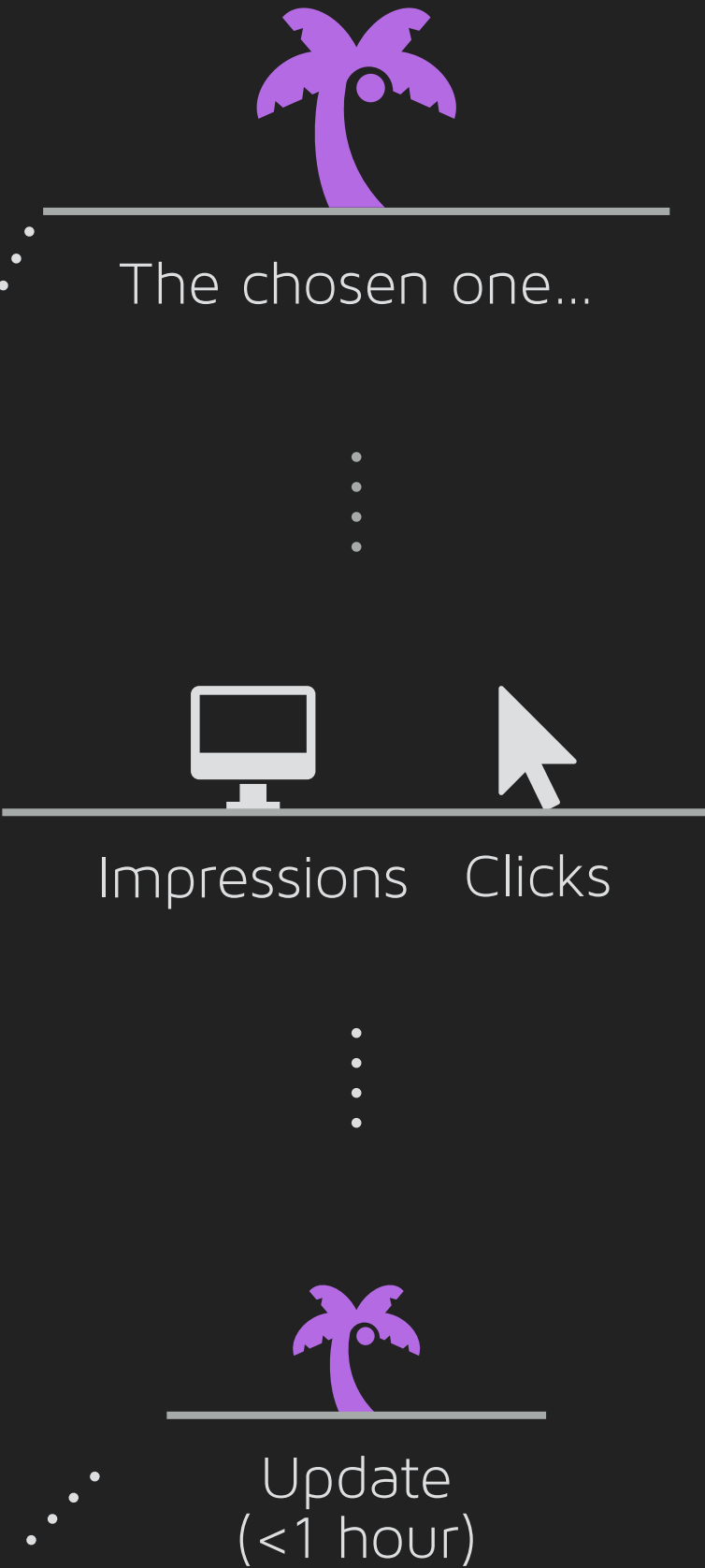


Bayesian Beta Bandits

More impressions

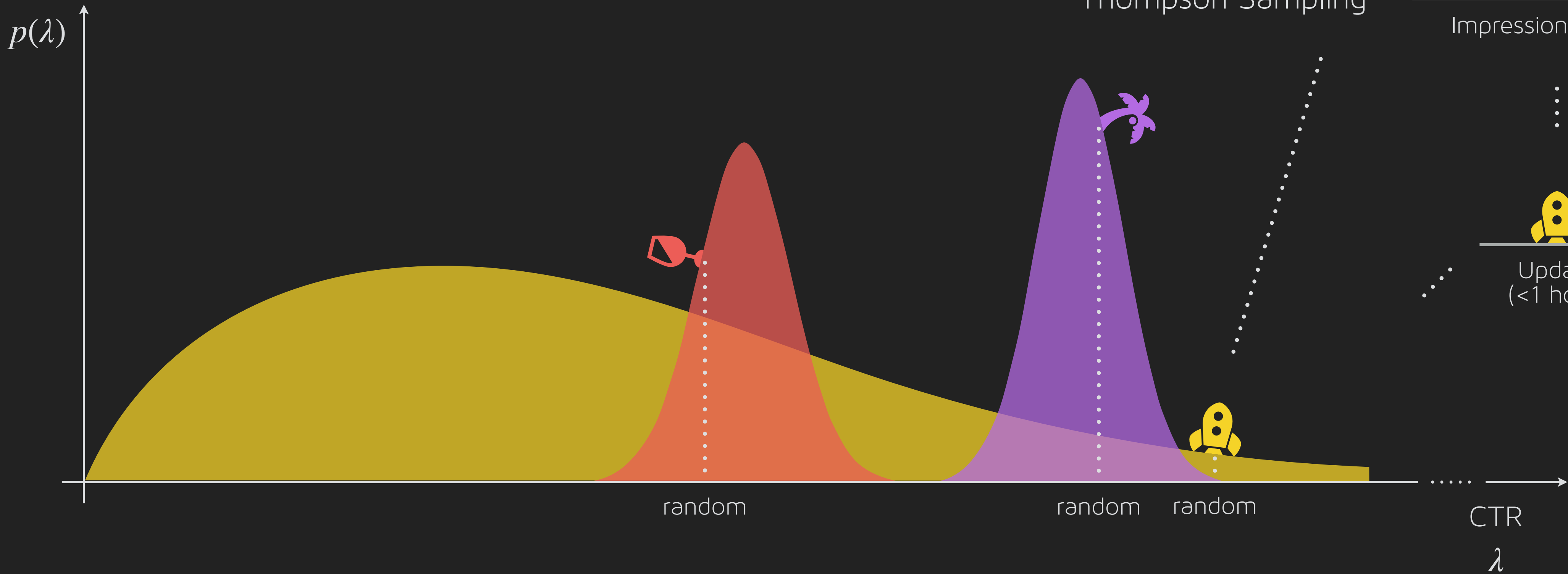


Thompson Sampling

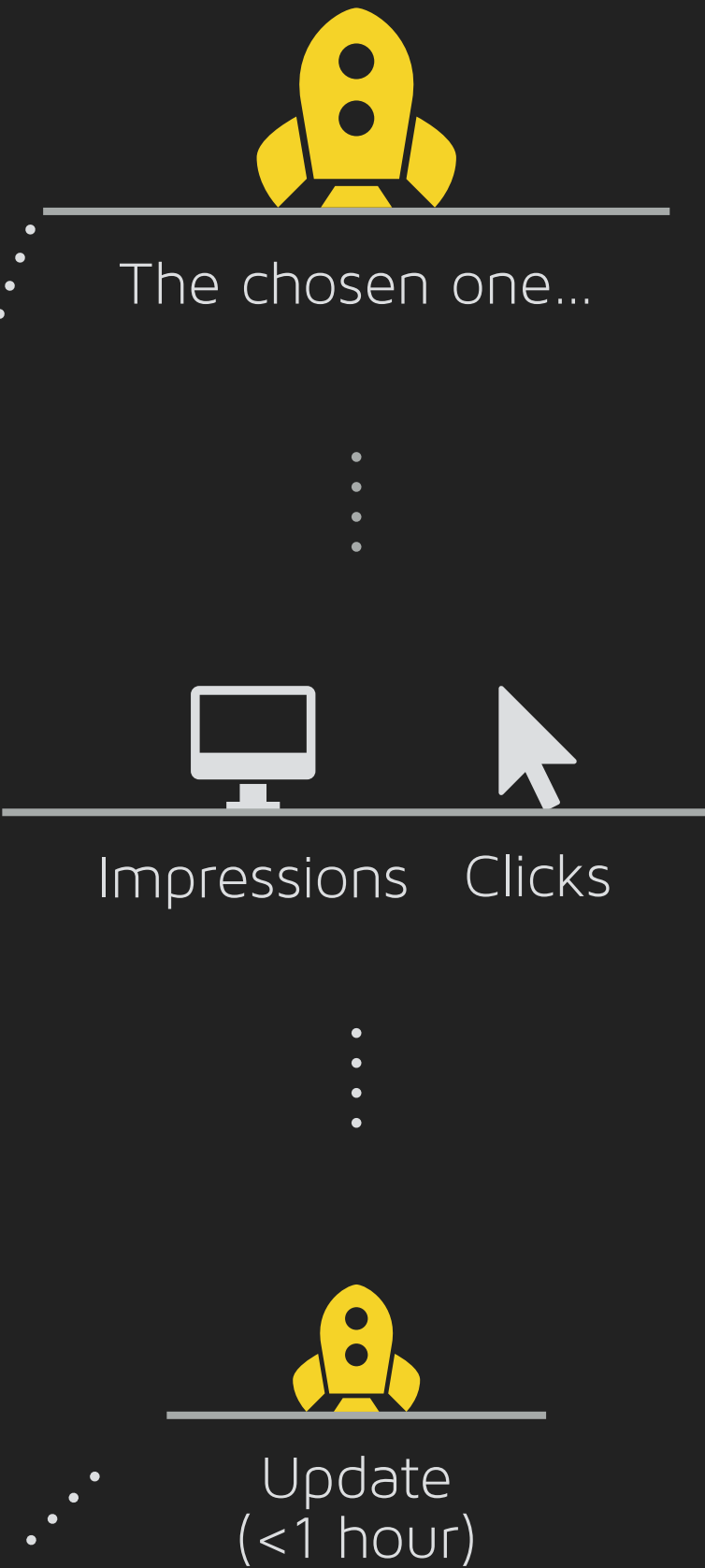


Bayesian Beta Bandits

More impressions

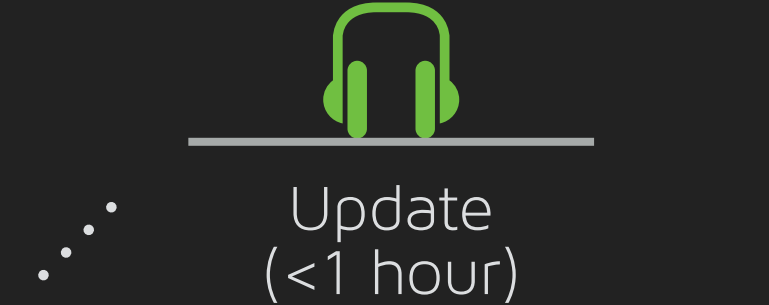


Thompson Sampling

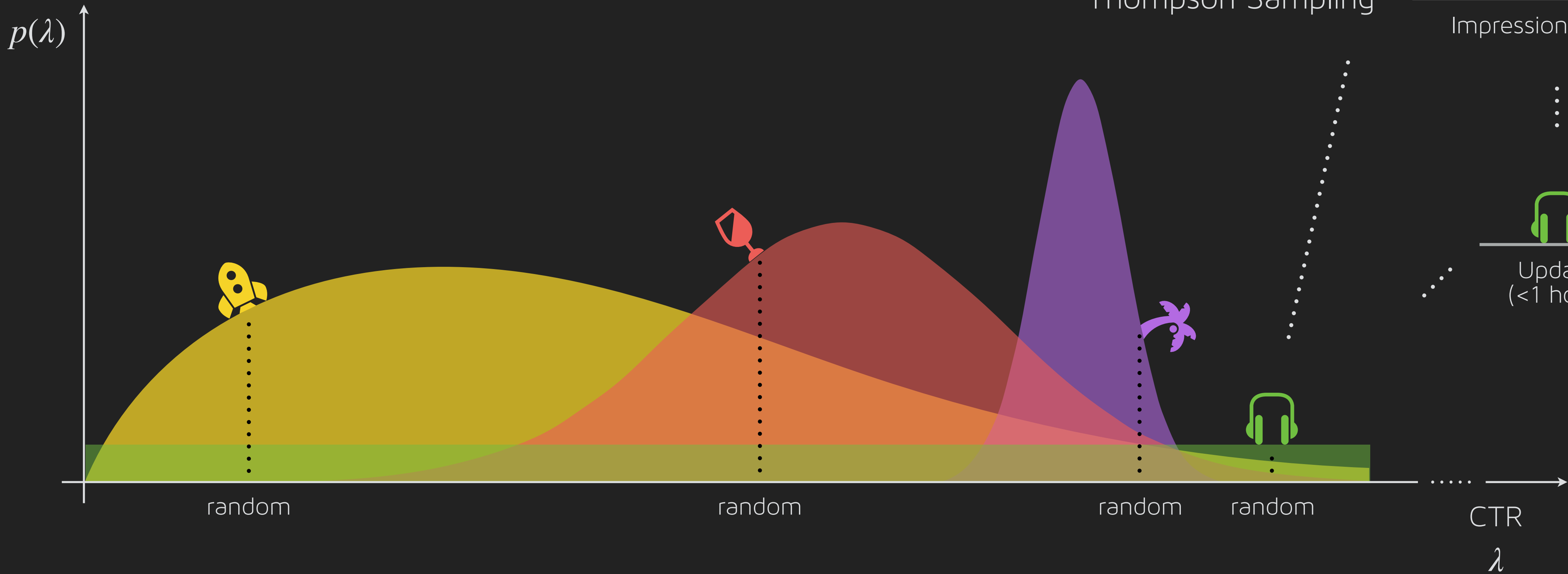


Bayesian Beta Bandits

Cold start



Thompson Sampling



Exploration



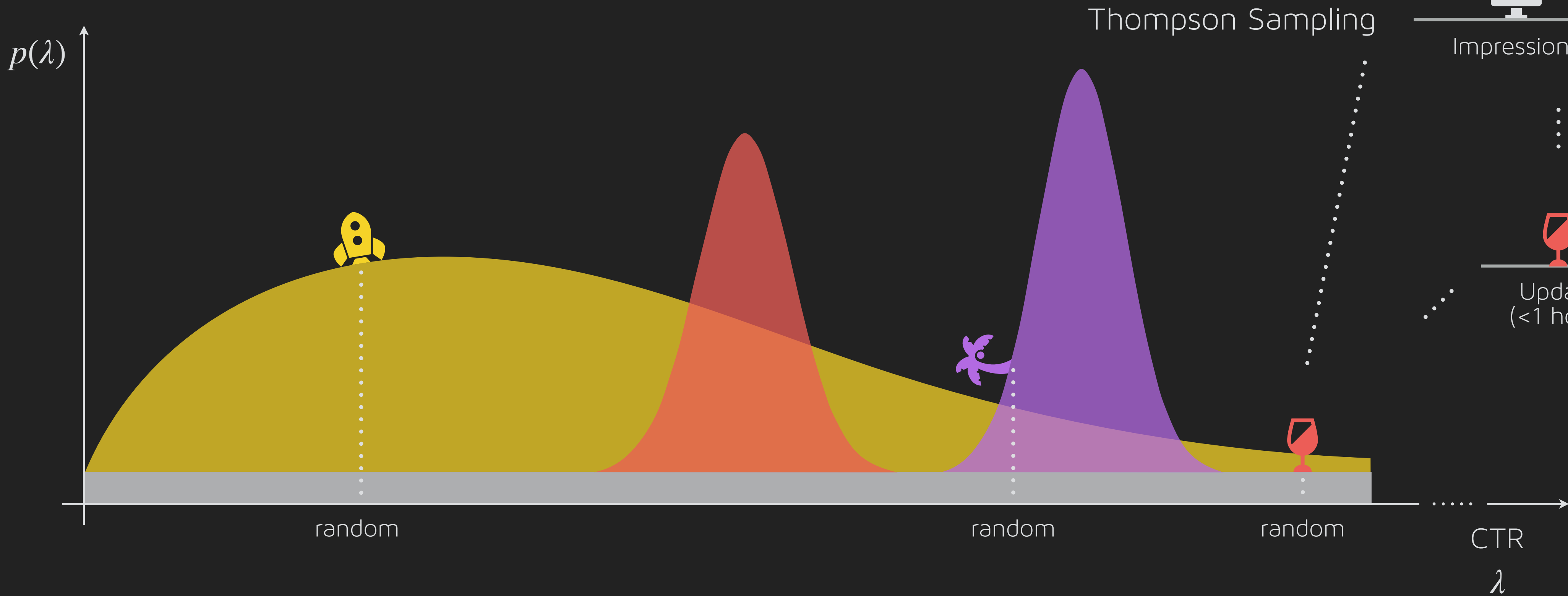
vs



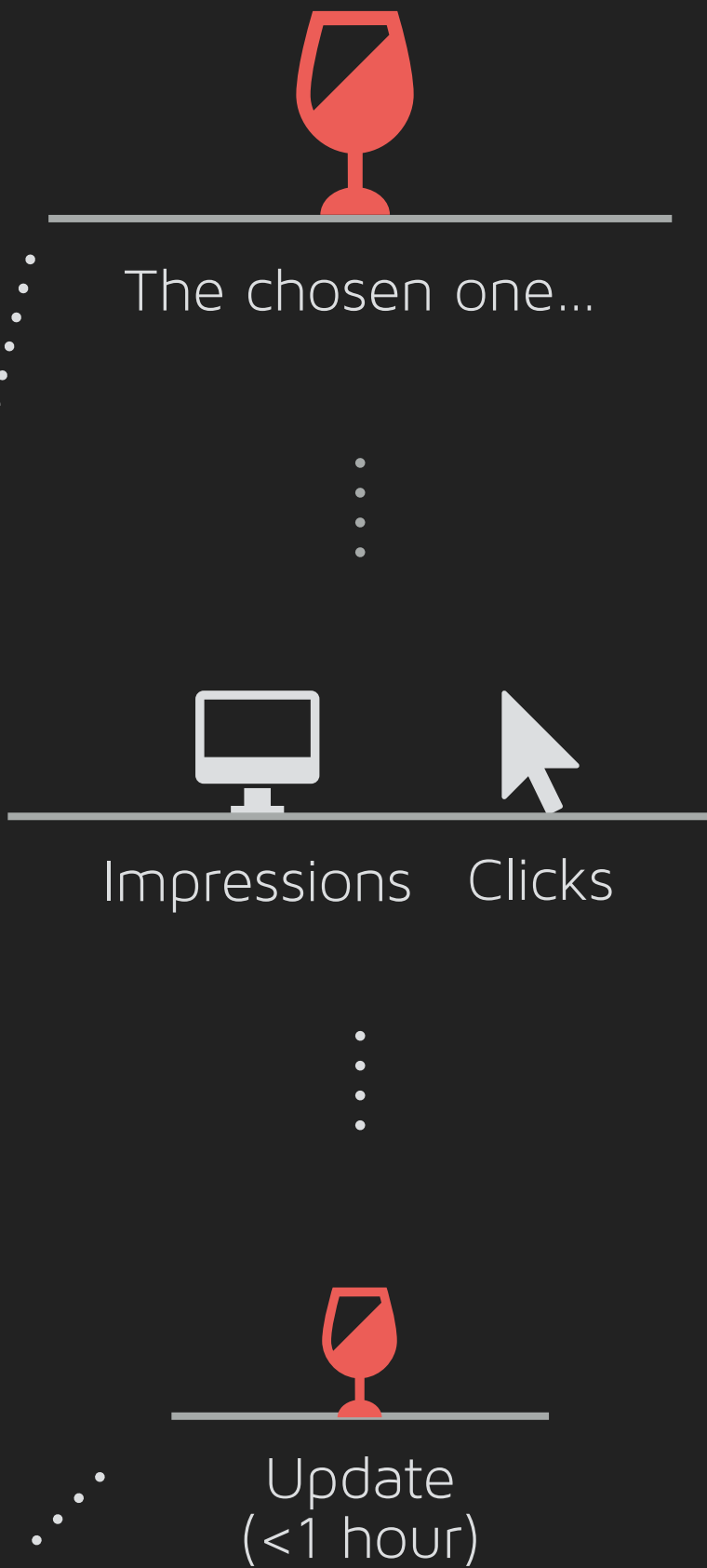
Exploitation

Bayesian Beta Bandits

Naive Exploration
 ϵ - greedy

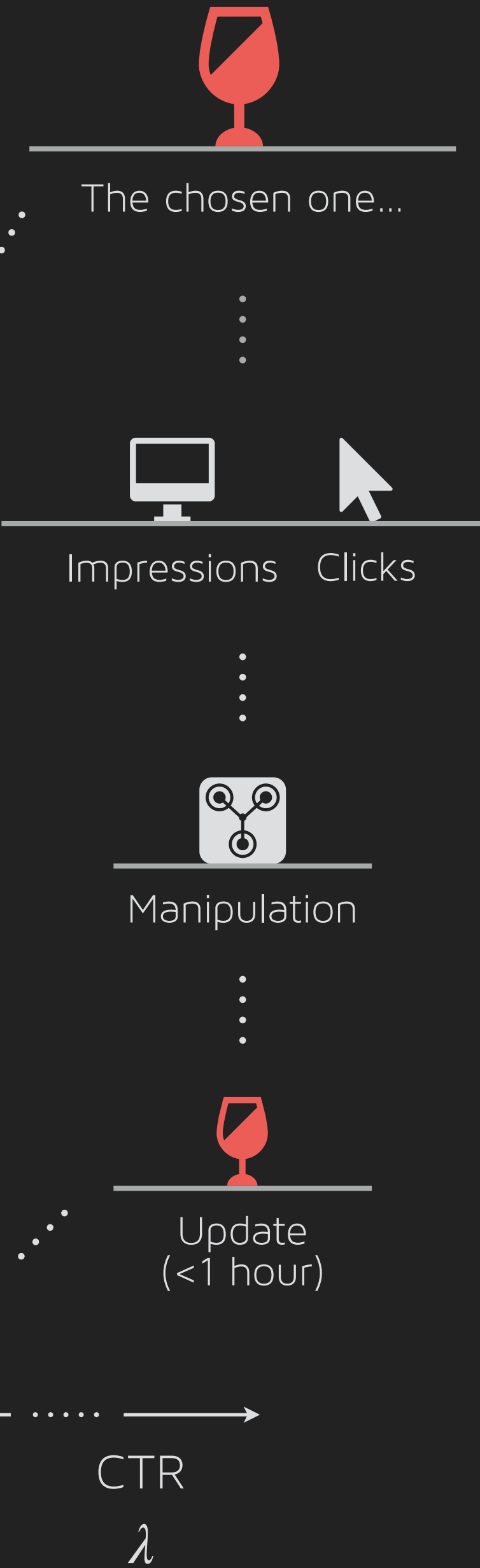


Thompson Sampling

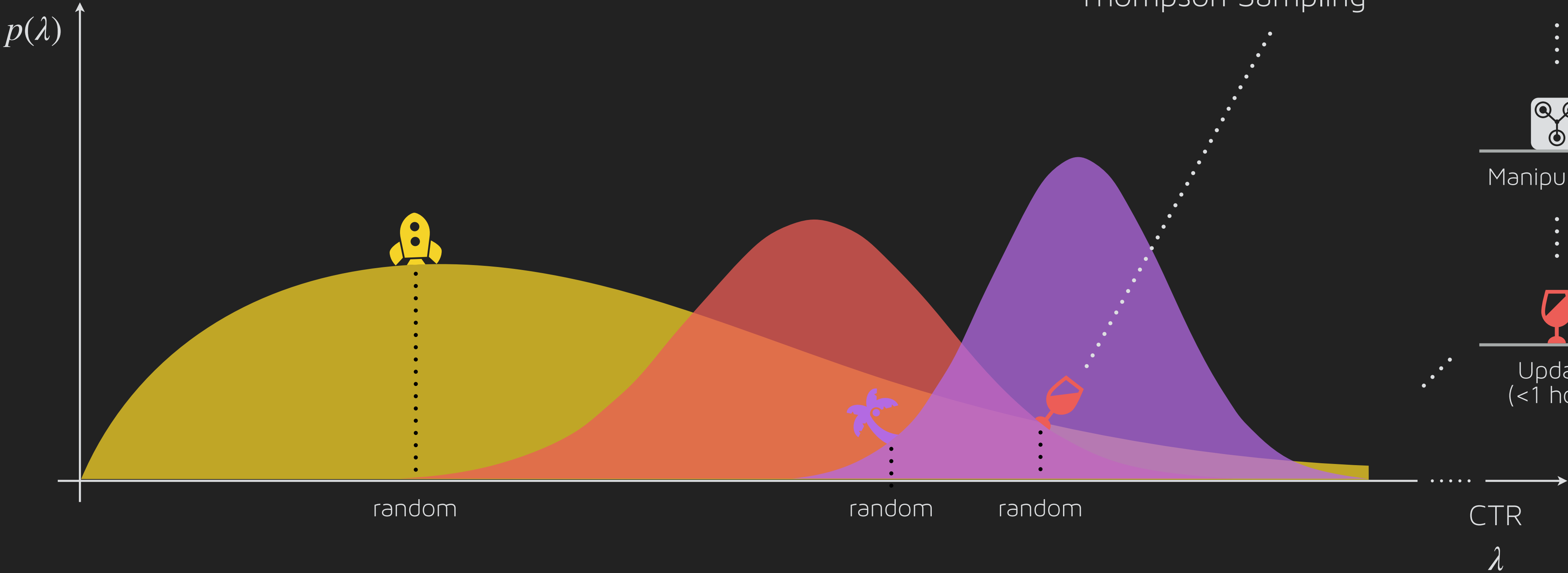


Bayesian Beta Bandits

More impressions



Thompson Sampling



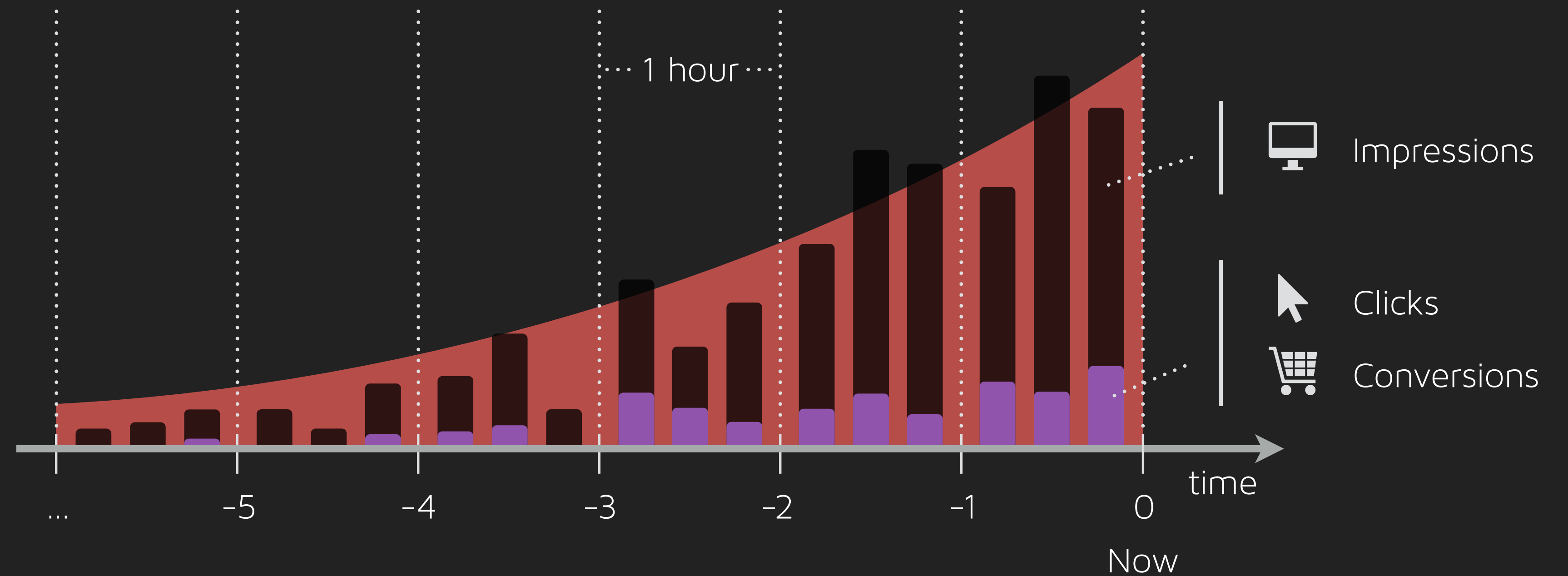


Recency
forgetting the past...

Recency - forgetting the past

Weight all counts
too limit the
influence of
historical data

Increases the
epistemic
uncertainty



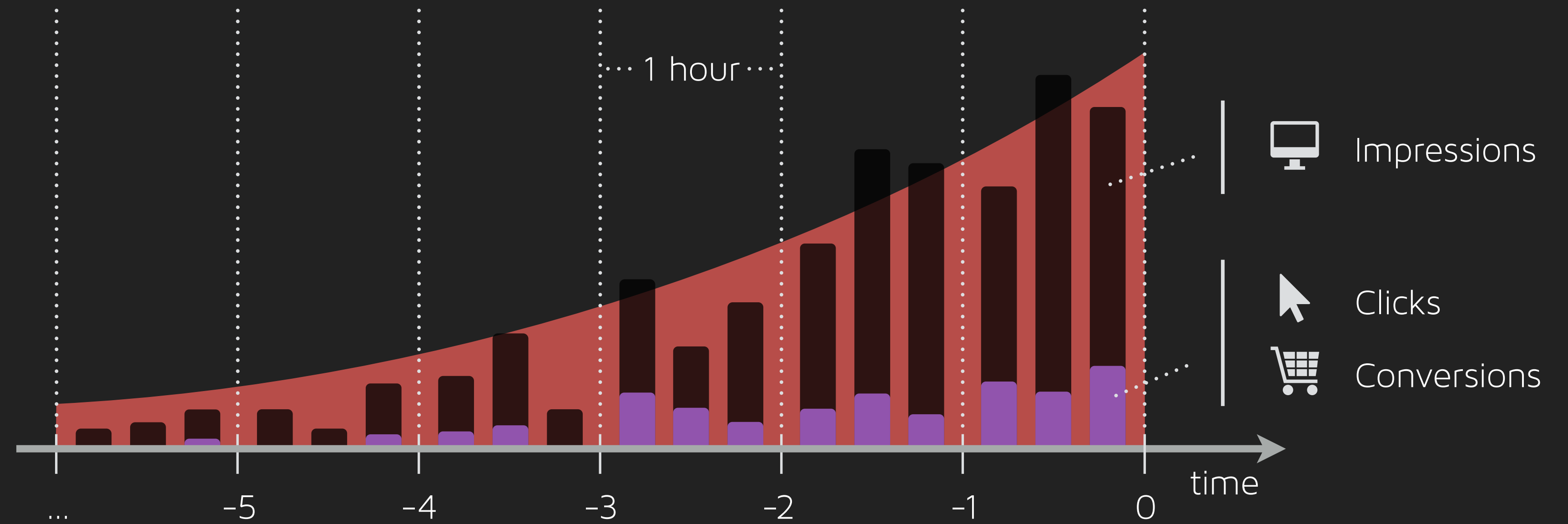
decay
function

$$f(t) = \alpha^t, \text{ where } 0 \leq \alpha \leq 1$$

Explore / Exploit trade-off

Weight all counts
too limit the
influence of
historical data

Increases the
epistemic
uncertainty



balance between
adapting to the
environment

or

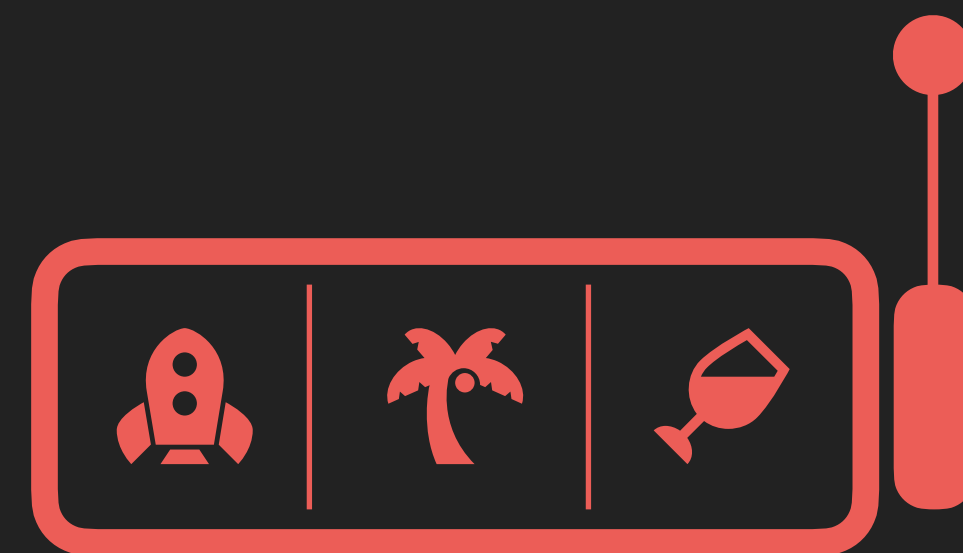
just going for it...



exploiting



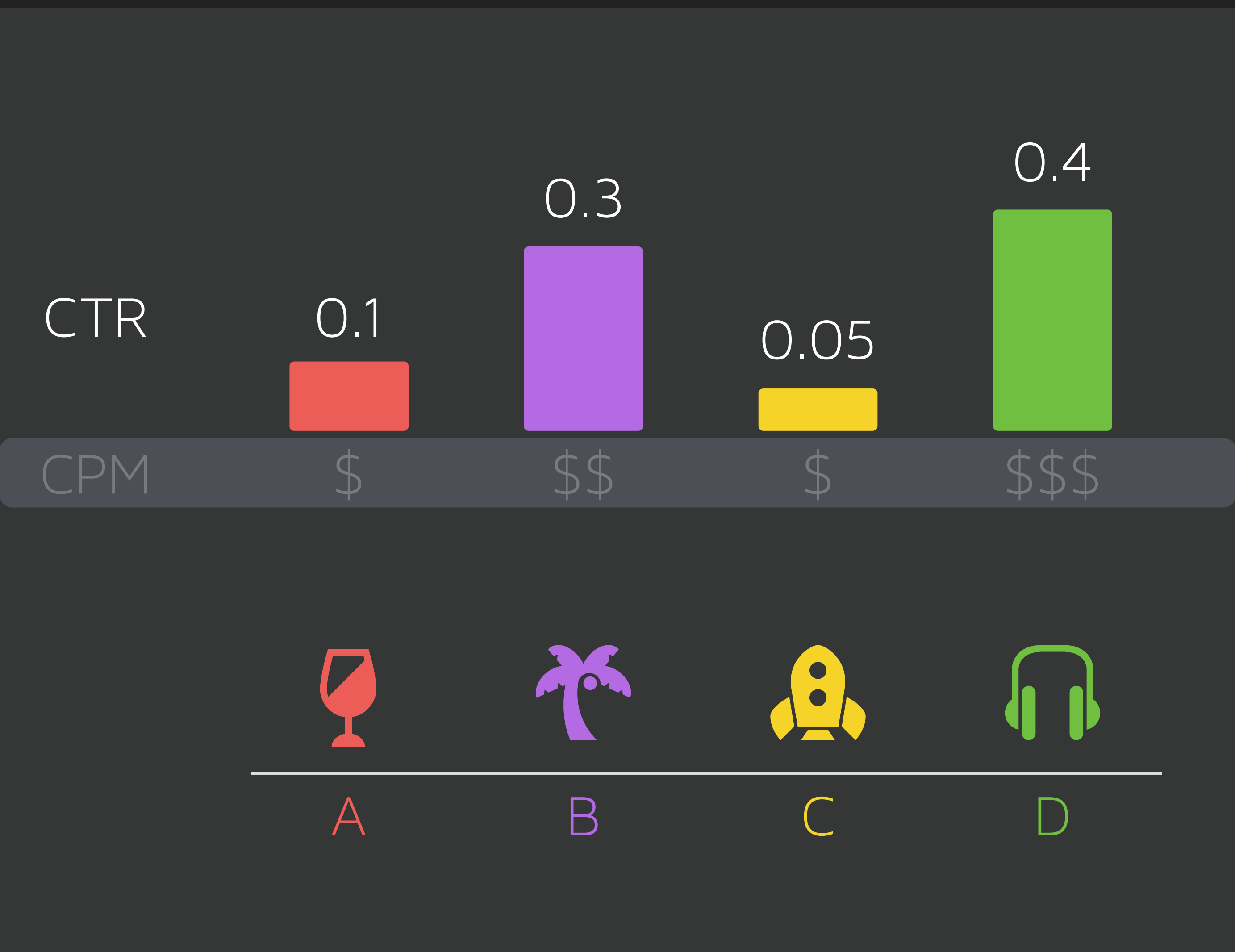
exploring



Simulation

Bandit Architecture

environment



metric

$$\text{regret} - \text{average}$$

in words

Average number of times the
optimal variant was not selected.

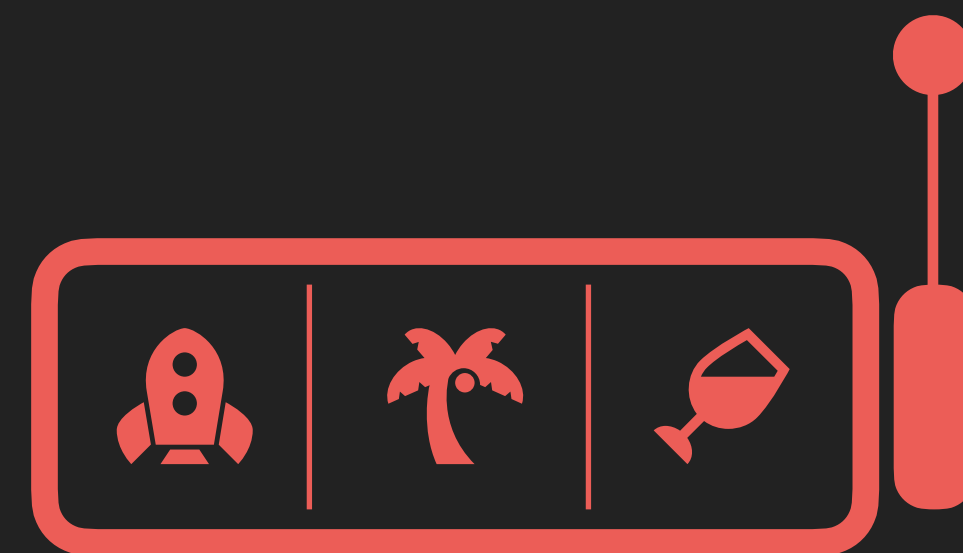
lower is better

in math

$$\rho = T\mu^* - \sum_{t=1}^T \hat{r}_t \qquad \mu^* = \max_k(\mu_k)$$

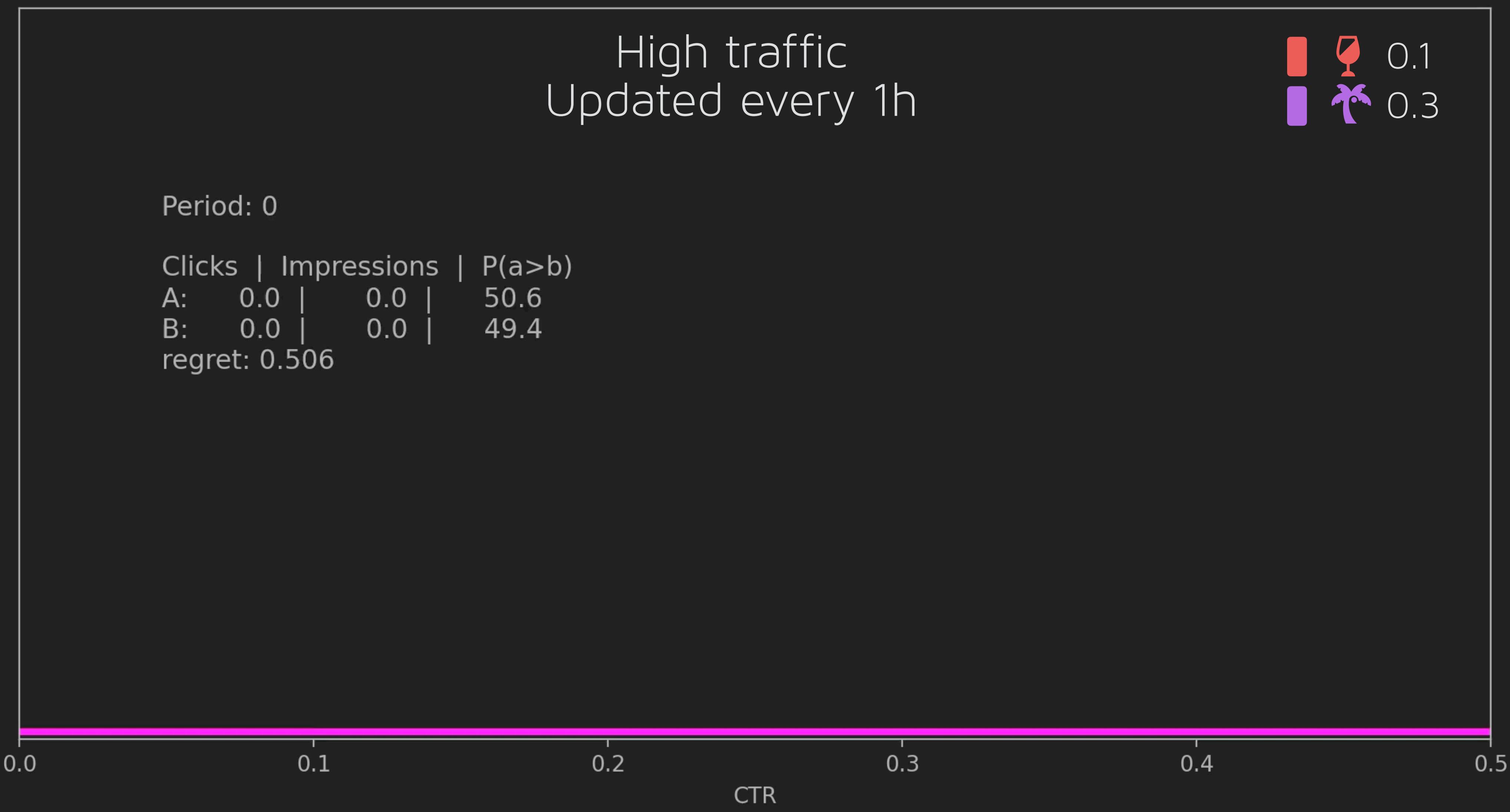
in code

```
(max_reward_mean - n_optimal_selections_b) / imp_per_period
```



Bayesian Bandit A/B testing

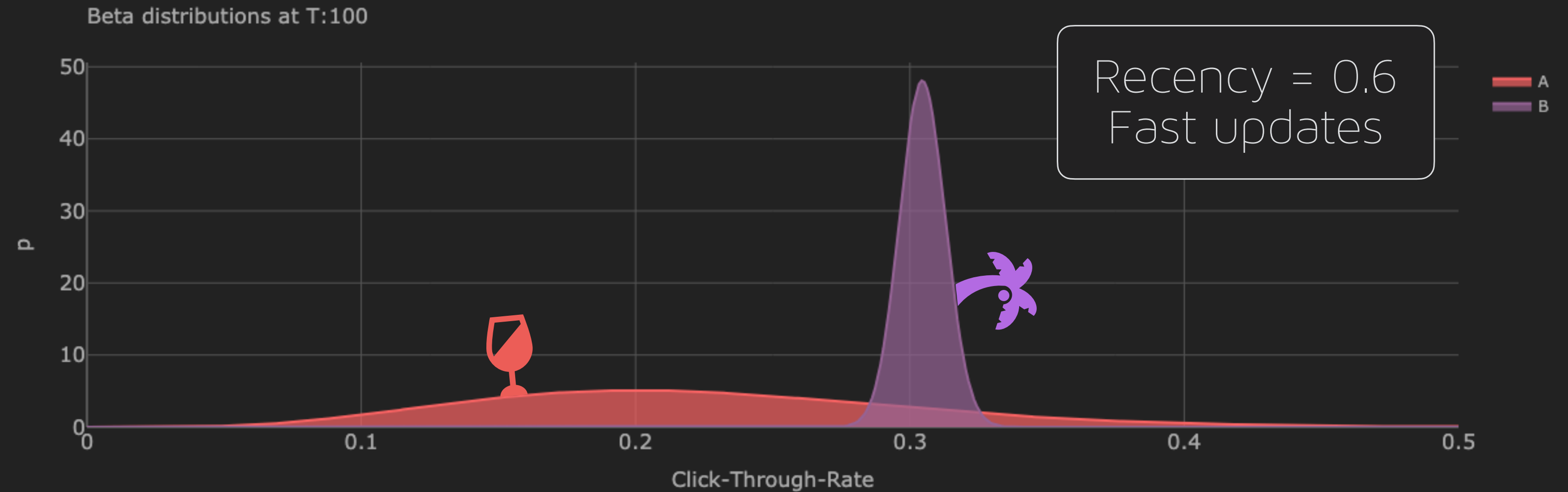
Bandit A/B-test



Bandit A/B-test

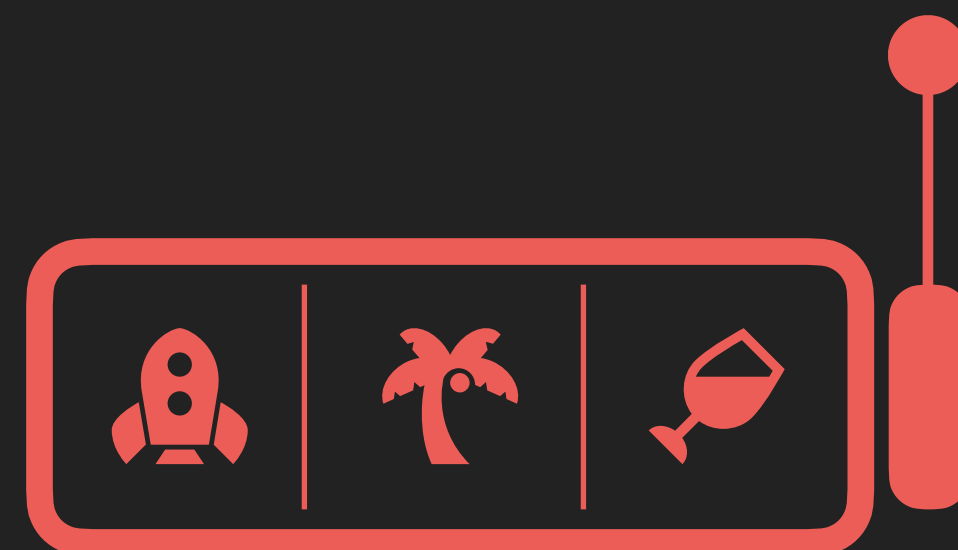
Beta Distributions

Example of the
recency making A
wide



This makes A get
selected more
often and we're
exploring...

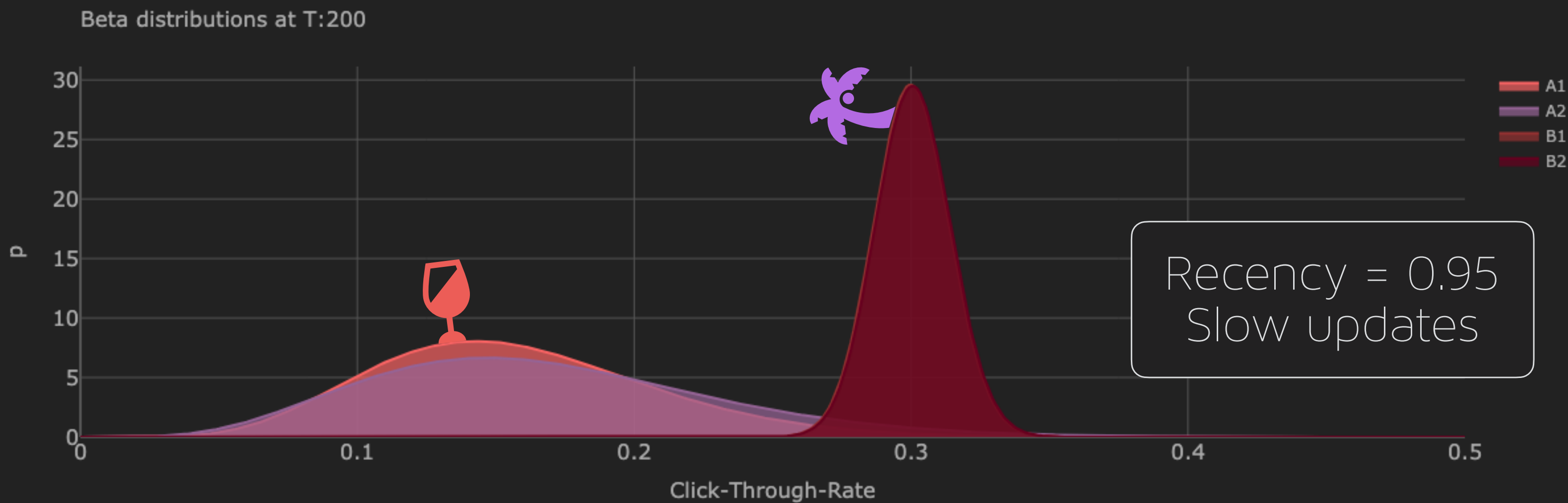




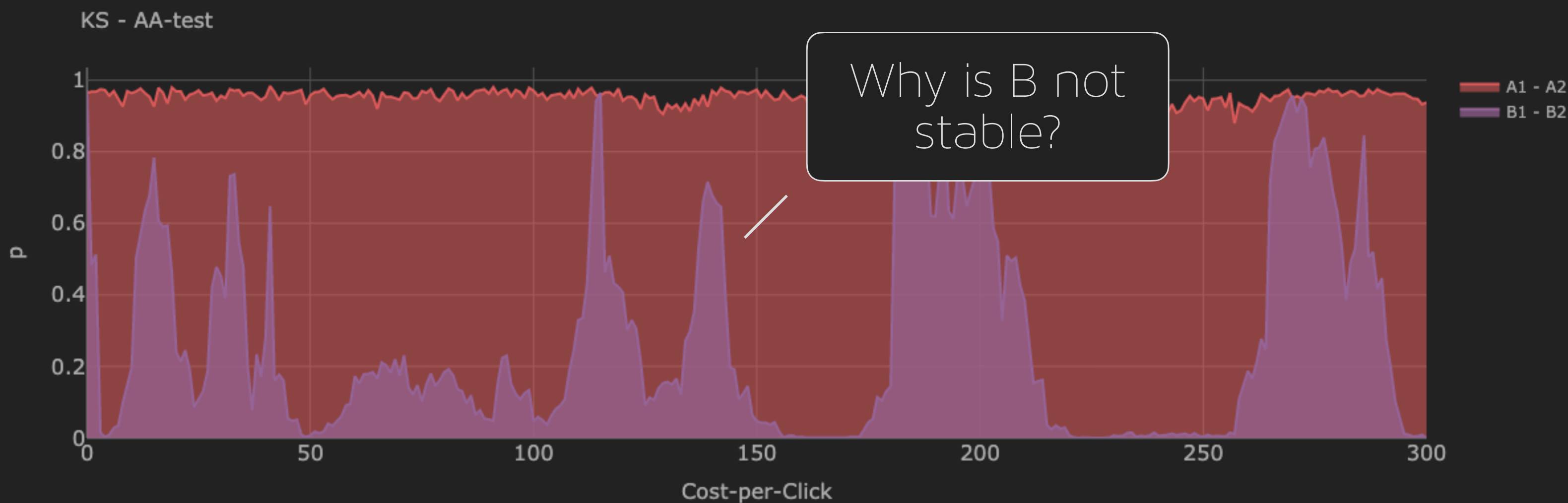
Bayesian Bandit A/A testing

Bandit A/A-test

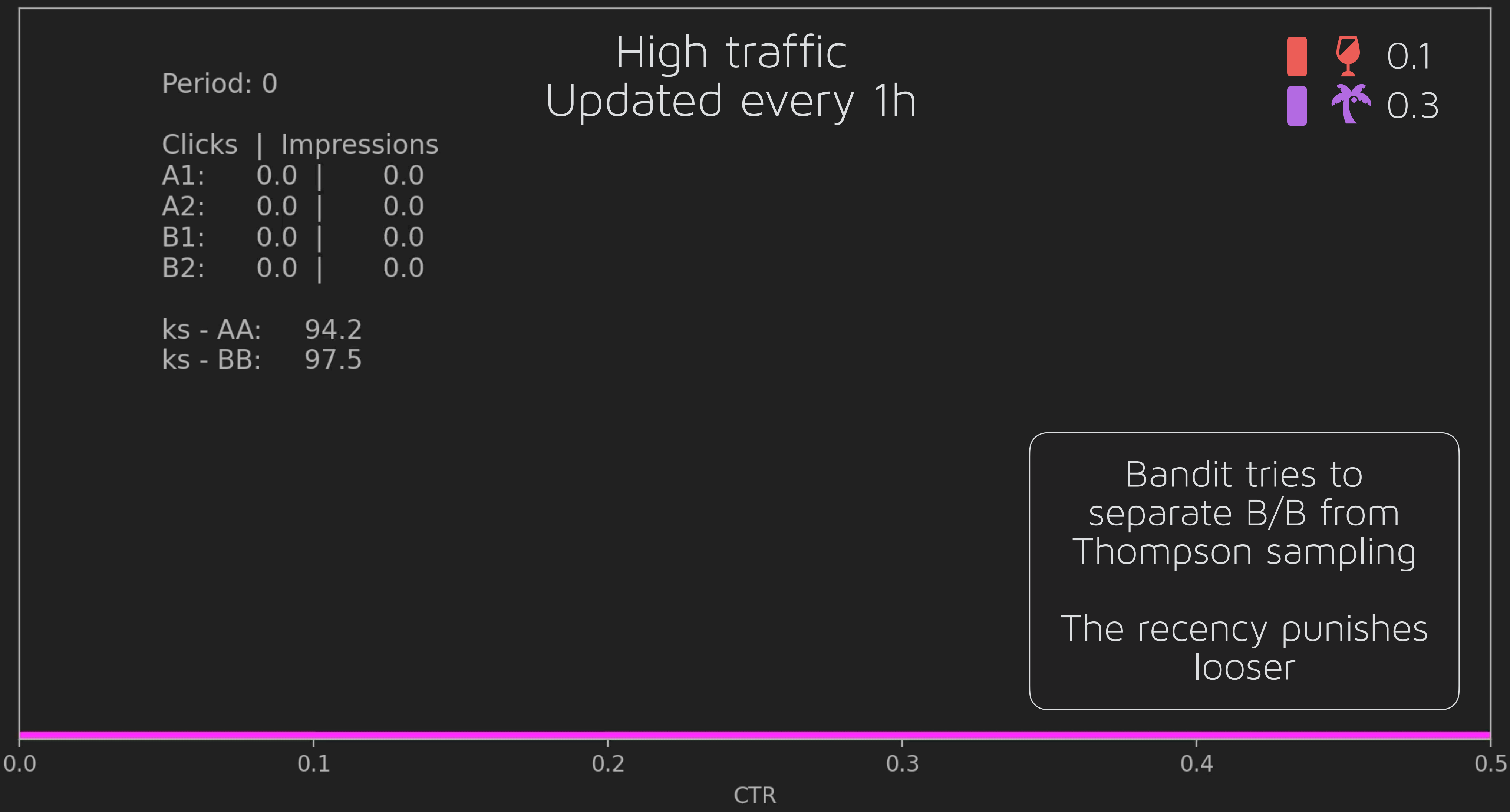
Both A/A and B/B seems similar

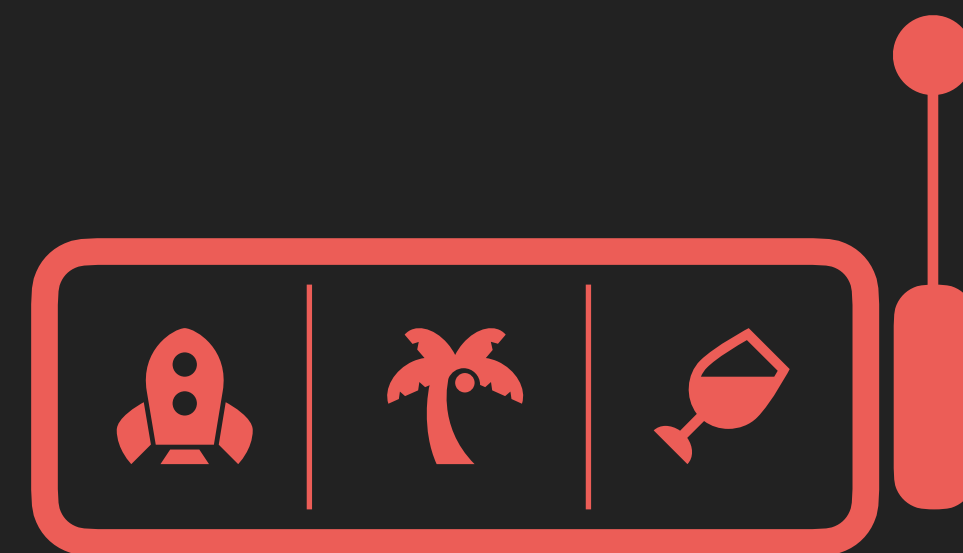


Kolmogorov-Smirnov
Similarity score



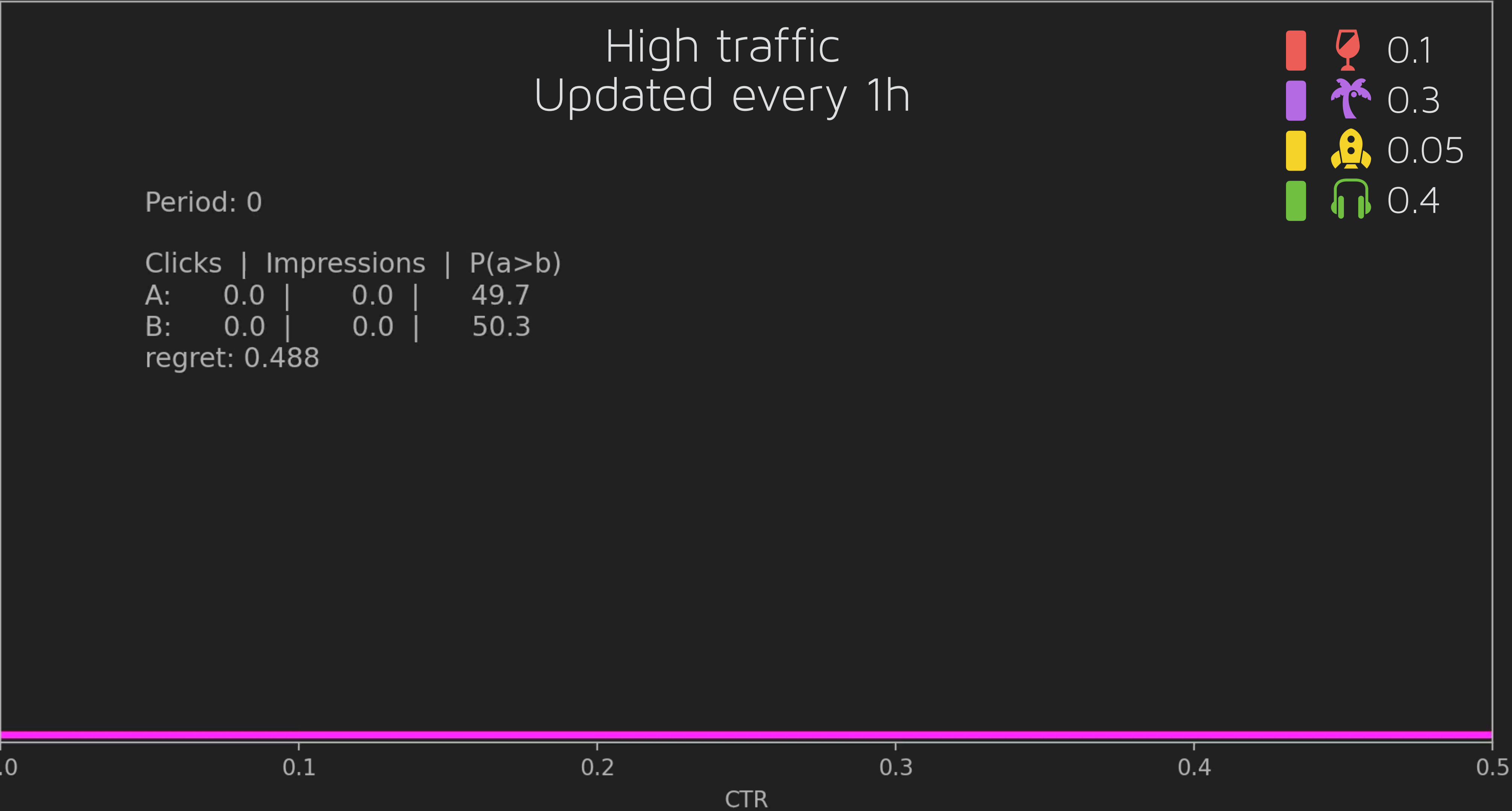
Bandit A/A-test





Cold starting your engine...

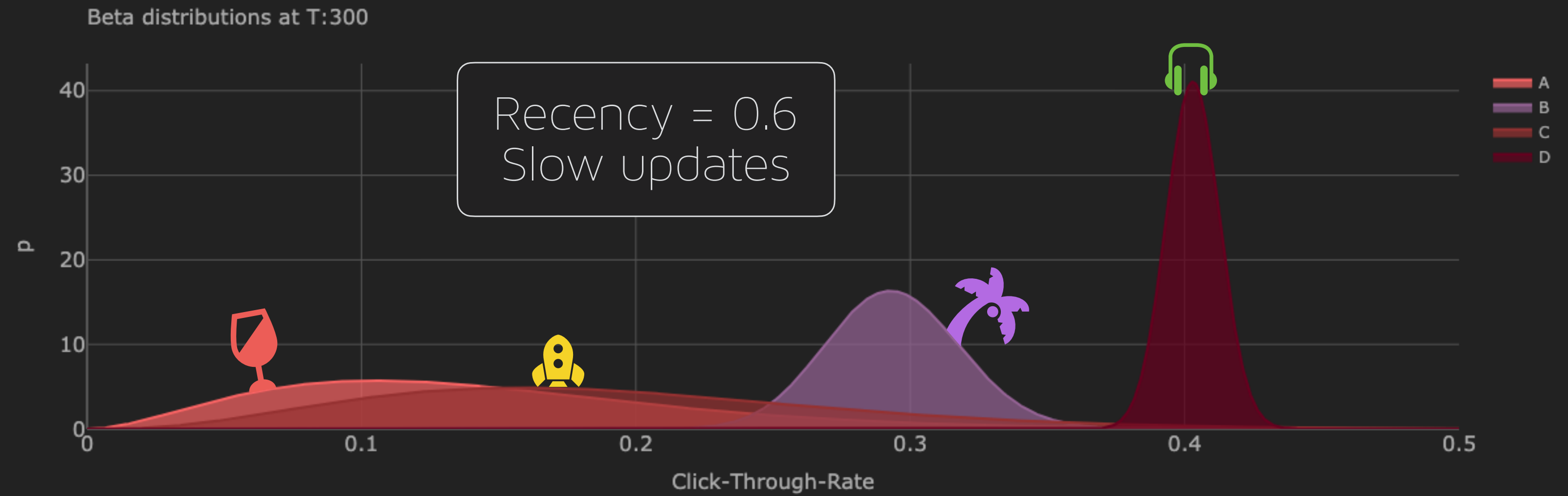
Bandit A/B/C/D-test



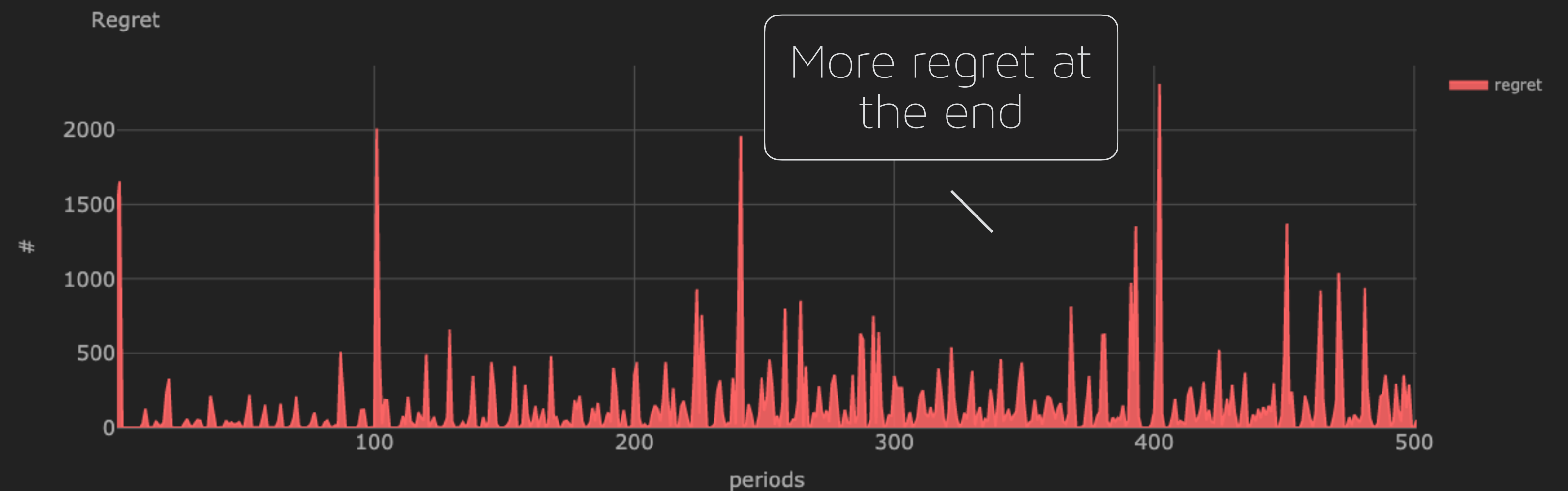
Bandit A/B/C/D-test

Beta Distributions

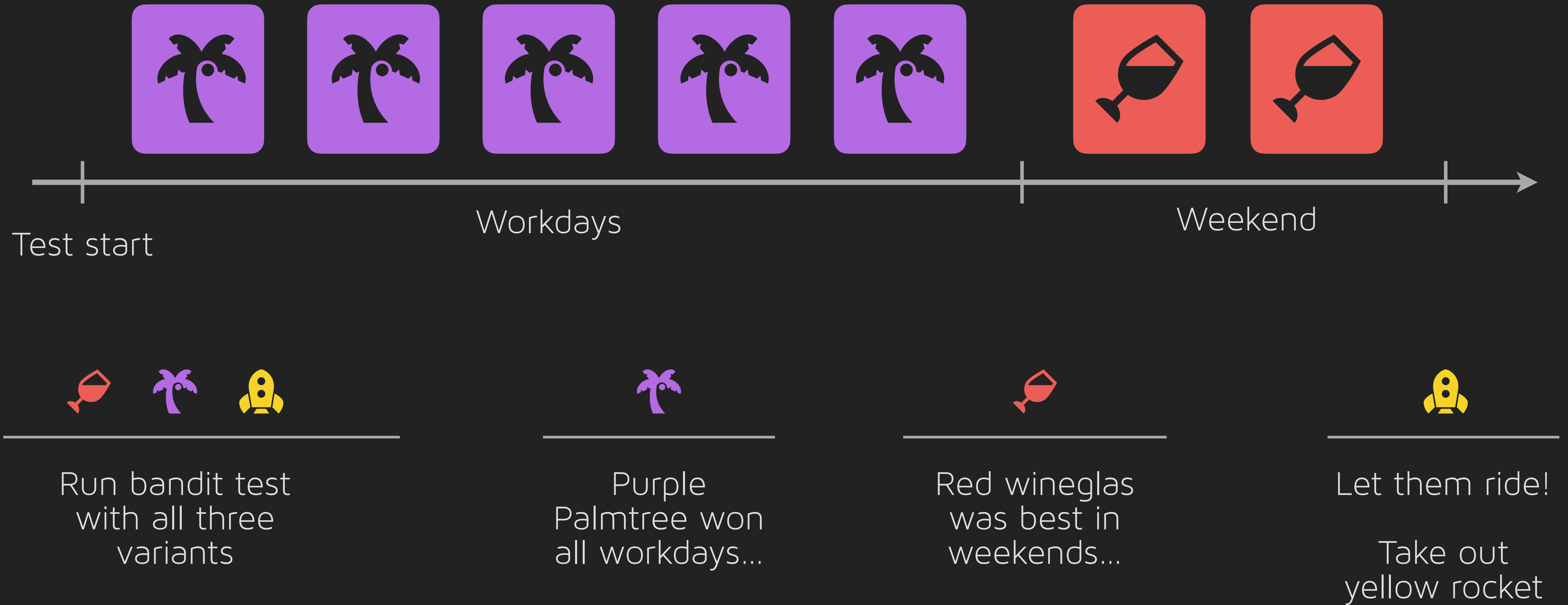
Example of the
recency making A
wide



This makes A get
selected more
often and we're
exploring...

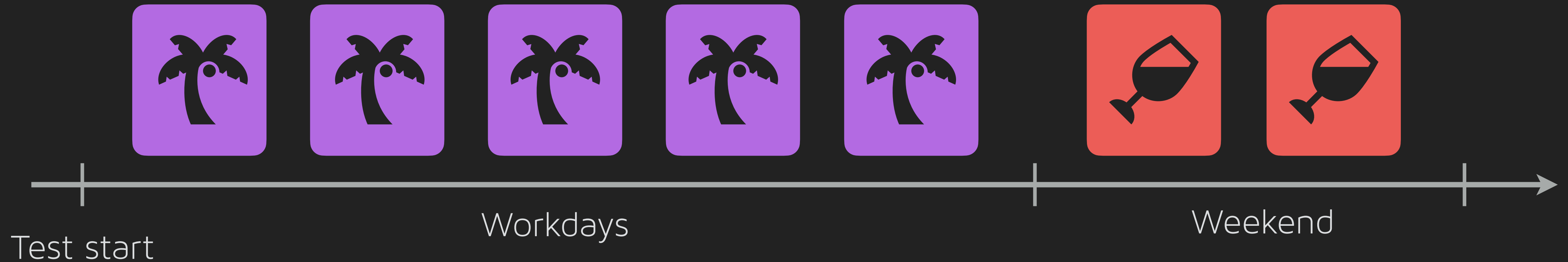


Look for losers instead...



In a normal A/B test - expensive to let it ride

However - in practice



Align with users / clients!

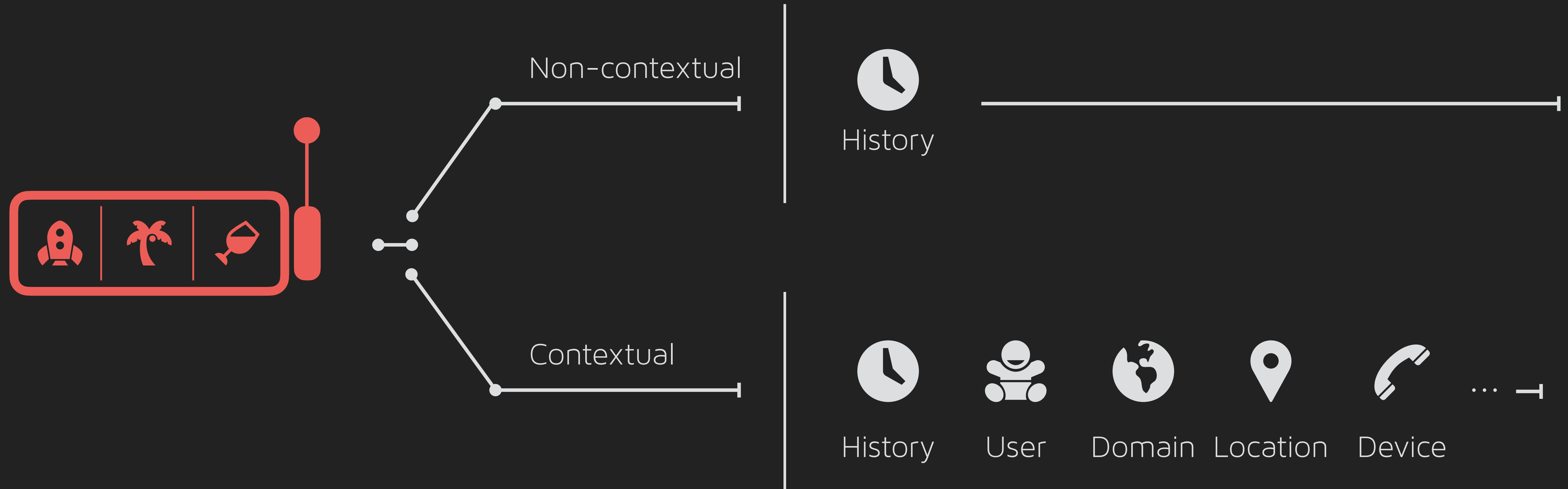
Keeping variants in the game
and not taking out losers
will make the bandit look like it
is not working.

Two use cases:

Traditional A/B-test - use to
find single winner

Bandit - keep running and
remove losers...

Things I didn't cover...



Other Metrics

Bayes Factor | False Detection Rate |



Stopping Criteria

ROPE metrics -

Bayesian Bandit End-of-Level summary

Practical



Know what you are doing!
You are the king/queen...



Be critical - visualise the
progression and assess,
not just the final numbers



Automated system, so let it ride
- not necessarily one winner.
Look for losers instead...

Business



Optimises while you sleep -
saving costs for running test.



Tests continuously...



Fire and forget....almost

Easy to inject new variants on
the fly.



Future

Bayesian Causal A/B-test

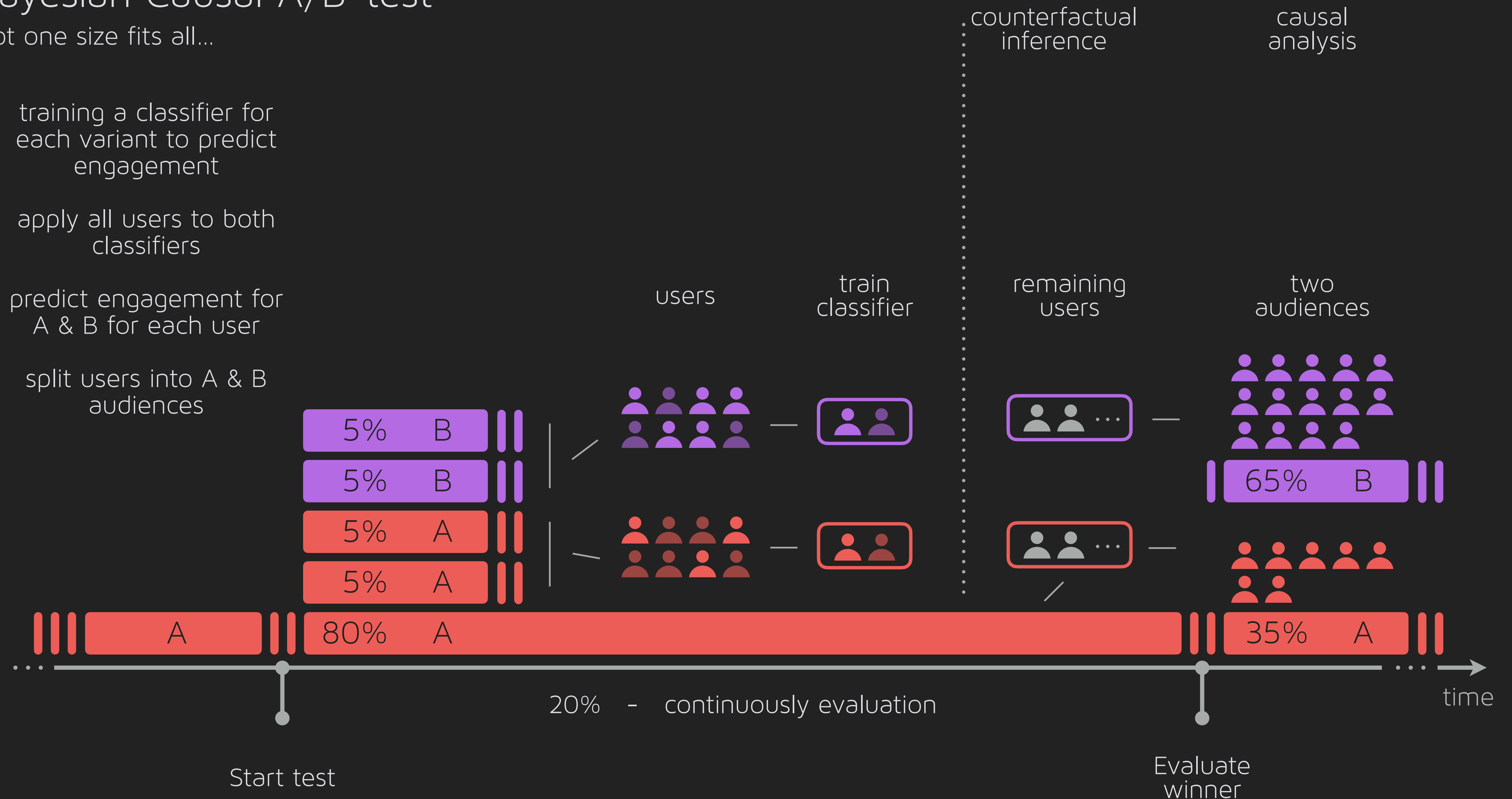
Not one size fits all...

training a classifier for each variant to predict engagement

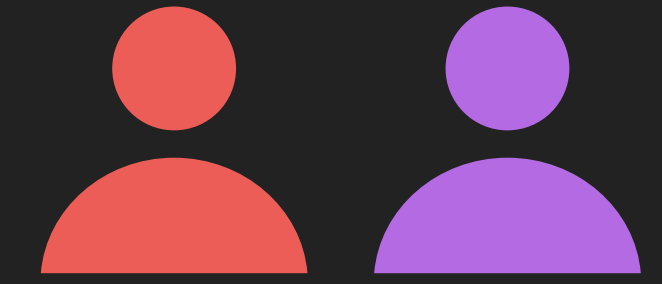
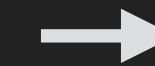
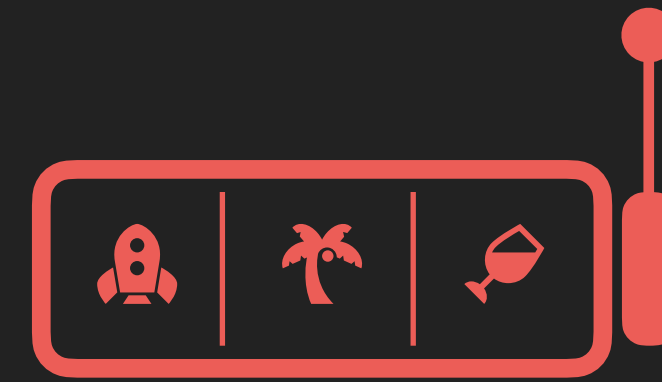
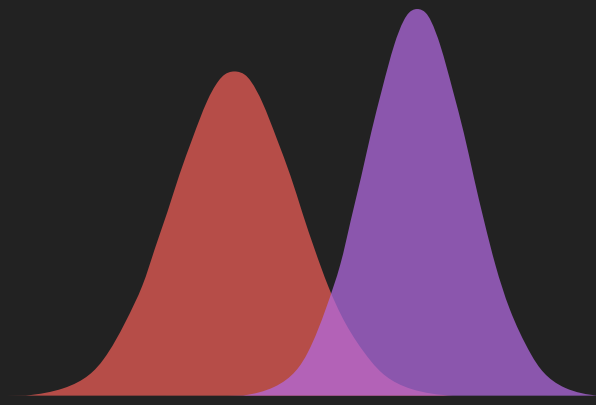
apply all users to both classifiers

predict engagement for A & B for each user

split users into A & B audiences



Started with a test and end with a complex AI system



AB-test

Bayesian AB-test

Bayesian Bandit

Counter Factual

Manual

Intuitive metrics

Automated – fire
and forget

Personalised

Winner takes
all

Faster convergence

Look for losers

More a full-blown
AI system



References

References

Calculating Sample Size for A/B Testing: Formulas, Examples & Errors

<https://splitmetrics.com/blog/mobile-a-b-testing-sample-size/> (paywall)

Kees Schippers

How to Calculate A/B Testing Sample Sizes?

<https://vwo.com/blog/how-to-calculate-ab-test-sample-size/>

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Probabilistic Programming and Bayesian Methods for Hackers

<https://github.com/CamDavidsonPilon/Probabilistic-Programming-and-Bayesian-Methods-for-Hackers>

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Alicia A. Johnson, Miles Q. Ott, Mine Dogucu

Bayes Rules! An Introduction to Applied Bayesian Modeling

<https://www.bayesrulesbook.com>

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Bayesian A/B-Testing

<https://www.evanmiller.org/bayesian-ab-testing.html>

John Kruschke

Rejecting or Accepting Parameter Values in Bayesian Estimation

<https://jkkweb.sit host.iu.edu/articles/Kruschke2018RejectingOrAcceptingParameterValuesWithSupplement.pdf>

Michael Frasco

The Power of Bayesian A/B Testing

<https://medium.com/convoy-tech/the-power-of-bayesian-a-b-testing-f859d2219d5>

Urteaga et al.

Bayesian bandits: balancing the exploration-exploitation tradeoff via double sampling

<https://arxiv.org/pdf/1709.03162.pdf>

John Cook

Exact Calculation of Beta Inequalities

https://www.johndcook.com/exact_beta_inequalities.pdf

Chris Stucchio

Bayesian A/B Testing at VWO

https://vwo.com/downloads/VWO_SmartStats_technical_whitepaper.pdf

kruschke

Bayesian Estimation Supersedes the t Test

<https://jkkweb.sit host.iu.edu/articles/Kruschke2013JEPG.pdf>

Hass et al.

Exact Bayesian Inference for A/B testing

<https://gist.github.com/lucidyan/89eee0db8ce353d91e6bfbc51b5dcf19>

Sureshkumar

Bayesian experimentation methods for products

<https://towardsdatascience.com/bayesian-experimentation-methods-for-products-636514951e43>

“Know what you
are doing”

A/B-TESTING

BAYESIAN - AND BANDITS

